

Clarkson Institute Summer School:

Group theory

Notes by James Moore
Email: jmm232@cam.ac.uk

I Group theory

1	Groups: definitions, examples, and isomorphism	5
1.1	Definition of a group	6
	<i>Definition of a group; statement of group axioms. Order of a group. Abelian groups.</i>	
1.2	Examples of groups from arithmetic	7
	<i>Additive and multiplicative groups of numbers. Modular addition and multiplication; group tables. Quaternions.</i>	
1.3	Examples of groups from geometry	11
	<i>Groups of symmetries. Rotation groups. Dihedral groups. Chiral and full tetrahedral groups.</i>	
1.4	Examples of groups from matrix algebra	19
	<i>Additive groups of matrices. The general linear group. Finite groups of matrices.</i>	
1.5	Isomorphic groups	20
	<i>Definition of isomorphism; examples from this chapter. Identities map to identities. Non-isomorphism.</i>	
2	Basic properties of group elements	24
2.1	Proving properties of abstract groups	25
	<i>Uniqueness of identity and inverses. Equation solving. Generalised associativity (without proof).</i>	
2.2	Properties of group tables	27
	<i>Latin square property of group tables. Classification of groups up to order four.</i>	
2.3	Power laws and periodicity	29
	<i>Exponent laws for group elements. Periodic elements; proof that in a finite group, elements have finite order.</i>	
2.4	Generators and relations	32
	<i>Generators of a group. Applications to the algebra of cyclic and dihedral groups. Group presentations.</i>	
3	Subgroup theory	35
3.1	Definition of a subgroup	36
	<i>Definition of a subgroup. Examples.</i>	
3.2	Cosets and Lagrange's theorem	37
	<i>Definition of left and right cosets. Proof that cosets partition a group. Statement and proof of Lagrange's theorem. Basic corollaries.</i>	
3.3	Extended example: cyclic and dihedral groups	39
	<i>Subgroup structure of cyclic and dihedral groups.</i>	
3.4	A converse to Lagrange's theorem: Cauchy's theorem	42
	<i>Cauchy's theorem as a partial converse of Lagrange's theorem. Falsity of the converse of Lagrange's theorem.</i>	

3.5	Classification of finite groups up to order seven	44
	<i>Use of Lagrange's theorem and Cauchy's theorem to classify groups to order seven.</i>	
4	Product groups	45
4.1	Direct products	46
	<i>Definition of the direct product. Examples. Basic properties.</i>	
4.2	The direct product theorem	48
	<i>Statement and proof of the direct product theorem. Example applications.</i>	
4.3	Classification of finite Abelian groups	51
	<i>Inductive proof of the fundamental theorem of finite Abelian groups.</i>	
4.4	Classification of finite groups up to order 11	53
	<i>Use of direct products to classify groups up to and including order 11.</i>	
5	Conjugacy and normality	55
5.1	Conjugate elements	56
	<i>Definition of conjugacy. Geometric intuition. Extended examples in D_4. Conjugates of identity; conjugates in Abelian groups.</i>	
5.2	Properties of conjugate elements	58
	<i>Basic properties. Conjugacy classes partition a group. Size of a conjugacy class divides the group order.</i>	
5.3	Normal subgroups	61
	<i>Normal subgroups defined through conjugacy classes. Examples. Equivalent definitions of normality.</i>	
5.4	Groups modulo subgroups; or, quotient groups	62
	<i>Motivation for quotient groups; necessity of normal subgroups. Formal definition. Examples.</i>	
5.5	Extended example: cyclic and dihedral groups	66
	<i>Conjugacy, normality, and quotients in cyclic and dihedral groups.</i>	
5.6	Simple groups	70
	<i>Definition of a simple group. Examples. Brief discussion of classification of finite simple groups.</i>	
6	Maps between groups	71
6.1	Homomorphisms	72
	<i>Definition and examples. Basic properties: identity, inverses, orders. Example of all homomorphisms $C_4 \rightarrow C_6$.</i>	
6.2	Homomorphisms, generators, and relations	74
	<i>Homomorphisms are uniquely determined by the image of generators, modulo relations. Example of $C_n \rightarrow C_m$.</i>	
6.3	Kernel and image of a homomorphism	75
	<i>Definition of kernel and image; examples. Proof that kernel is a normal subgroup, image is a subgroup.</i>	
6.4	The isomorphism theorem	77
	<i>Statement and proof of the isomorphism theorem. Example applications.</i>	
6.5	Automorphisms	78
	<i>Definition and examples. The automorphism group. Conjugacy as an automorphism.</i>	
7	Permutation groups	79
7.1	Permutations and Cayley's theorem	80
	<i>Definition of a permutation of a set. Proof that permutations form a group. Statement and proof of Cayley's theorem.</i>	
7.2	Permutations of a finite set	81
	<i>The symmetric group S_n. Decomposition of a permutation into disjoint cycles & into transpositions. The alternating group A_n.</i>	
7.3	Conjugacy in the permutation groups	86
	<i>Conjugacy in S_n. Condition for splitting of S_n conjugacy classes in A_n.</i>	

II Representation theory

8	Permutation representations, or group actions	89
8.1	Permutation representations: basic definitions and examples	90
	<i>Definition of a permutation representation. Examples.</i>	
8.2	Orbits, stabilisers, and the orbit-stabiliser theorem	91
	<i>Definition of orbit and stabiliser. Examples. Statement and proof of the orbit-stabiliser theorem.</i>	
9	Linear representations: definitions, examples, and equivalence	93
9.1	Definition of a representation	94
	<i>Definition of a representation. Faithfulness.</i>	
9.2	Examples of representations	94
	<i>Trivial representation. Proof that every finite group has a representation.</i>	
9.3	Equivalence of representations	100
	<i>Equivalence of representations; intertwining matrices. Examples.</i>	
9.4	Inequivalence of representations: a first look at characters	102
	<i>Inequivalence of representations through invariants.</i>	
10	Reducibility to direct sums	104
10.1	Complex vector spaces, scalar products, and unitary matrices	105
	<i>Vector subspaces in $\mathbb{C}^n$. Complex scalar products, abstract scalar products, and Gram-Schmidt orthogonalisation. Unitary matrices.</i>	
10.2	Reducibility, invariant subspaces, and subrepresentations	110
	<i>Reducible representations and invariant subspaces. Subrepresentations. Equivalence to upper triangular form.</i>	
10.3	Direct sums	112
	<i>Direct sum of two representations.</i>	
10.4	Unitarity and Maschke's theorem	113
	<i>Unitarity of a representation. Unitary representations are completely reducible to direct sums. Maschke's theorem.</i>	
11	The great orthogonality theorems	117
11.1	Schur's lemmas	118
	<i>Statement and proof of Schur's lemmas. Abelian groups only have one-dimensional irreducible representations.</i>	
11.2	The great orthogonality theorem	119
	<i>Statement and proof of the great orthogonality theorem.</i>	
11.3	Character tables	123
	<i>Character tables of irreducible representations.</i>	
11.4	Exercises	124
12	Direct sums and Maschke's theorem	125
12.1	Direct sums	126
	<i>Direct sums. Examples. Behaviour of characters under direct sums.</i>	
12.2	Inner products and unitary representations	127
	<i>Complex inner products; the Gram-Schmidt procedure. Examples of weighted inner products. Unitary representations.</i>	
12.3	Maschke's theorem	128
	<i>The group-invariant inner product. Statement and proof of Maschke's theorem.</i>	

Part I

Group theory

Chapter 1

Groups: definitions, examples, and isomorphism

Many objects in mathematics have similar properties under certain operations, despite seeming very different. For example, consider the set of complex numbers (excluding zero) under multiplication, and the set of n -dimensional real vectors under vector addition. These two sets and operations both satisfy very similar rules:

Complex numbers, excluding zero	n -dimensional vectors
If z, w are complex numbers, then their product is also a complex number: $z \cdot w$.	If $\mathbf{v}, \mathbf{w}$ are vectors, then their sum is also a vector: $\mathbf{v} + \mathbf{w}$.
Given three complex numbers z, w, u , it does not matter where we put the brackets in a product. We have: $z \cdot (w \cdot u) = (z \cdot w) \cdot u.$	Given three vectors $\mathbf{v}, \mathbf{w}, \mathbf{u}$, it does not matter where we put the brackets in a sum. We have: $\mathbf{v} + (\mathbf{w} + \mathbf{u}) = (\mathbf{v} + \mathbf{w}) + \mathbf{u}.$
There is a complex number, 1, satisfying: $1 \cdot z = z = z \cdot 1$ for all complex numbers z .	There is a vector, $\mathbf{0}$, satisfying: $\mathbf{0} + \mathbf{v} = \mathbf{0} = \mathbf{v} + \mathbf{0}$ for all vectors $\mathbf{v}$.
For every complex number $z \neq 0$, there is a complex number $1/z$ satisfying: $z \cdot (1/z) = 1 = (1/z) \cdot z.$	For every vector $\mathbf{v}$, there is a vector $-\mathbf{v}$ satisfying: $\mathbf{v} + (-\mathbf{v}) = \mathbf{0} = (-\mathbf{v}) + \mathbf{v}.$

The modern mathematical subject of *abstract algebra* unifies seemingly distinct sets and operations such as these. It does so by studying the *algebraic rules* that structures have in common, rather than studying the structures individually. By studying the algebraic rules in their own right, we can deduce properties of many different structures in mathematics at the same time (e.g. properties applying to complex number multiplication, and also to vector addition).

The most basic example of a collection of algebraic rules are the four *group axioms*;¹ when we study the group axioms we are doing *group theory*, the subject of this course. The properties we compared above are the group axioms, which we shall formalise shortly.

Perhaps most surprisingly, the group axioms are satisfied by all sets of *symmetries* of objects. This means that people often refer to group theory as the ‘study of symmetry’, and this is a key idea which will emerge throughout the text.

¹More advanced common examples include the axioms for rings, fields, vector spaces, and modules.

In this introductory chapter, we shall cover the following:

- First, we shall define a *group*, a set with an algebraic operation which satisfies the *group axioms*. We shall define the terms *order of a group* (referring to the group's size) and *Abelian groups* (which satisfy an additional, important axiom).
- In the main part of the chapter, we shall give many examples of groups, some familiar and some unfamiliar. The examples we shall cover are the following:
 - Groups of numbers: the integers, rationals, reals, and complex numbers under addition and multiplication; modular groups of integers under addition and multiplication; quaternions.
 - Groups of symmetries: the cyclic group of rotations; the dihedral group of rotations and reflections; the chiral and full tetrahedral groups.
 - Groups of matrices: $n \times n$ matrices under addition and matrix multiplication; some finite groups of matrices.
- Finally, we shall introduce the idea of two groups being the 'same' - *isomorphism*. This helps to motivate one of the central tasks for the first part of the course: classifying groups of small order up to isomorphism.

1.1 Definition of a group

Following on from the introductory motivation, we shall begin with the formal mathematical definition of a group.

Definition 1.1: A *group* is a set G together with a *binary operation* $*$ (a rule for combining two elements of G to produce something new) satisfying the following four properties, called the *group axioms*:

(G1) *Closure*. For all $x, y \in G$,² we have $x * y \in G$. That is, when we combine two elements x, y under the binary operation, then we stay in the original set.

(G2) *Associativity*. For all $x, y, z \in G$, we have:

$$(x * y) * z = x * (y * z).$$

That is, it does not matter where we place the brackets in the multiplication of three elements, so we can unambiguously write $x * y * z$.

(G3) *Identity*. There exists some special element, $e \in G$, such that for all other elements $x \in G$, we have:

$$e * x = x \quad \text{and} \quad x * e = x.$$

(G4) *Inverses*. For each element $x \in G$, there exists a special element $x^{-1} \in G$ such that:

$$x * x^{-1} = e \quad \text{and} \quad x^{-1} * x = e,$$

where e is the identity element from axiom (G3).

²The symbol $\in$ means 'is a member of the set'.

These four basic rules, whilst deceptively simple, contain lots of rich mathematics which we shall explore in subsequent chapters. We start here by focussing on examples though - this helps us understand the definition further, and also gives us lots of examples to play with when we study the general theory in later chapters.

We have now already seen two examples: the set of complex numbers excluding zero forms a group under multiplication, and the set of vectors forms a group under addition. We will see many more throughout this chapter.

Before diving in further, though, it is useful to make two further definitions about groups.

Definition 1.2: The *order* of a group G is the number of elements in the group. If there are infinitely many elements, the group is said to have *infinite order*. The order is often written as $|G|$.

The two groups we have seen so far, the complex numbers excluding zero, and the set of vectors, are both *infinite order* groups. We will see lots of examples of both infinite order and finite order groups later in this chapter.

Definition 1.3: An *Abelian group* is a group G satisfying the addition fifth axiom:

(G5) *Commutativity*. For all $x, y \in G$, we have:

$$x * y = y * x.$$

That is, it does not matter the order in which we multiply elements in the group.

The two groups we have seen so far are also both Abelian groups. This is because for any two complex numbers z, w , we have $z \cdot w = w \cdot z$. For any two vectors $\mathbf{v}, \mathbf{w}$, we have $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$. Not all groups are Abelian though - we will see an example in the next section.

1.2 Examples of groups from arithmetic

Similar to our complex number example, many familiar examples of groups come from basic arithmetic:

- **The integers, rationals, reals, and complex numbers under addition.** The set of integers, written $\mathbb{Z}$, is a group under the operation of addition. Checking each of the axioms in turn, we have:

(G1) *Closure*. Given two integers a, b , we have $a + b$ is an integer.

(G2) *Associativity*. Given three integers a, b, c , we have $(a + b) + c = a + (b + c)$.

(G3) *Identity*. The identity is 0, since for all integers a , we have $0 + a = a = a + 0$.

(G4) *Inverses*. The inverse of the element a is $-a$, since $a + (-a) = 0 = (-a) + a$.

Similarly, the set of rational numbers $\mathbb{Q}$ is a group under addition, the set of real numbers $\mathbb{R}$ is a group under addition, and the set of complex numbers $\mathbb{C}$ is a group under addition.

All of these groups have infinite order. They are also Abelian groups, since $a + b = b + a$ in all of these sets.

- **The rationals, reals, and complex numbers under multiplication.** The set of rationals excluding zero, written $\mathbb{Q}^*$, is a group under multiplication. Checking each of the axioms in turn, we have:

(G1) *Closure.* Given two rationals p, q , we have pq is a rational.

(G2) *Associativity.* Given three rationals p, q, r , we have $(pq)r = p(qr)$.

(G3) *Identity.* The identity is 1, which is a rational, since for all rationals p , we have $1 \cdot p = p = p \cdot 1$.

(G4) *Inverses.* The inverse of the element p is $1/p$, which exists since p is not zero. This satisfies $(1/p) \cdot p = 1 = p \cdot (1/p)$.

Similarly, the set of real numbers excluding zero $\mathbb{R}^*$ is a group under multiplication, and the set of complex numbers excluding zero $\mathbb{C}^*$ is a group under multiplication. Again, all of these groups have infinite order and are Abelian groups.

Note that $\mathbb{Q}$, including zero, is *not* a group under multiplication because 0 has no inverse.³ Similarly, the integers excluding zero are not a group under multiplication, because - for example - 2 has no inverse.

- **The integers modulo n under modular addition.** A perhaps less familiar example of a group comes in the form of the integers modulo n , written $\mathbb{Z}_n$, under addition. The set $\mathbb{Z}_n$ is defined to be the set of all possible remainders when an integer is divided by n ,

$$\mathbb{Z}_n = \{0, 1, 2, 3, \dots, n-2, n-1\}.$$

Whenever we perform an arithmetic operation on the set $\mathbb{Z}_n$, we only keep the remainder after division by n .

More concretely, we can consider the example of $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$. In this set, we have $2 + 2 = 4$, but we also have $2 + 3 = 0$, since $2 + 3$ has zero remainder when divided by 5. Similarly, we have $2 + 4 = 1$. A nice way of visualising this kind of arithmetic is to imagine that $\mathbb{Z}_5$ is represented as a clock face, as shown in Fig. 1. When we add 2 and 3, we are 'adding 3 hours to 2 o'clock' on the clock face, taking us back to 0, for example.

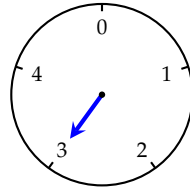


Figure 1: A clock is a useful way of visualising the set $\mathbb{Z}_n$; the clock face above shows the set $\mathbb{Z}_5$.

We can check that $\mathbb{Z}_n$ under addition is indeed a group by verifying the group axioms:

(G1) *Closure.* If $a, b \in \mathbb{Z}_n$, then $a + b \in \mathbb{Z}_n$ by definition.

(G2) *Associativity.* This is inherited from the associativity of addition in the full set of integers.

(G3) *Identity.* The identity is 0, since $0 + a = a = a + 0$ for all $a \in \mathbb{Z}_n$.

(G4) *Inverses.* The inverse of the element a is $n - a$ (for all $a = 1, \dots, n - 1$), since $a + (n - a) = 0 = (n - a) + a$ (and the inverse of 0 is just 0).

³The particularly strident reader may wish to define a new element $\infty := 0^{-1}$, to extend the rationals to include infinity too. However, this approach is fraught with peril. To see why, consider the resulting group element $\infty \cdot \infty$. If $\infty \cdot \infty = \infty$, then we could multiply on the left by ∞^{-1} to give $\infty = 1$, which is a contradiction. On the other hand, if $\infty \cdot \infty = x$ for some finite rational, multiplying on the left by ∞^{-1} we would get $\infty = \infty^{-1} \cdot x = 0 \cdot x = 0$, which is also a contradiction.

For finite groups like $\mathbb{Z}_n$, there is a useful visual aid to help us understand the effect of the group operation on the elements of the group.

Definition 1.4: Let G be a finite group with operation $*$. The *group table* (or *Cayley table*) of the group G is defined to be a table with rows labelled by each of the group elements, and columns labelled by each of the group elements. In the (g, h) -entry of the table, we write the product $g * h$ of the group elements.

As an example, the group tables of $\mathbb{Z}_3$ and $\mathbb{Z}_4$ are shown in Fig. 2.

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Figure 2: The group tables for $\mathbb{Z}_3$ and $\mathbb{Z}_4$ under addition.

Some of the group axioms are naturally seen from a group table's structure: note that the closure of the operation is represented in the table by the entries all being drawn from the row and column labels, and the existence of an identity is represented by a row whose entries are identical to the column labels, and a column whose entries are identical to the row labels. For both of the above group tables, the symmetry of the table about the leading diagonal represents the fact that the group is Abelian.

- **The integers modulo n under modular multiplication.** Similarly, we can consider the set $\mathbb{Z}_n$ under multiplication of integers. However, here we must be more careful - usually, this set does *not* form a group!

Let's begin by considering $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ under multiplication. Constructing the candidate multiplication table, we have:

$\cdot$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

The table shows that if $\mathbb{Z}_4$ was a group, we would have that 1 was the identity. But then 0, for example would be non-invertible, because 1 does not appear in the 0 row or 0 column. Similarly, 2 would be non-invertible, because 1 does not appear in the 2 row or 2 column.

To fix this problem, we introduce the set $\mathbb{Z}_n^*$, which is the set of all possible remainders when an integer is divided by n , which *also possess a multiplicative inverse*. For example, in the above we have $\mathbb{Z}_4^* = \{1, 3\}$.

We claim that $\mathbb{Z}_n^*$ is indeed a group under multiplication; we can check the axioms as follows:

- (G1) *Closure.* Suppose that $a, b \in \mathbb{Z}_n^*$. Certainly we have $ab \in \mathbb{Z}_n$. Further, we claim that ab has a multiplicative inverse; this follows because a, b both have multiplicative inverses a^{-1}, b^{-1} . We can then construct $a^{-1}b^{-1}$ as the inverse of ab , since $(ab)(a^{-1}b^{-1}) = 1$.
- (G2) *Associativity.* This is inherited from associativity of multiplication in the full set of integers.
- (G3) *Identity.* We have that 1 is the identity, since $1 \cdot a = a = a \cdot 1$ for all $a \in \mathbb{Z}_n^*$.

(G4) *Inverses.* By definition of $\mathbb{Z}_n^*$, every element has an inverse. That is, for each $a \in \mathbb{Z}_n^*$, there exists some $a^{-1} \in \mathbb{Z}_n^*$ such that $a \cdot a^{-1} = 1 = a^{-1} \cdot a$.

It can be shown generally that an element a of $\mathbb{Z}_n$ is in $\mathbb{Z}_n^*$ if and only if a and n are coprime (i.e. they possess no common factors, except one); this requires a bit of number theory, though, and takes us slightly off course!⁴ This fact tells us that the order of $\mathbb{Z}_n^*$ is the number of positive integers less than n which are coprime to n ; this is the *Euler totient function* $\phi(n)$.

- **The quaternions.** To finish our survey of groups of numbers, we will give a completely unfamiliar example: the set of *quaternions*. Like the complex numbers, the quaternions extend the real numbers with some new symbols - but unlike the real numbers, the quaternions introduce *three* new formal square roots of negative one, written i, j, k , satisfying:

$$i^2 = j^2 = k^2 = -1.$$

The general form of a quaternion is $a + bi + cj + dk$ where a, b, c, d are real numbers. Now, note that unlike complex numbers, in order to work out the product of two quaternions, e.g. $(1 + i)(2 + j) = 2 + 2i + j + ij$, we need more than just i^2, j^2, k^2 to be known; we additionally need to know the products ij, ik, jk , etc.

The key relation is:⁵

$$ijk = -1. \quad (\dagger)$$

Multiplying $(\dagger)$ on the right by k , we have $ijk^2 = -k$, which gives $ij = k$. Multiplying $(\dagger)$ on the left by ji , we have:

$$ji \cdot ijk = -ji \quad \Rightarrow \quad k = -ji.$$

So we see that $ij = k$, but $ji = -k$. Thus, *multiplication of quaternions is not commutative*. This should surprise you, since multiplication of complex numbers *is* commutative!

Repeatedly using the relation $(\dagger)$, we can similarly derive the following multiplication table for the basic quaternions i, j, k .

$\cdot$	i	j	k
i	-1	k	$-j$
j	$-k$	-1	i
k	j	$-i$	-1

This allows us to compute arbitrary products of quaternions, for example the above product can be computed as $(1 + i)(2 + j) = 2 + 2i + j + k$.

It is true that the set of quaternions forms a group under addition, and the set of non-zero quaternions forms a group under multiplication, just like the complex numbers from earlier. The proof in the multiplicative case is quite long and tedious though, and involves a lot of nasty algebra - we therefore omit it from this text (you may like to attempt it as an exercise though).

⁴This can be established quickly using *Bézout's identity*, if you know it.

⁵A bit of history: the discovery of quaternions was first published in 1843 by William Rowan Hamilton (although Gauss wrote privately about them in 1819), in his quest to generalise the geometric properties of complex numbers (as scaled rotations of the plane) to three dimensions. He discovered that it was in fact impossible to introduce just two new complex units (i, j), but that three were required (i, j, k), obeying the fundamental relations given above. Three-dimensional rotations are therefore 'hidden' inside some four-dimensional space. Hamilton discovered this fact when he was walking with his wife along the Royal Canal in Dublin. He carved the relations given above with a pen-knife into a bridge on the way, which you can still see today!

More generally, Hamilton was an exceptionally interesting character, who contributed not only to the theory of quaternions, but also to fundamental physics - the *Hamiltonian* is an absolutely fundamental concept in both classical and quantum physics. He is also famous for his contribution to the variational principle governing optics. See https://www.youtube.com/watch?v=SZXH0WwBcDc&list=RDSZXH0WwBcDc&start_radio=1 for a fun summary!

However, it is quite easy to identify a small, interesting, finite subset of quaternions which form a group: the set of quaternions $\{1, -1, i, j, k, -i, -j, -k\}$ form a group under multiplication called Q_8 . Its (rather lengthy) multiplication table is given below.

$\cdot$	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	$-j$	j
$-i$	$-i$	i	1	-1	$-k$	k	j	$-j$
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Figure 3: The set of eight basic quaternions $Q_8 = \{1, -1, i, j, k, -i, -j, -k\}$ forms a group under multiplication, with the above group table. Note that it is non-Abelian, since e.g. $ij = k$ but $ji = -k$.

1.3 Examples of groups from geometry

Next, we shall see groups arise in an area seemingly unrelated to algebra: sets of *symmetry transformations* of geometric objects. As alluded to in the introduction, the set of symmetry transformations of *any* geometric object forms a group, which is one of the reasons people often call group theory the ‘study of symmetry’.⁶

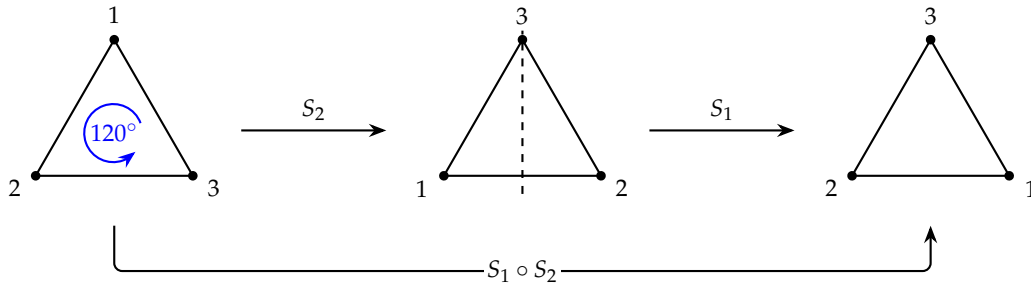


Figure 4: An example of composition of two symmetry transformations. Here, S_2 is a rotation of an equilateral triangle by an angle 120° anticlockwise about the origin; the vertices are labelled to help track the transformation. The transformation S_1 is a reflection of the triangle in the vertical axis of symmetry. This is equivalent to some new symmetry transformation $S_1 \circ S_2$ (in this case, a reflection about the axis through the initial 2-vertex).

In order to prove this rather remarkable fact, we must first decide on the operation which combines symmetry transformations. Whenever we deal with groups of symmetries, the operation is called *composition of symmetries*. Given one symmetry transformation, S_1 , and another symmetry transformation, S_2 , the *composition* $S_1 \circ S_2$ is the symmetry transformation that results from applying S_2 first, then S_1 (note the order carefully!). An example of this is given in Fig. 4.

⁶Perhaps even more surprisingly, the converse is also true: *any* group can be regarded as the set of symmetries of *some* object. This result is called *Cayley's theorem*, and we shall prove it later in the course. For example, the group $\mathbb{R}$ under addition is the set of all translational symmetries of a straight line - the real number $x \in \mathbb{R}$ tells you how much to translate by!

Now we have identified the group operation, we can prove that the set of symmetry transformations of any object form a group under composition of symmetries.

- *Closure.* If we perform a symmetry transformation S_2 followed by another symmetry transformation S_1 , the resulting overall transformation $S_1 \circ S_2$ must also be a symmetry transformation (at all stages, the geometric object is left invariant). Hence the composition $S_1 \circ S_2$ is in the set, so the operation is closed.
- *Associativity.* Suppose we have three symmetry transformations S_1, S_2, S_3 . Then the sequence of doing S_3 first, then S_2 , then S_1 can be factored in two ways:

$$S_1 \circ (S_2 \circ S_3) = (S_1 \circ S_2) \circ S_3.$$

The first way regards S_2 and S_3 as a single combined transformation, while the second regards S_1 and S_2 as a single combined transformation. In either case, exactly the same three transformations are performed in exactly the same order, so the overall result is identical. Hence composition of symmetry transformations is associative.

- *Identity.* Doing nothing is a symmetry transformation - the identity, I .
- *Inverses.* A symmetry transformation is necessarily invertible; if a map S is a symmetry, it maps all points in an object to other points in the object. Each point has to be mapped to exactly once, otherwise two distinct points would occupy the same position after the transformation, or some point of the object would fail to be reached. Hence, symmetry transformations are *invertible maps*; indeed S^{-1} must exist. Moreover, S^{-1} is also a symmetry transformation, since it simply undoes the action of S , returning every point of the object to its original position. Thus every symmetry transformation has an inverse which is also a symmetry.

Now, we can give some examples:

- **Rotation groups.** The set of all rotational symmetries of a regular n -gon forms a group, called the *cyclic group of order n* , and written C_n . The elements of the group can be written as:

$$C_n = \{I, R_{2\pi/n}, R_{4\pi/n}, R_{6\pi/n}, \dots, R_{2(n-1)\pi/n}\},$$

where I is the identity symmetry (i.e. doing nothing to the n -gon), and R_θ is an anticlockwise rotation of the n -gon about its centre by an angle θ (measured in radians).

We know that this forms a group, because it is the set of all possible symmetries of a *directed* regular n -gon; see Fig 5. To see this, label the vertices 1 to n . A symmetry must map a vertex to a vertex, so vertex 1 has n choices as to where it can go. After this, the entire symmetry is determined, because the orientation of the n -gon must stay the same (as it has a direction).⁷

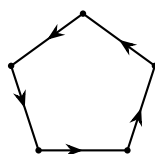


Figure 5: A directed regular pentagon (i.e. a regular pentagon with arrows on each side); rotations are symmetry transformations, but reflections are not.

⁷The eagle-eyed reader will have noticed that there is a bit of an issue with this argument. We could, for example, do a transformation that just scales one of the sides non-linearly - it stretches one part of the edge more than another. If we demand that distances between points are fixed by the transformation, i.e. it is an *isometry*, then the only possibilities become rotations and reflections as discussed in this chapter.

Alternatively, we can verify the group axioms directly. We notice that the composition of symmetries in this case can be expressed as the group law:

$$R_\theta R_\phi = R_{\theta+\phi},$$

where $I = R_0$, and we adjust the angle $\theta + \phi$ so that it is taken in the range $0 \leq \theta + \phi < 2\pi$, if necessary.

Verifying the group axioms, we have:

- (G1) The set is clearly closed under the operation.
- (G2) Note that $(R_\theta R_\phi)R_\tau = R_{\theta+\phi}R_\tau = R_{(\theta+\phi)+\tau}$ and $R_\theta(R_\phi R_\tau) = R_\theta R_{\phi+\tau} = R_{\theta+(\phi+\tau)}$, so associativity follows from the associativity of real numbers.
- (G3) The identity is $I = R_0$, since $R_\theta I = R_\theta = IR_\theta$.
- (G4) The inverse of R_θ is $R_{2\pi-\theta}$, since $R_\theta R_{2\pi-\theta} = R_0 = I = R_0 = R_{2\pi-\theta} R_\theta$.

The group has finite order, n , and is Abelian.

As a more concrete example, the elements of the cyclic group C_4 are shown in Fig. 6. The group table is given in Fig. 7.

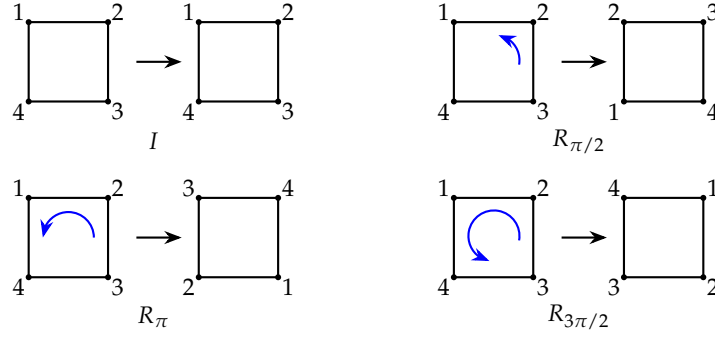


Figure 6: The elements of the cyclic group C_4 are the rotational symmetries of a square. They consist of the identity I (doing nothing), an anticlockwise rotation $R_{\pi/2}$ by $\pi/2$, an anticlockwise rotation R_π by π , and an anticlockwise rotation $R_{3\pi/2}$ by $3\pi/2$. The transformations are shown in each of the four panels.

$\circ$	I	$R_{\pi/2}$	R_π	$R_{3\pi/2}$
I	I	$R_{\pi/2}$	R_π	$R_{3\pi/2}$
$R_{\pi/2}$	$R_{\pi/2}$	R_π	$R_{3\pi/2}$	I
R_π	R_π	$R_{3\pi/2}$	I	$R_{\pi/2}$
$R_{3\pi/2}$	$R_{3\pi/2}$	I	$R_{\pi/2}$	R_π

Figure 7: The group table of the cyclic group of rotations C_4 . Note that the reflectional symmetry about the leading diagonal indicates the fact that the group is Abelian.

- **Dihedral groups.** The set of all rotational *and* reflectional symmetries of a regular n -gon form a group, called the *dihedral group of order n* , and written D_n .⁸ The elements of the group can be written as:

$$D_n = \left\{ I, R_{2\pi/n}, R_{4\pi/n}, R_{6\pi/n}, \dots, R_{2(n-1)\pi/n}, m_1, m_2, m_3, m_4, \dots, m_n \right\},$$

where the elements $I, R_{2\pi/n}, \dots, R_{2(n-1)\pi/n}$ are the same rotations as those of the cyclic group, and the elements $m_1, \dots, m_n$ are each of the reflectional symmetries.

Again, we know this is a group because it is the set of all possible symmetries of a regular n -gon (this time, without direction, as oppose to the cyclic group case). Indeed, if we label the vertices of an n -gon from 1 to n , and think about where the 1-vertex goes, we have n choices. After that, we know that the 2-vertex must be connected to the 1-vertex after the symmetry; however, there is no requirement that the original orientation is preserved, so there are two choices for where it can go. Once we have decided that, there is no more freedom; this gives the $2n$ symmetries above. Hence, the group has finite order, $2n$, as expected.

The group is also *non-Abelian* when $n > 2$. To see this, consider the effect of a rotation followed by a reflection, versus the same reflection followed by the same rotation. These give *different results in general*, as can be seen in Fig. 8. In order for agreement to be reached, if we rotate by an angle θ anticlockwise first, then reflect, this is equivalent to reflecting, then rotating by an angle θ *clockwise* instead, establishing the identity:

$$m_i R_\theta = R_{2\pi-\theta} m_i.$$

When $n = 2$, the only rotation available is R_π , and since $R_\pi = R_{2\pi-\pi}$, this problem does not occur - indeed, we shall shortly see that the group is in fact *Abelian* in this case.

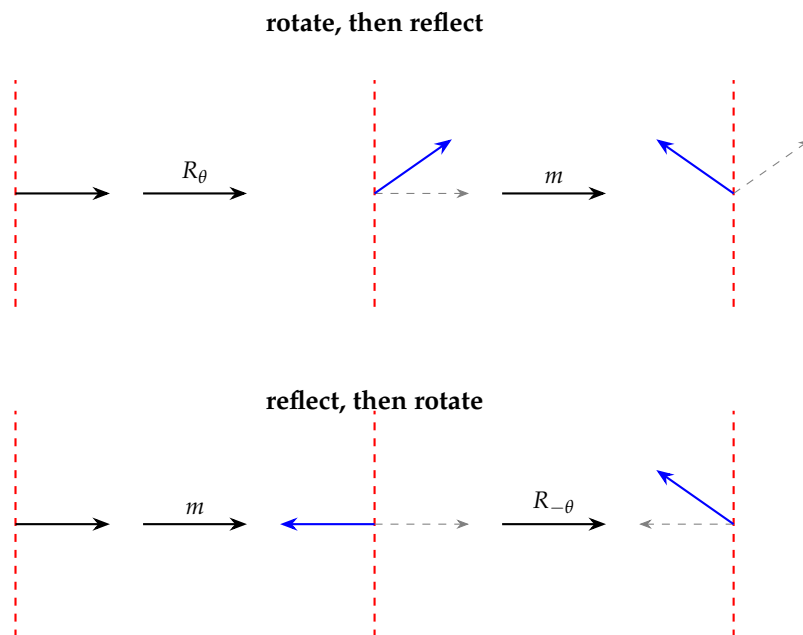


Figure 8: A rotation by angle θ anticlockwise followed by a reflection is equivalent to first performing the reflection, but then rotating by the angle θ clockwise instead. Note that if we had used an anticlockwise rotation in the lower panel, we would have obtained inequivalent results. Hence the dihedral group is non-Abelian for $n > 2$.

⁸Chemists use a slightly different notation; D_n is instead called C_{nv} (understood to be the group C_n together with reflections in vertical mirror planes, perpendicular to the plane of the polygon - which is the plane of a molecule in chemistry, usually).

Unhelpfully, some mathematicians also write this group as D_{2n} , rather than D_n , to emphasise the order of the group rather than the number of vertices of the n -gon. Make sure you understand which convention is being used!

A more concrete example can be constructed if we consider the case D_3 , the set of reflectional and rotational symmetries of a triangle. The symmetries in D_3 are shown in Fig. 9, where the vertices of the triangle are labelled as 1, 2, 3 to make clear the effect of the symmetry transformations in each case.

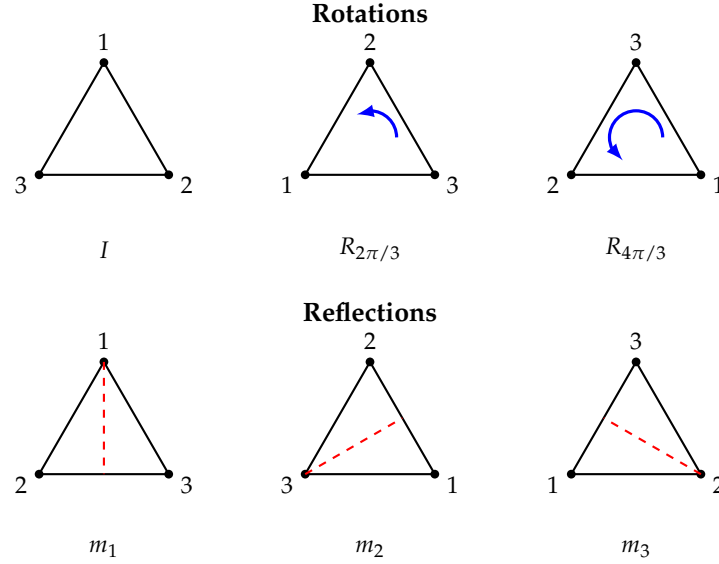


Figure 9: The elements of the dihedral group D_3 , the symmetry group of the equilateral triangle. The top row consists of the three rotations of the triangle, while the bottom row consists of the three reflections. The transformed vertex labels show the action of each symmetry. Rotations are indicated by blue arrows and reflections by dashed red axes.

The combination of rotations follows the same rule as the cyclic group, namely $R_\theta R_\phi = R_{\theta+\phi}$, with all angles ψ restricted to lie in the range $0 \leq \psi < 2\pi$, and $R_0 \equiv I$. Meanwhile, the discussion above tells us that $R_\theta m_i = m_i R_{-\theta}$; that is, a reflection followed by a rotation is equivalent to the same rotation, but in the *opposite sense*, followed by the same reflection. Finally, we can observe that $m_i^2 = I$, since a rotation repeated twice is equivalent to doing nothing.

These facts leaves only $R_\theta m_1, R_\theta m_2, R_\theta m_3$ and $m_i m_j$ for $i \neq j$ to compute. For example, we can calculate:

$$R_{2\pi/3} m_1 = m_3$$

by considering the effect of the transformations on the triangle carefully. First, we perform the reflection m_1 which swaps the original 2,3 vertices. On that transformed triangle, we then rotate by an angle $2\pi/3$ anticlockwise to produce a configuration identical to that which would have been obtained by just m_3 alone. This composition is shown in Fig. 10 below.

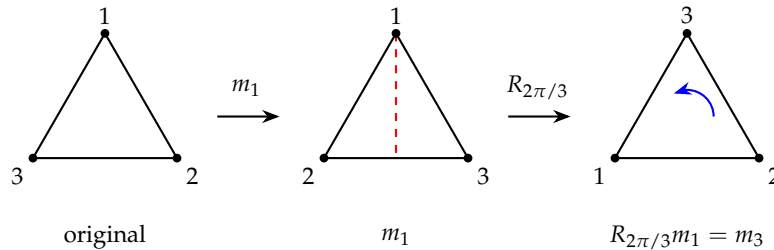


Figure 10: The composition $R_{2\pi/3} m_1 = m_3$. First, the reflection m_1 swaps vertices 2 and 3. The resulting configuration is then rotated anticlockwise by $2\pi/3$, producing the same permutation of vertices as the reflection m_3 .

Calculating the remaining products then, we can produce the complete group table:

$\circ$	I	$R_{2\pi/3}$	$R_{4\pi/3}$	m_1	m_2	m_3
I	I	$R_{2\pi/3}$	$R_{4\pi/3}$	m_1	m_2	m_3
$R_{2\pi/3}$	$R_{2\pi/3}$	$R_{4\pi/3}$	I	m_3	m_1	m_2
$R_{4\pi/3}$	$R_{4\pi/3}$	I	$R_{2\pi/3}$	m_2	m_3	m_1
m_1	m_1	m_2	m_3	I	$R_{2\pi/3}$	$R_{4\pi/3}$
m_2	m_2	m_3	m_1	$R_{4\pi/3}$	I	$R_{2\pi/3}$
m_3	m_3	m_1	m_2	$R_{2\pi/3}$	$R_{4\pi/3}$	I

Observe that constructing this table was an extremely tedious, laborious task, given we had to keep thinking about the geometry of the square. In Chapter 2 we will introduce the concept of *generators* and *relations* for groups, and hence derive a much more efficient way of calculating products of elements in dihedral groups. This is the way that we should normally construct these group tables.

- **The Klein four-group, D_2 .** A particular special case of interest within the dihedral groups, that will occur repeatedly throughout the text, is the dihedral group D_2 . This group is often called the *Klein four-group*, and written K_4 , or the *Vierergruppe*, written V_4 .

Using our previous interpretation, this group consists of the symmetries of a ‘regular 2-gon’. Evidently, this is a degenerate case - a 2-gon is a line segment. To keep things consistent with the other dihedral groups D_n though, we should think of it as two vertices 1, 2 joined by *two* edges (mirroring the structure of the other polygons). The group then contains four symmetry transformations:

$$D_2 = \{I, R_\pi, m_1, m_2\}$$

which are respectively doing nothing, a rotation by π , a reflection in the horizontal m_1 , and a reflection in the vertical m_2 . The transformations are shown in Fig. 11 below.

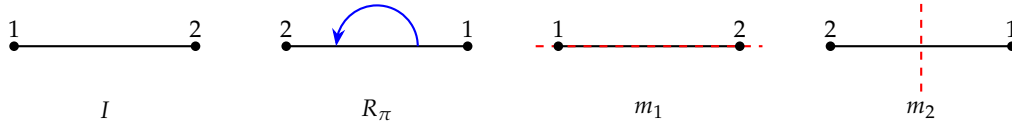


Figure 11: The four symmetry transformations of the dihedral group D_2 , also known as the Klein four-group. The object is a line segment with labelled endpoints, so that the effect of each transformation can be seen as a permutation of the labels.

Superficially, the symmetries I and m_1 seem identical, for example - however, if we think of the vertices 1, 2 as being connected by two edges, then I does nothing whilst m_1 swaps the edges. See Fig. 12 for a visual explanation.

Note that the viewpoint of having ‘two edges’ also ensures that our ‘symmetry counting’ arguments don’t break down - here, the vertex 1 must be mapped to the vertex 1 or 2 by the transformation, so there are only two options. But, the ‘two edges’ can also be swapped, resulting in a total of four choices, and hence four symmetries overall.

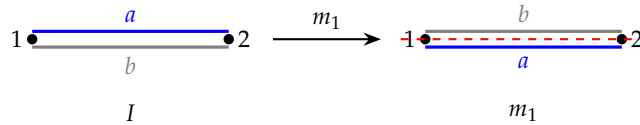


Figure 12: Although the identity I and the horizontal reflection m_1 appear identical when the 2-gon is drawn as a line segment, they act differently on the two edges of the degenerate polygon. The reflection m_1 fixes the vertices but swaps the two edges.

By keeping track of both the vertices, *and* the edges, the group table can be constructed as:

$\circ$	I	R_π	m_1	m_2
I	I	R_π	m_1	m_2
R_π	R_π	I	m_2	m_1
m_1	m_1	m_2	I	R_π
m_2	m_2	m_1	R_π	I

We note that D_2 is an *Abelian group*, since its group table is reflectionally symmetric about its leading diagonal. The group is of order four.

- **Chiral tetrahedral group.** The set of all rotational symmetries of a regular tetrahedron form a group, called the *chiral tetrahedral group*, written T . The set of all rotations and reflections forms a group because of the general symmetry argument we made at the start of this section. However, the set of all *rotations* can be identified as a group because they are the symmetries that *preserve the handedness of the tetrahedron*.

By this, we mean that if we label the vertices of the tetrahedron 1 to 4, we have two options generally. One way of labelling is to choose a vertex to be the vertex 1, then put our right hand at vertex 1, point our index finger in the direction of vertex 2, our middle finger in the direction of vertex 3, and our thumb in the direction of vertex 4. Alternatively, we could do this with our left hand, putting our left hand at vertex 1, then point our *left* index finger in the direction of vertex 2, our middle finger in the direction of vertex 3, and our thumb in the direction of vertex 4. The two respective labellings are shown in Fig. 13 shown below.

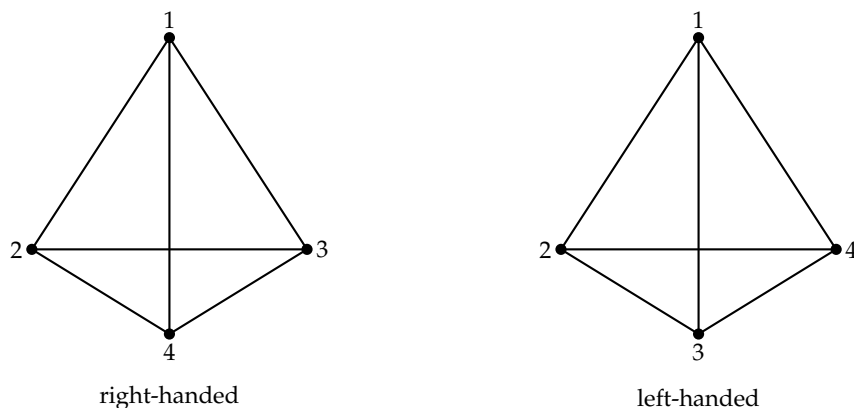


Figure 13: The two possible handedness choices for a labelled tetrahedron. The right-handed and left-handed configurations cannot be transformed into one another by rotations alone; a reflection is required to reverse the handedness.

If we are only allowed to perform rotations, then we *cannot* transform a left-handed tetrahedron into a right-handed tetrahedron, or vice-versa. If we could, we would suddenly have been able to transform our right-hand into our left-hand just via rotations, which is impossible! Therefore, the set of rotations alone forms a group, since it is the set of symmetries of a tetrahedron with a *fixed handedness* chosen.

To count the order of the group, we label the vertices 1 to 4. Vertices must be mapped to vertices, so the vertex 1 has four options as to where it can go. Then, vertex 2 has a choice of three options as to where it can go; after this, the handedness of the tetrahedron is fixed, as we are only allowed to use rotations, and hence the complete configuration is determined. Thus there are $4 \cdot 3 = 12$ total symmetries.

What do the elements actually look like? We can enumerate them as follows:

- There is of course the identity symmetry, I , doing nothing.
- For each vertices, we can perform either a $2\pi/3$ or $4\pi/2$ about the axis that passes through that vertex and the centre of the opposite face. This gives a total of eight symmetries. See Fig. 14 for a visualisation of the axis of rotation. There are three possible such rotations, given there are three pairs of opposite vertices.

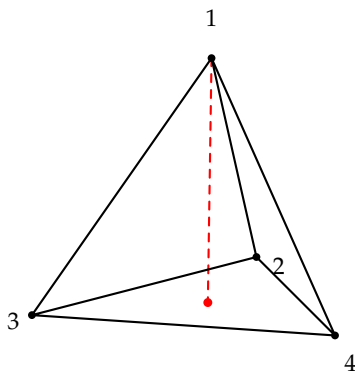


Figure 14: An axis about which there is a three-fold rotational symmetry of the tetrahedron. Rotations by $2\pi/3$ or $4\pi/2$ around these axes give eight possible rotational symmetries of the tetrahedron, when we account for the four possible vertices through which such axes can pass.

- The remaining symmetries are rotations by an angle π about axes which pass through the mid-points of opposite edges of the tetrahedron (these require a little bit more thinking about in your head as to why they work!). See Fig. 15 for a visualisation of the axis of rotation.

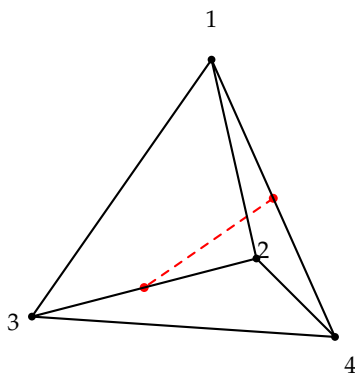


Figure 15: An axis about which there is a two-fold rotational symmetry of the tetrahedron. A rotation by π about this axis gives one of three additional possible rotational symmetries of the tetrahedron, when we account for the three pairs of opposite vertices.

Having identified all the symmetries, we could in principle construct the group table. This is rather tedious though, and we won't spend further time on this. Indeed, we shall actually see in Chapter 7 (when we study *permutation groups*) that there is an extremely efficient representation of the elements of this group that allow for fast calculations, bypassing the geometry altogether.⁹

⁹We shall see that the chiral tetrahedral group is in fact the same as (isomorphic to) the *alternating group* A_4 , consisting of the even permutations of four symbols.

- **Full tetrahedral group.** The set of all rotational *and* reflectional symmetries of a regular tetrahedron form a group, called the *full tetrahedral group*, written T_h . This time, we don't worry about the handedness being preserved - thus, once we have mapped 1 to one of four vertices, then 2 to one of the three adjacent vertices, we have complete freedom to map 3 to either of the remaining two vertices (using a reflection to swap handedness if possible). This gives an order $4 \cdot 3 \cdot 2 = 24$ group.¹⁰

1.4 Examples of groups from matrix algebra

Matrix algebra also provides a plentiful sources of groups. We shall only see a couple of examples in this course; an advanced course in *Lie group theory* is the place to go for more information on matrix groups.

- **Matrices under addition.** The set of all $n \times m$ (real or complex) matrices forms a group under matrix addition. Verifying each of the axioms in turn, we have:

(G1) *Closure.* Let A, B be $n \times m$ matrices. Then $A + B$ is an $n \times m$ matrix, so the operation is closed.

(G2) *Associativity.* Given three $n \times m$ matrices A, B, C , we have $A + (B + C) = (A + B) + C$.

(G3) *Identity.* The identity is the $n \times m$ zero matrix 0 , since we have $A + 0 = A = 0 + A$ for all $n \times m$ matrices A .

(G4) *Inverses.* The inverse of the matrix A is the matrix $-A$, whose entries are the negatives of all the corresponding entries of the matrix A . This follows since $A + (-A) = 0 = (-A) + A$.

- **The general linear group.** The set of all invertible $n \times n$ real matrices forms a group under matrix multiplication, called the *real general linear group*, and written $GL(n, \mathbb{R})$. Verifying each of the axioms in turn, we have:

(G1) *Closure.* The product of two invertible real $n \times n$ matrices A, B is necessarily a real $n \times n$ matrix AB . This matrix is invertible, with inverse $B^{-1}A^{-1}$ since $AB(B^{-1}A^{-1}) = I$, so the set is closed under the operation.

(G2) *Associativity.* Matrix multiplication is associative; this can be verified directly in the case of 2×2 matrices, but can be proved generally using an argument with summation notation.

(G3) *Identity.* The identity matrix I is clearly the identity of the operation, since $AI = A = IA$ for all matrices A .

(G4) *Inverses.* By definition, every element of the set is invertible, with $AA^{-1} = I = A^{-1}A$.

Similarly, the set of all invertible $n \times n$ complex matrices forms a group under matrix multiplication, called the *complex general linear group*, and written $GL(n, \mathbb{C})$. Both groups $GL(n, \mathbb{R}), GL(n, \mathbb{C})$ obviously have infinite order, but are non-Abelian.

- **Finite matrix groups.** It is also possible to have *finite* groups of matrices. Consider for example the set of matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Calculating all possible products of the matrices, we observe that

$$\begin{array}{lll} A^2 = B, & A^3 = C, & A^4 = I, \\ B^2 = I, & C^2 = B, & C^3 = A, \\ AB = BA = C, & AC = CA = I, & BC = CB = A. \end{array}$$

¹⁰We shall see that this group is in fact the same as (isomorphic to) the *symmetric group* S_4 , consisting of all possible permutations of four symbols.

Hence, the set is closed under matrix multiplication. Further, matrix multiplication is associative, the identity is included in the set, and clearly all the matrices are invertible from the above calculation. Hence this set forms a group under matrix multiplication. The group multiplication table is:

$\cdot$	I	A	B	C
I	I	A	B	C
A	A	B	C	I
B	B	C	I	A
C	C	I	A	B

1.5 Isomorphic groups

Some of the groups we have seen in this chapter ‘feel’ inherently the same as one another. For example, consider the groups $\mathbb{Z}_4$, C_4 , and the finite matrix group from the last section. They have the following group tables:

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\mathbb{Z}_4$

$\circ$	I	$R_{\pi/2}$	R_{π}	$R_{3\pi/2}$
I	I	$R_{\pi/2}$	R_{π}	$R_{3\pi/2}$
$R_{\pi/2}$	$R_{\pi/2}$	R_{π}	$R_{3\pi/2}$	I
R_{π}	R_{π}	$R_{3\pi/2}$	I	$R_{\pi/2}$
$R_{3\pi/2}$	$R_{3\pi/2}$	I	$R_{\pi/2}$	R_{π}

C_4

$\cdot$	I	A	B	C
I	I	A	B	C
A	A	B	C	I
B	B	C	I	A
C	C	I	A	B

Matrix group

The group tables are identical, except for the names of the elements; if we were to identify:

$$0 \leftrightarrow I \leftrightarrow I, \quad 1 \leftrightarrow R_{\pi/2} \leftrightarrow A, \quad 2 \leftrightarrow R_{\pi} \leftrightarrow B, \quad 3 \leftrightarrow R_{3\pi/2} \leftrightarrow C,$$

then the tables are exactly the same! Therefore, it seems reasonable that we should consider them to be ‘equivalent’ groups.

More formally, we say that two groups G_1, G_2 are *isomorphic* (this is the word we use for ‘equivalence’ of group structures) if there is a mapping $f : G_1 \rightarrow G_2$ between them such that:

- f is an invertible mapping. In particular, this means that for every element $x \in G_1$, we can assign a unique element $f(x) \in G_2$. Similarly, for every element $y \in G_2$, we can assign a unique element $f^{-1}(y) \in G_1$. This gives a ‘one-to-one correspondence’ between the groups - our idea of ‘renaming all the group elements’.

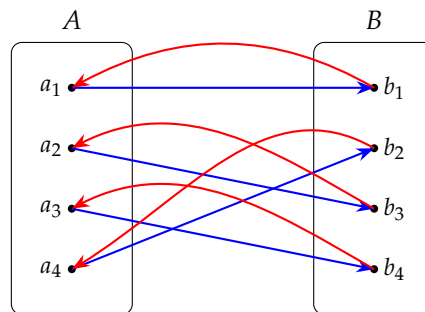


Figure 16: An invertible function provides a one-to-one correspondence between sets. Every element of the set A is mapped to a unique element of the set B by the function f (shown in blue), and every element of B is mapped back to a unique element of A by the inverse function f^{-1} (shown in red).

- Of course, it is not enough just to be able to rename some elements; the group tables must be the same after we have performed the renaming.

If we have the element $x * y$ in the (x, y) position of the group table of G_1 , it must be the case that in the $(f(x), f(y))$ position of the group table of G_2 , we must have the renamed version of the element $x * y$. That is, we must have the key property:

$$f(x) \circ f(y) = f(x * y)$$

for all elements x, y . We can visualise this with Fig. 17 below.

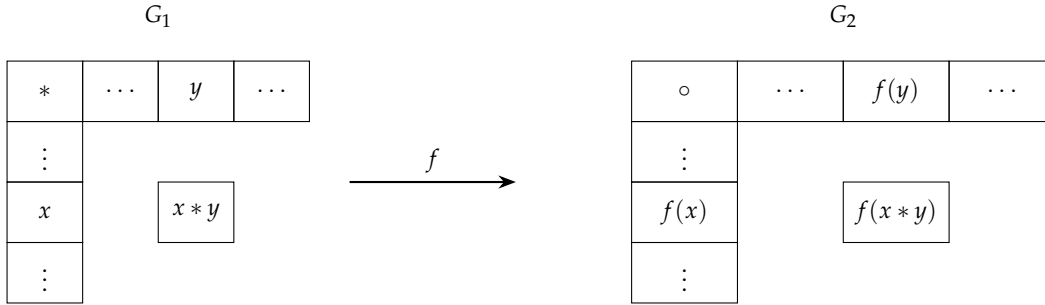


Figure 17: An isomorphism preserves the group multiplication table. The element in the (x, y) position of the first table is $x * y$. After relabelling the elements by the bijection f , the corresponding entry in the second table must be the relabelled element $f(x * y)$, so that $f(x) \circ f(y)$ must be the same as $f(x * y)$.

Now we have motivated and understood the idea, for clarity, let us restate this as a formal mathematical definition with the two conditions that we need to check:

Definition 1.5: Let G_1, G_2 be groups with operations $*, \circ$ respectively. We say the two groups are *isomorphic* if there exists a function $f : G_1 \rightarrow G_2$ such that:

- (i) f is invertible;
- (ii) $f(x * y) = f(x) \circ f(y)$ for all $x, y \in G_1$.

The map f is called an *isomorphism* between the groups. If two groups G_1, G_2 are isomorphic, we write $G_1 \cong G_2$.

Let's see some examples of this formal definition in action:

- Certainly it is very straightforward to check that $\mathbb{Z}_4$ and C_4 are isomorphic groups according to this definition, then: we define $f : \mathbb{Z}_4 \rightarrow C_4$ through:

$$f(0) = I, \quad f(1) = R_{\pi/2}, \quad f(2) = R_{\pi}, \quad f(3) = R_{3\pi/2}.$$

This is evidently a one-to-one correspondence, and hence invertible. We then have

$$f(x + y) = R_{(x+y)\pi/2} = R_{x\pi/2} \circ R_{y\pi/2} = f(x) \circ f(y),$$

thus verifying the second condition. Alternatively, we can simply check that once we have performed the relabelling according to f , the group tables coincide, as we describe above. We conclude that $\mathbb{Z}_4 \cong C_4$ is an isomorphism.

- For another example, let us note that the *positive* real numbers under multiplication form a group, $\mathbb{R}^+$. We can check each of the axioms in turn:

(G1) *Closure*. Certainly if x, y are positive reals, then their product xy is a positive real.

(G2) *Associativity*. Multiplication of real numbers is clearly associative.

(G3) *Identity*. The identity is clearly 1, which is a positive real number, and hence included in the set.

(G4) *Inverses*. The inverse of x is $1/x$, since $x \cdot 1/x = 1 = 1/x \cdot x$. If x is a positive real number, then $1/x$ exists and is a positive real number, so inverses are all included in the set.

We claim that $\mathbb{R}$, the real numbers under addition, and $\mathbb{R}^+$, the positive real numbers under multiplication, are isomorphic groups.

We propose the isomorphism $f : \mathbb{R} \rightarrow \mathbb{R}^+$ given by:

$$f(x) = e^x,$$

where e^x is the standard exponential function. This is certainly invertible, since $f^{-1}(x) = \ln(x)$ is well-defined. Further, we have:

$$f(x + y) = e^{x+y} = e^x e^y = f(x)f(y),$$

so the addition of real numbers is correctly translated across to multiplication of positive real numbers by the mapping f . Thus $\mathbb{R} \cong \mathbb{R}^+$ is an isomorphism.

Some general properties of isomorphisms are immediate from the definition; one we might expect is that the identity of the group G_1 must be mapped to the identity of the group G_2 by an isomorphism (we cannot ‘relabel the identity’ to a non-identity element!). We prove this formally as follows:

Proposition 1.6: If $f : G_1 \rightarrow G_2$ is an isomorphism, and e_1 is an identity element in G_1 , then $f(e_1)$ is an identity element in G_2 .

Proof: We have:

$$\begin{aligned} f(e_1) &= f(e_1 * e_1) && \text{(identity in } G_1) \\ &= f(e_1) \circ f(e_1) && \text{(property (ii) of isomorphism).} \end{aligned}$$

Now multiply on the left by $f(e_1)^{-1}$. On the left hand side, we have $f(e_1)^{-1} \circ f(e_1)$, which is equal to e_2 by the inverses axiom in G_2 . Hence we have:

$$\begin{aligned} e_2 &= f(e_1)^{-1} \circ (f(e_1) \circ f(e_1)) \\ &= (f(e_1)^{-1} \circ f(e_1)) \circ f(e_1) && \text{(associativity in } G_2) \\ &= e_2 \circ f(e_1) && \text{(inverses in } G_2) \\ &= f(e_1) && \text{(identity in } G_2) \end{aligned}$$

Hence, we’re done. $\square$

Non-isomorphism of groups

We have now seen some examples of when groups *are* isomorphic. How can we prove that two groups are *not* isomorphic?

In general, this is a much more difficult task. There are many, many ways we could relabel the elements of a group G_1 as the elements of a group G_2 , corresponding to many possibilities for invertible mappings $f : G_1 \rightarrow G_2$. It is very difficult to check all of them, to see that *none* of those possible relabellings could satisfy the key property $f(x * y) = f(x) \circ f(y)$.

Throughout some of the subsequent chapters, we will begin to develop some more sophisticated tools for telling whether two groups are isomorphic or not. For now though, we shall prove two of the most basic results:

- two finite groups are non-isomorphic if they have unequal size;
- if two groups are isomorphic, then they must both be Abelian, or must both be non-Abelian. That is, an Abelian group can never be isomorphic to a non-Abelian group.

The first result is almost trivial to prove:

Proposition 1.7: If the group G_1, G_2 have unequal size, they are non-isomorphic.

Proof: Suppose for a contradiction that $f : G_1 \rightarrow G_2$ is an isomorphism. Then the map f is invertible, providing a one-to-one correspondence between G_1 and G_2 as we saw in Fig. 16. Obviously this may only occur if the sets G_1, G_2 have the same size. This is a contradiction, hence G_1, G_2 could not have been isomorphic. $\square$

This result proves that, for example, C_4 is not isomorphic to C_6 (which you probably wouldn't have expected anyway).

The second result is more involved:

Proposition 1.8: If the groups G_1, G_2 are isomorphic, then G_1 is Abelian if and only if G_2 is Abelian.

Proof: Let $f : G_1 \rightarrow G_2$ be an isomorphism. Suppose first that G_1 is Abelian. Then:

$$f(y) \circ f(x) = f(y * x) = f(x * y) = f(x) \circ f(y)$$

for all $x, y \in G_1$, using the isomorphism property (ii). As a result, G_2 must be Abelian, since all elements in G_2 can be expressed in the form $f(x), f(y)$ for some $x, y \in G_1$.

Conversely, suppose that G_2 is Abelian. Then:

$$y * x = f^{-1}(f(y * x)) = f^{-1}(f(y) \circ f(x)) = f^{-1}(f(x) \circ f(y)) = f^{-1}(f(x * y)) = x * y,$$

for all $x, y \in G_1$, using the isomorphism property (ii). The result follows. $\square$

This result proves that, for example, C_6 is not isomorphic to D_3 , even though they have the same size.

Chapter 2

Basic properties of group elements

We have now seen many examples of groups, emphasising their ubiquity in mathematics. In this chapter, we shall begin to understand the advantages of studying the *abstract properties* of groups, rather than the groups themselves.

In this chapter, we shall discuss the following:

- We shall begin by proving that the identity of a group is unique (that is, it cannot have two identities), and that the inverse of an element in a group is also unique. We shall also discuss how to solve equations in groups, using left and right cancellation.
- Next, we shall investigate properties of finite group tables. We shall show that all finite group tables are *Latin squares*, meaning that every element appears exactly once in every row and column of the group table. This heavily restricts the form of group tables; we will use this property to classify finite groups with order up to and including four, showing that all such groups are either cyclic or dihedral.
- Next, we shall state and prove exponent laws for group multiplication. We shall introduce the idea of *periodic elements*, and prove the surprising fact that all elements of finite groups are periodic (that is, by raising them to a high enough power, we can eventually reach the identity). We shall also prove that the period of a group element is preserved by isomorphism, giving us a test for when two groups are *non-isomorphic*.
- Finally, we introduce the idea of *generators* and *relations* for describing a group. Here, we try to express all the elements of a group in terms of powers and products of a small subset of group elements (the *generators*). This greatly simplifies the presentation of much of the algebra we do in groups - in particular, we shall introduce generators and relations for cyclic and dihedral groups, which helps us perform computations in later chapters.

2.1 Proving properties of abstract groups

This chapter is all about learning how to *prove* things abstractly in a group. We start with some basic results: the identity of a group is unique, and the inverse of an element in a group is unique. To prove these results, we are *only allowed to use the group axioms*, and we must clearly state where we use them.

Proposition 2.1: The identity of a group is unique.

Proof: Suppose that e, f are both identity elements. Then:

$$\begin{aligned} e &= e * f && \text{(since } f \text{ is an identity)} \\ &= f && \text{(since } e \text{ is an identity)} \end{aligned}$$

Hence $e = f$, so the identity is indeed unique. $\square$

Note that this proof *only* used the identity axiom for groups; it follows that any set with a (closed) binary operation that has an identity necessarily has a *unique* identity.

The fact we have proved uniqueness of an identity in *any* group means we can deduce many facts about many seemingly unrelated mathematical structures. For example, in matrix algebra, we can deduce that if a matrix A satisfies $AB = B = BA$ for all matrices B , then $A = I$.

Similarly, we can prove that the inverse of a give element in a group is unique.

Proposition 2.2: The inverse of an element in a group is unique.

Proof: Suppose that x has inverses y, z . Then:

$$\begin{aligned} y &= y * e && \text{(identity axiom)} \\ &= y * (x * z) && \text{(} z \text{ is an inverse of } x \text{)} \\ &= (y * x) * z && \text{(associativity axiom)} \\ &= e * z && \text{(} y \text{ is an inverse of } x \text{)} \\ &= z && \text{(identity axiom)} \end{aligned}$$

Hence $y = z$, so the inverse of x is indeed unique. $\square$

This time, we needed to use all of the group axioms in order to prove the uniqueness of inverses. And again, we can use this fact to deduce facts about many mathematical structures. For example, in matrix algebra again, we can deduce that there is *at most* one matrix A such that $AB = I = BA$ for some fixed matrix B .

We can also think about solving equations in groups. You will see a further example in the problem set; for now, we shall show how to use the axioms to deduce the solutions of 'linear' group equations.

Proposition 2.3: Let $a, b \in G$ be fixed, given elements of a group G , and suppose that $x \in G$ is an unknown element. Then the equations:

$$a * x = b, \quad \text{and} \quad x * a = b$$

have unique solutions in the group.

Proof: Suppose that x satisfies $a * x = b$. Then multiplying on the left by a^{-1} , we have:

$$\begin{aligned} a^{-1} * (a * x) &= a^{-1} * b &\Rightarrow (a^{-1} * a) * x &= a^{-1} * b && \text{(associativity)} \\ &&\Rightarrow e * x &= a^{-1} * b && \text{(inverses)} \\ &&\Rightarrow x &= a^{-1} * b && \text{(identity)} \end{aligned}$$

So if there is a solution, then it is given by $x = a^{-1} * b$ (this is the *uniqueness* of a solution). Now let's assume that $x = a^{-1} * b$. Then:

$$\begin{aligned} a * (a^{-1} * b) &= (a * a^{-1}) * b && \text{(associativity)} \\ &= e * b && \text{(inverses)} \\ &= b && \text{(identity)} \end{aligned}$$

Hence, $x = a^{-1} * b$ solves the equation (this is the *existence* of a solution).

The same can be done with $x * a = b$, with unique solution $x = b * a^{-1}$; we leave this as an exercise to the reader. $\square$

We shall shortly use this property to prove something interesting about finite group tables.

On the problem sheet, you will continue to prove properties of abstract groups; one of the most important that you shall see is the *generalised associative law*. We already know that one of the group axioms is that it does not matter where brackets are placed in the product:

$$a * b * c.$$

This generalises to n elements; it does not matter where brackets are placed in the product:

$$a_1 * a_2 * a_3 * \dots * a_n.$$

Proposition 2.4: (The generalised associative law) In a group, all possible placements of brackets in the product:

$$a_1 * a_2 * a_3 * \dots * a_n,$$

where $a_1, \dots, a_n$ are group elements, leads to the same resulting group element.

Proof: By induction; see problem sheet. $\square$

2.2 Properties of group tables

In the first section of this chapter, we proved properties that hold true in *all* groups. By restricting to finite groups, we can say more about the properties of groups.

One way we can do this is by considering the allowed structure of group tables. For example, we already know that a group needs a unique identity, so there must be one row and one column of a group table which matches the header row and header column, respectively. Without loss of generality, we usually take these to be the first row and first column respectively.

We can say more by using the ‘equation solving’ proposition we proved above; using this property, we can show that every group table is necessarily a *Latin square*:

Proposition 2.5: (The Latin square property) Every group table is a *Latin square*: each element appears exactly once in each row and column.

Proof: Let g, h be any two group elements. We shall show that h appears in row g exactly once. We seek a column x such that:

$$g * x = h.$$

But, by the ‘equation solving’ proposition from earlier, we know that this equation has a unique solution. So indeed there is exactly one column index for which h appears in row g . Similarly, every element appears exactly once in every column. $\square$

You can readily verify that every group table we have seen so far is indeed a Latin square. Importantly, however, the converse is *not* true: not every Latin square is a group table. The smallest counterexample occurs at order five.

Remarkably, using the Latin square property, we can *classify* all possible finite groups up to order four (up to isomorphism). It turns out that the only possibilities are cyclic and dihedral groups.

Proposition 2.6: The complete list of finite groups, up to isomorphism, up to and including order four is the following:

- Order 1. C_1 .
- Order 2. C_2 .
- Order 3. C_3 .
- Order 4. C_4 and D_2 .

Proof: For order one, the only possible Latin square is obviously:

$$\begin{array}{c|c} * & e \\ \hline e & e \end{array}$$

This is clearly isomorphic to the cyclic group C_1 , consisting only of the identity element.

Similarly, for order two, the only possible Latin square (with an identity row and column) is:

$$\begin{array}{c|cc} * & e & a \\ \hline e & e & a \\ a & a & e \end{array}$$

This is clearly isomorphic to the cyclic group C_2 , which has elements $\{I, R_\pi\}$.¹¹

For order three, we can start constructing a Latin square with an identity row and column as:

*	e	a	b
e	e	a	b
a			
b			

Note that in the (a, a) entry, we could choose to put e or b . But if we put e , we would be forced to put b in the (a, b) entry, which would contradict the Latin square property. This implies that the completed Latin square is:

*	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

This is clearly isomorphic to the cyclic group C_3 , with $e \mapsto I$, $a \mapsto R_{2\pi/3}$ and $b \mapsto R_{4\pi/3}$, for example.

Finally, for order four, assuming an identity row and column, the Latin square must take the initial form:

*	e	a	b	c
e	e	a	b	c
a				
b				
c				

Let's now think about the position of b in the a row: it has two possibilities, either in the a column or the c column. Breaking into sub-cases then:

- If we put it in the a column, then the positions of c, e are forced in the row too, and indeed the a column. We are left with the table:

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c		
c	c	e		

Now, the positions of the remaining elements are also forced by the Latin square property. We end up with:

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

This is isomorphic to the cyclic group C_4 , with $e \mapsto I$, $a \mapsto R_{\pi/2}$, $b \mapsto R_{\pi}$, $c \mapsto R_{3\pi/2}$.

- If we put b in the c column, we have two choices for filling in the remaining parts of the Latin square. Skipping some steps, we have the three possibilities:

$*$	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

To finish, we note that the second group table represents a group that is isomorphic to D_2 . Further, by swapping the b, c columns and rows in the third group table, we again get a group that is isomorphic to D_2 . Finally, by swapping the b, c columns and rows in the first group table, we get a group that is isomorphic to C_4 .

Technically, we still need to show that C_4 is *not* isomorphic to D_2 ; we shall shortly show this by considering *element orders*, however. $\square$

We will continue to classify finite groups in the next chapter; in fact, we shall see that up to order seven, there are only cyclic and dihedral groups. Order eight is when things first begin to get interesting!

2.3 Power laws and periodicity

Have you noticed how annoying it is to continuously write $a * b$ and $x * y$, where $*$ is a group operation? From now on, let's leave it implied:

Definition 2.7: In a group G with operation $*$, if it is clear from context, we write the product $x * y$ of two elements $x, y \in G$ as:

$$xy,$$

where the $*$ is left implied. Note, we do *not* usually do this for groups where the operation is 'addition' $+$ (like the real numbers under addition, or a set of vectors under addition).

Similarly, we also introduce notation for the *powers* of a group element, involving repeated multiplication of a group element with itself.

Definition 2.8: We define the *powers* of a group element g in a group G with operation $*$ via:

$$g^n := \begin{cases} \underbrace{g * g * \dots * g}_{n \text{ times}}, & \text{if } n > 0 \text{ is a positive integer,} \\ e, & \text{if } n = 0, \\ \underbrace{g^{-1} * g^{-1} * \dots * g^{-1}}_{-n \text{ times}}, & \text{if } n < 0 \text{ is a negative integer.} \end{cases}$$

Note, this definition requires the expressions $g * g * \dots * g$ and $g^{-1} * g^{-1} * \dots * g^{-1}$ to be unambiguous; that is, we require the generalised associative property.

¹¹Famously, C_2 is the most romantic group. You can find out more here: https://www.youtube.com/watch?v=p84FPXA1_Jk&list=RDp84FPXA1_Jk&start_radio=1.

This basic definition immediately implies the standard exponent laws:

Proposition 2.9: (Exponent laws in groups) For any integers n, m and any group element g , we have:

$$g^n g^m = g^{n+m}, \quad (g^n)^m = g^{nm}.$$

Proof: If n, m are positive integers, in the first case we have:

$$g^n g^m = \underbrace{(g * g * \dots * g)}_{n \text{ times}} * \underbrace{(g * g * \dots * g)}_{m \text{ times}} = \underbrace{g * g * \dots * g}_{n+m \text{ times}} = g^{n+m},$$

using generalised associativity. The other cases follow similarly. $\square$

One interesting question we can ask about powers in groups is whether multiplying a group element by itself repeatedly eventually ends up at the identity. This is motivated by symmetry groups, for example, where performing enough of the same rotation or reflection is equivalent to doing nothing; see for example Fig. 18.

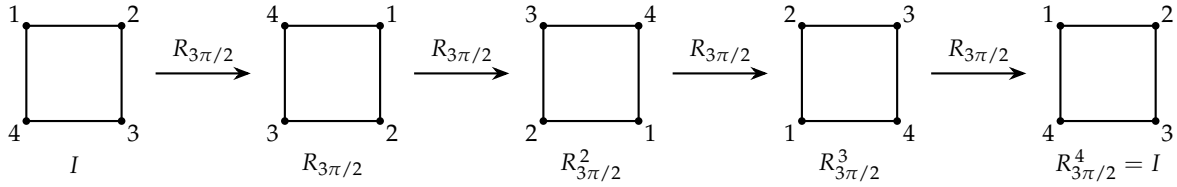


Figure 18: Performing the rotation $R_{3\pi/2}$ on a square sufficiently many times is equivalent to doing nothing; hence the rotation $R_{3\pi/2}$ is a ‘periodic’ group element. In the above figure, this is shown by a sequence of $R_{3\pi/2}$ being performed multiple times to a square with vertices labelled 1 to 4.

Making this a formal definition, we have:

Definition 2.10: Let $g \in G$ be an element of the group G , and let $e \in G$ be the identity of the group. We say that g is *periodic* if $g^n = e$ for some positive integer n . The smallest such integer is called the *period* (or more commonly, *order*) of $g \in G$.

The more common word for period is order in the mathematical literature - it is important not to get this confused with the *order* of a group, which is its size!

Some examples are the following:

- In the set of integers $\mathbb{Z}$ under addition, the only periodic element is 0. This is because:

$$\underbrace{x + x + \dots + x}_{n \text{ times}} = nx = 0$$

if and only if $x = 0$ or $n = 0$ (not a positive integer). Note that ‘powers’ here correspond to adding things n times, because the group operation is *addition*, not multiplication! The order of 0 is 1.

- In the set of integers modulo 5 under multiplication, $\mathbb{Z}_5^* = \{1, 2, 3, 4\}$, all elements are periodic. The element 1 has order 1. For 2, we have:

$$2^2 = 4, \quad 2^3 = 3, \quad 2^4 = 1,$$

implying that the order of 2 is 4. Similarly, we can check that 3 has order 4, and 4 has order 2.

Surprisingly, it turns out that in a finite group, every element is periodic:

Proposition 2.11: In a finite group, every element is periodic.

Proof: Let x be any element, and consider the list:

$$x, \quad x^2, \quad x^3, \quad x^4, \quad \dots$$

Since the group is finite, this list must contain only finitely many distinct entries. Hence, we must have $x^n = x^m$ for some $n < m$. Thus $x^{m-n} = e$, using the exponent laws. It follows that there is some positive integer N such that $x^N = e$; therefore, there is a least such positive integer. $\square$

Further, the order of an element is something that is intrinsic to the element, and cannot be changed by isomorphism. This provides a useful criterion for proving that two groups are *not* isomorphic.

Proposition 2.12: Let $f : G \rightarrow H$ be an isomorphism between groups. Given any element $x \in G$, we have that x and $f(x)$ have the same order.

Proof: Let x have order n . Consider:

$$f(x^k) = f(x^{k-1})f(x) = \dots = f(x)^k.$$

If $k < n$, then $x^k \neq e_G$, hence $f(x^k) \neq e_H$ (since an isomorphism must map the unique identity to the unique identity). However, when $k = n$, we have $f(x^n) = f(e) = e$, so $f(x)^n = e$. Thus $f(x)$ has order n . $\square$

This is a final verification that C_4 and D_2 are not isomorphic groups; their elements have orders $\{1, 2, 4, 4\}$ and $\{1, 2, 2, 2\}$ respectively, so they cannot be isomorphic.

On the other hand, we *cannot* use this result to conclude that two groups whose elements have the same orders *are* isomorphic. The smallest counterexample happens at order 16 though, and requires a bit more machinery to construct it!

2.4 Generators and relations

The groups C_n and D_n defined in the first chapter have quite unwieldy definitions - particularly D_n , where multiplying the reflections $m_1, \dots, m_n$ and the rotations requires some rather annoying geometry, for example.

This problem can be dealt with by introducing a more *algebraic* way of thinking about these groups. For example, in the cyclic group C_n :

$$C_n = \{I, R_{2\pi/n}, R_{4\pi/n}, \dots, R_{2(n-1)\pi/n}\},$$

instead of writing all the elements using different symbols, we could define $r = R_{2\pi/n}$. Then all elements could be written as *powers* of r instead:

$$C_n = \{r^0 = e, r, r^2, \dots, r^{n-1}\}.$$

The group multiplication becomes extremely simple now; it is just the standard exponent law:

$$r^n r^m = r^{n+m}.$$

Generators: definitions, and cyclic and dihedral examples

The above idea generalises well. Given a complicated group, we often choose a small set of elements in terms of which we write all the other elements as powers and products of these elements: this small set is called a *generating set*.

Proposition 2.13: We say that a group G has *generators* $\{g_1, \dots, g_n\}$ if every element of G can be written in the form:

$$x_1^{\alpha_1} \dots x_m^{\alpha_m},$$

where α_i are integers (which could be negative) and each x_i is one of $g_1, \dots, g_n$.

We have just seen that the cyclic group C_n is generated by one of the rotations, $r = R_{2\pi/n}$. Similarly, we can show that D_n is generated by one of the rotations *and* one of the reflections.

Proposition 2.14: The dihedral group D_n is generated by $\{r, m\}$ where $r = R_{2\pi/n}$ and $m = m_1$.

Proof: The key algebraic relation we require is that:

$$r^{-1} m = m r. \quad (*)$$

We saw why this was true in the first chapter; if we reflect, then rotate, this is equivalent to rotating in the opposite sense, then reflecting.

Using this relation, we note that any product of the form:

$$r^{\alpha_1} m^{\beta_1} \dots r^{\alpha_n} m^{\beta_n}$$

may be transformed to the form:

$$r^\alpha m^\beta.$$

This is because if a rotation appears to the right of a reflection, it can always be moved to the left of the reflection at the cost of turning it into its inverse (which is also another rotation), using the relation (*).

We now simply count how many possible elements there are of the form $r^\alpha m^\beta$. Since $r^n = e$ and $m^2 = e$, there are at most $2n$ distinct elements here. Further, if:

$$r^\alpha m^\beta = r^{\alpha'} m^{\beta'},$$

we have $r^{\alpha-\alpha'} = m^{\beta-\beta'}$. A rotation is only equal to a reflection if they are both the identity, implying that $\alpha = \alpha'$ and $\beta = \beta'$. So there are at least $2n$ distinct elements here, and indeed $\{r, m\}$ generates D_n . $\square$

It follows that D_n can be written in the form:

$$D_n = \{e, \quad r, \quad r^2, \quad \dots \quad r^{n-1}, \quad m, \quad rm, \quad r^2m, \quad \dots \quad r^{n-1}m\}.$$

The first n elements are obviously the rotations (together with the 'do nothing symmetry' e), and therefore the remaining elements must be the reflections.

Having this algebraic representation of D_n can be very useful for calculations, if we remember the key relationship $r^{-1}m = mr$. For example, we can show that $r^\alpha m$ has order 2 (which we expect because it is a reflection) by calculating:

$$\begin{aligned} (r^\alpha m)^2 &= r^\alpha m r^\alpha m \\ &= r^\alpha r^{-\alpha} m^2 && \text{(repeated use of } r^{-1}m = mr) \\ &= e. \end{aligned}$$

In fact, the general multiplication rule for any two elements of D_n can be deduced. We first observe the more general form of the 'reflection commuting past a rotation rule':

$$m^\beta r^\alpha = r^{\alpha(-1)^\beta} m^\beta.$$

This is because if $\beta = 1$, the rotation is transformed to its inverse, and if $\beta = 0$, there is no reflection to commute with in the first place! This implies that the general multiplication rule for two elements of D_n is given by:

$$\begin{aligned} (r^\alpha m^\beta)(r^{\alpha'} m^{\beta'}) &= r^\alpha r^{\alpha'(-1)^\beta} m^{\beta+\beta'} \\ &= r^{\alpha+\alpha'(-1)^\beta} m^{\beta+\beta'}. \end{aligned}$$

This will be useful for some of the Examples Sheet questions, and later in the course.

Groups generated by one element

Now we have seen the utility of generators, we might like to analyse groups in terms of them. The simplest possible case are *groups generated by one element*.

Definition 2.15: A *cyclic group* is a group which can be generated by a single element.

Unsurprisingly, the cyclic groups C_n are cyclic (hence the name). But, there is also a single infinite cyclic group, $\mathbb{Z}$.

Definition 2.16: All cyclic groups (i.e. groups generated by one element) are either isomorphic to cyclic rotation groups, C_n , or to $\mathbb{Z}$ under addition.

Proof: Let r be the generator of the group. If r has finite order n , then the elements are $\{e, r, \dots, r^{n-1}\}$ and the group is isomorphic to C_n . If r has infinite order, then the elements of the group are:

$$\{\dots, r^{-2}, r^{-1}, e, r, r^2, \dots\},$$

which is obviously isomorphic to $\mathbb{Z}$ via the isomorphism $f(r^k) = k$. $\square$

Group presentations

To generalise to more than one generator, we need to explain how they interact with one another. An equation constraining the multiplication of generators is called a *relation*, and we can use them to write groups in an elegant form:

Definition 2.17: A *group presentation* is a representation of a group G in the form:

$$G = \langle x_1, \dots, x_n \mid R_1, \dots, R_m \rangle,$$

where $x_1, \dots, x_n$ are generators, and $R_1, \dots, R_m$ are a set of constraints on the multiplication of the generators of the group. These constraints are called *relations* defining the group.

For example, we have now seen that:

- $C_n = \langle r \mid r^n = e \rangle;$
- $D_n = \langle r, m \mid r^n = e, m^2 = e, rm = mr^{-1} \rangle.$

We will not use group presentations in this text, but it is useful to know this notation as it appears fairly frequently in the mathematical literature on groups. You will examine one example of a group presentation on the Examples Sheet.

Chapter 3

Subgroup theory

We have now learned lots about the structure of general groups, and we have begun to classify all of the possible finite groups of small order (indeed, we have successfully classified all groups of orders up to and including four).

To make further progress in understanding the possible structure of groups, we now turn to a new concept: *subgroups*. Subgroups are ‘smaller groups hidden inside larger groups’; understanding the possible subgroups of a group is one way of understanding the group’s overall structure itself.

We shall see in this chapter that the possible subgroups of a finite group are highly constrained; the main result in this area is *Lagrange’s theorem*, which says that the order of a subgroup must be a factor of the group’s order. This key result will allow us to classify all groups of prime order (they are all cyclic), and allow us to classify groups of order six. We thus will know all finite groups up to order seven by the end of this chapter.

In more detail, in this chapter we shall cover the following:

- We shall introduce the definition of a subgroup, and give several examples.
- Next, we will define *cosets* of a subgroup, and explain why they *partition* the group. This is the key step in the proof of *Lagrange’s theorem*, the main result in this chapter. Lagrange’s theorem states that for a finite group, the order of a subgroup is a factor of the group’s order. Immediately after, we deduce some simple consequences of Lagrange’s theorem, namely that the order of an element in a finite group must divide the group’s order.
- Next, we focus on the cyclic and dihedral groups as extended examples. We use Lagrange’s theorem to classify all possible subgroups of cyclic and dihedral groups.
- We also discuss the converse of Lagrange’s theorem, and give an example of why it is *false* in general. We will state and prove a partial converse to Lagrange’s theorem, *Cauchy’s theorem*, which guarantees the existence of subgroups of certain sizes.
- Finally, we complete the classification of small finite groups up to order seven. We do this by using Lagrange’s theorem to classify all groups of prime order, and to investigate the order six case more closely.

3.1 Definition of a subgroup

As promised in the introduction to this chapter, we define a *subgroup* to be a smaller group hidden inside a larger group:

Definition 3.1: Let $H \subseteq G$ be a subset of a group G . If H is a group in its own right, with respect to the same operation as the group G , then H is called a *subgroup* of G . We write $H \leq G$.

Some basic examples include the following:

- Consider the integers, rationals, reals, and complex numbers under addition. Then:

$$\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}.$$

- Consider the non-zero rationals, non-zero reals, and non-zero complex numbers under multiplication. Then:

$$\mathbb{Q}^* \leq \mathbb{R}^* \leq \mathbb{C}^*.$$

- Both C_4 and D_2 have subgroups isomorphic to C_2 . One way of seeing this is to look at the group tables; we now write the group tables in terms of the generators, as per the end of the previous chapter:

*	e	r	r^2	r^3
e	e	r	r^2	r^3
r	r	r^2	r^3	e
r^2	r^2	r^3	e	r
r^3	r^3	e	r	r^2

*	e	r	m	rm
e	e	r	m	rm
r	r	e	rm	m
m	m	rm	e	r
rm	rm	m	r	e

In the first table, we can spot a ‘subtable’ which itself is the group table of C_2 ; this is indicated by highlighting the rows and columns corresponding to $\{e, r^2\}$.

In the second table, we can similarly spot a ‘subtable’ which itself is the group table of C_2 , indicated by highlighting the rows and columns corresponding to $\{e, r\}$. However, in this case, there are multiple subgroups of size two; for example, $\{e, m\}$ would have worked equally well to produce another subgroup isomorphic to C_2 . Thus, we see that subgroups of a given size are not unique (we cannot meaningfully talk about ‘the C_2 subgroup of D_2 ’, as a result).

There are also some ‘obvious’ subgroups of every group:

Definition 3.2: Every group has the subgroup $\{e\}$, where e is the identity of the group. This subgroup is called the *trivial subgroup*.

Every group G contains itself as a subgroup, G . This subgroup is called the *improper subgroup*.

Subgroups which are not the identity subgroup $\{e\}$ or the whole group G are called *proper, non-trivial subgroups*.

We will see many more examples of subgroups throughout this chapter, and you will explore many more examples of subgroups on the Examples Sheet too.

3.2 Cosets and Lagrange's theorem

One of the key properties of subgroups is that they can be used to 'split up a group' into smaller, more manageable pieces. These pieces are called *cosets*.

Definition 3.3: Let $H \leq G$ be a subgroup of a group G . The *left coset of H with representative $g \in G$* is the set:

$$gH = \{gh : h \in H\}.$$

Similarly, the *right coset of H with representative $g \in G$* is the set:

$$Hg = \{hg : h \in H\}.$$

As some examples of left cosets, we have:

- Let $H \leq C_4$ be the subgroup $H = \{e, r^2\}$ from above. The possible left cosets are:

$$eH = \{e, r^2\}, \quad rH = \{r, r^2\}, \quad r^2H = \{r^2, e\}, \quad r^3H = \{r^3, r\}.$$

We see that of the four left cosets, only two are distinct, since $H = r^2H$ and $rH = r^3H$. Further, all the left cosets have the same size, and if they are distinct, they are non-overlapping. We could break up C_4 as:

$$C_4 = H \cup rH,$$

for example.

- Let $H \leq D_2$ be the subgroup $H = \{e, r\}$ from above. The possible left cosets are:

$$eH = \{e, r\}, \quad rH = \{r, e\}, \quad mH = \{m, rm\}, \quad rmH = \{rm, m\}.$$

Once again, of the four left cosets, only two are distinct, since $H = rH$ and $mH = rmH$. Again, the left cosets have the same size, and if they are distinct, they are non-overlapping. We could break up D_2 as:

$$D_2 = H \cup mH.$$

This phenomenon is completely general; we prove it as follows.

Proposition 3.4: The left cosets of a subgroup $H \leq G$ of a finite group G satisfy the following properties:

- All left cosets have the same size.
- Every element of G belongs to exactly one left coset of H ; we say that the left cosets *partition* G .

Proof: (i) Let $H = \{h_1, h_2, \dots, h_n\}$, and let $gH = \{gh_1, \dots, gh_n\}$ be a left coset. We claim that gH has the same size as H . The only way that this could fail is if:

$$gh_i = gh_j$$

for some $i \neq j$. But then multiplying on the left by g^{-1} , we would have $h_i = h_j$, so the elements would not have been distinct in H in the first place. Hence $|H| = |gH|$ and all cosets have the same size.

(ii) Let $g \in G$ be an arbitrary element of G . Then $g \in gH$, since the identity $e \in H$, so gH contains $ge = g$. Thus every element is in *at least one* left coset.

On the other hand, suppose that $xH \cap yH$ is non-empty. Then for some elements $h_1, h_2 \in H$, we can write:

$$xh_1 = yh_2.$$

Then if $g \in xH$, for some h we can write $g = xh = xh_1h_1^{-1}h = yh_2h_1^{-1}h \in yH$. It follows that xH is completely contained in yH . Similarly, yH is completely contained in xH , and thus $xH = yH$. So two left cosets overlap if and only if they are the same coset. $\square$

The picture of left cosets we should therefore have in our minds is that they ‘cut up’ a group into equal sized parcels, as shown in Fig. 19.

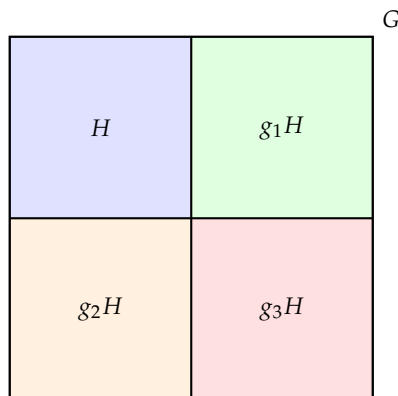


Figure 19: The left cosets of a subgroup $H \leq G$ partition a group into subsets of equal size, covering the whole group. This is represented in the above diagram by the group G (represented as a square) being divided into four equal left cosets. The left cosets shown are H , g_1H , g_2H , and g_3H , but note that the representatives of given cosets are not unique.

An immediate consequence of this result is the famous theorem of Lagrange:

Theorem 3.5: (Lagrange’s theorem) In a finite group G , the order of a subgroup H is a factor of the order of the group.

Proof: Immediate from the fact that left cosets all have the same size (and $H = eH$ itself is a coset), and they partition the group. $\square$

Lagrange’s theorem places a powerful restriction on the types of subgroups that a group can have. For example, the dihedral group D_3 of a triangle can only have subgroups of size 1, 2, 3 or 6 according to Lagrange’s theorem.

An immediately useful consequence of Lagrange's theorem is the following:

Proposition 3.6: In a finite group, the order of a group element is a factor of the order of the group.

Proof: Let $g \in G$ be any element of the group G , with order n . Then:

$$\{e, g, g^2, \dots, g^{n-1}\}$$

is a subgroup of G of order n . Hence n must divide the order of G , by Lagrange's theorem. $\square$

We have already seen this is the case for some small groups, for example the orders of the elements in $C_4 = \{e, r, r^2, r^3\}$ are $\{1, 4, 2, 4\}$ respectively, and the orders of the elements in $D_2 = \{e, r, m, rm\}$ are $\{1, 2, 2, 2\}$ respectively.

3.3 Extended example: cyclic and dihedral groups

Armed with Lagrange's theorem, we can begin to understand the subgroup structure of the cyclic and dihedral groups more generally. The cyclic groups have a particularly simple subgroup structure:

Proposition 3.7: In the cyclic group C_n , there is exactly one subgroup of order m for each factor m of n , and these are all the subgroups. This subgroup is isomorphic to C_m .

Proof: Let r be a generator of C_n , so that the group can be written as:

$$C_n = \{e, r, r^2, \dots, r^{n-1}\}.$$

Now suppose that m is a factor of n . Then $r^{n/m}$ is an element of C_n . Further, it generates a subgroup:

$$\{e, r^{n/m}, r^{2n/m}, \dots, r^{n(m-1)/m}\}.$$

This is clearly isomorphic to C_m , as required. Hence, we have shown that there is at least one subgroup of size m for each factor of n .

By Lagrange's theorem, there can only be subgroups of order m where m is a factor of n . It remains only to show that the subgroup above is the unique subgroup of size m .

Suppose H is any subgroup of size m . If $m = 1$, then the subgroup is obviously just the identity, and hence is unique. Now suppose that $m > 1$, and let $r^k \in H$ be a non-identity element in H with k the smallest possible positive integer we can choose.

Now take any other element $r^l \in H$. Since $k < l$, we can write $l = kq + s$, for some $0 \leq s < k$ and some $q \geq 1$. Therefore, we have:

$$r^l = r^{kq+s} = r^{kq}r^s.$$

Since $r^l \in H$ and $r^{kq} = (r^k)^q \in H$, we must have $r^s = r^l r^{-kq} \in H$. But then $s = 0$ by the minimality of k . So all elements of H can be written in the form $(r^k)^q$. Thus H is a cyclic group of order m , with generator r^k . In particular, we need $r^{km} = r^n = e$, so that $k = n/m$, as required. $\square$

Geometrically, the subgroups correspond to being able to ‘break’ the symmetry of our original direct polygon. For example, if we insert smaller regular polygons inscribed in the larger directed polygon, we break the symmetry. For example, the group C_6 is the group of rotational symmetries of the hexagon. It has unique subgroups isomorphic to C_1, C_2, C_3 and C_6 . The trivial subgroups C_1, C_6 correspond to breaking all the symmetry of the hexagon, or keeping all the symmetries of the hexagon, respectively. On the other hand, the group C_2 corresponds to being able to inscribe a straight line in the hexagon, between two of its vertices. The group C_3 corresponds to being able to inscribe a triangle in the hexagon, between three of its vertices. See Fig. 20 for a visualisation of this.

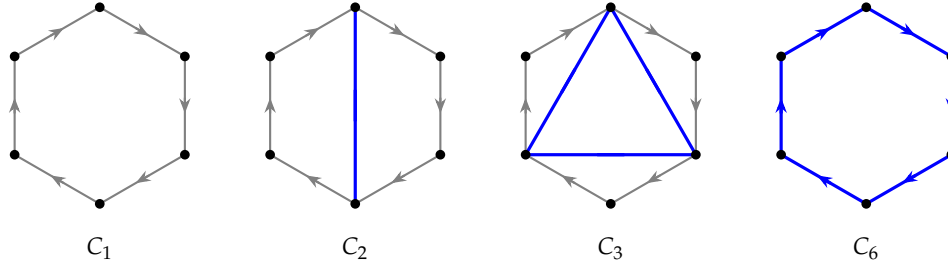


Figure 20: The subgroups of the cyclic group C_6 correspond geometrically to regular polygons that can be inscribed in a regular hexagon using its vertices. The subgroup C_2 corresponds to an inscribed line joining opposite vertices, C_3 to an inscribed equilateral triangle, and C_6 to the hexagon itself. The trivial subgroup C_1 corresponds to the identity symmetry alone.

Proposition 3.8: Consider the dihedral group $D_n = \{e, r, r^2, \dots, r^{n-1}, m, rm, \dots, r^{n-1}m\}$, written in terms of a generating rotation r and a generating reflection m . Every subgroup of D_n is one of the following:

- a cyclic group generated by $r^{n/d}$ for a factor d of n ;
- a dihedral group generated by $r^{n/d}$ and $r^k m$ for a factor d of n , and $0 \leq k < n/d$ (when $d = 1$, the group is actually cyclic, and isomorphic to C_2).

Proof: If a subgroup contains no reflections, it is a subgroup of $\{e, r, r^2, \dots, r^{n-1}\}$, i.e. the cyclic group C_n . This gives precisely the cyclic subgroups in the classification, using the proposition we proved above.

Now suppose that a subgroup H does contain a reflection, $r^k m \in H$, where k is the smallest non-negative integer for which this is the case. You will show on Examples Sheet 2, Question 4, that $H \cap \{e, r, r^2, \dots, r^{n-1}\}$ is a subgroup (the intersection of two subgroups is always a subgroup); further, this must be a cyclic subgroup, so is generated by some $r^{n/d}$ for d a factor of n . This consists of all the rotations contained in H . Now note that by closure, H needs to contain all the elements:

$$\{e, r^{n/d}, r^{2n/d}, \dots, r^{n(d-1)/d}, r^k m, r^{n/d+k} m, \dots, r^{n(d-1)/d+k} m\}.$$

Suppose now that H contains some other reflection $r^l m$, for $l > k$. Then by closure, we have that:

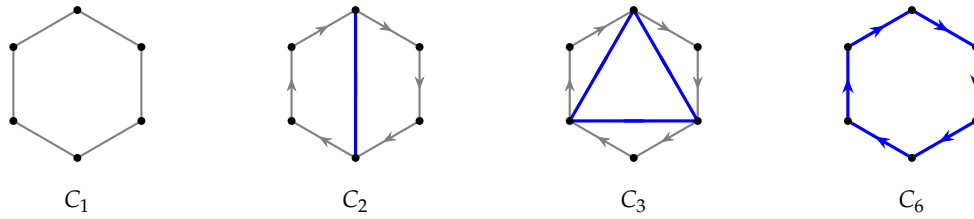
$$r^l m r^k m = r^{l-k}$$

is in H . This implies that $l - k = qn/d$ for some $q \geq 0$, which gives $l = qn/d + k$. Thus the only reflections contained in H are those already listed. From the list of reflections in H , we see that $0 \leq k < n/d$. Finally, observe that this is isomorphic to a dihedral group, since $(r^{n/d})^d = e$, $(r^k m)^2 = e$ and $(r^k m)r^{n/d} = r^{-n/d}(r^k m)$, which are the defining relations of $D_{n/d}$ (or C_2 if $d = 1$). $\square$

What do these subgroups look like concretely? Let's consider D_6 , the dihedral group of the hexagon, in some detail. By the classification result, the possible subgroups are:

- The cyclic subgroups C_1, C_2, C_3, C_6 of rotations of the equilateral hexagon. These correspond to inserting inscribed polygons into a directed version of the hexagon, just as before.
- The dihedral subgroups are generated by the following combinations:
 - The trivial rotation (i.e. the identity), and each of the possible reflections $m, rm, r^2m, r^3m, r^4m, r^5m$. These give six subgroups of order two, isomorphic to C_2 .
 - The 180° rotation r^3 , and any of the reflections m, rm or r^2m . All of these dihedral groups are isomorphic to the dihedral group D_2 .
 - The 120° rotation r^2 , and either the reflection m or the reflection rm . Both of these are the dihedral groups D_3 of the triangle.
 - The 60° rotation r , and the reflection m . This is simply the full dihedral group D_6 of the hexagon again.

Cyclic subgroups of rotations



Dihedral subgroups (and C_2 reflection groups)

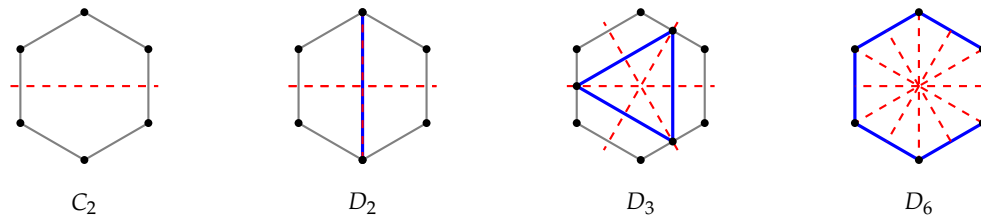


Figure 21: The subgroups of the dihedral group D_6 . The cyclic subgroups consist solely of rotations, corresponding to polygons inscribed in the directed version of the hexagon. The dihedral subgroups additionally contain reflections, indicated by dashed reflection axes. In each case we choose to use the reflection m as the generating reflection. In the first panel, this corresponds to the subgroup $\{e, m\}$. In the second panel, it corresponds to the subgroup $\{e, r^2, m, r^2m\}$. In the third panel, this corresponds to the subgroup $\{e, r, r^2, m, rm, r^2m\}$. In the final panel, the full group is the subgroup.

3.4 A converse to Lagrange's theorem: Cauchy's theorem

So far, we have stated and proved Lagrange's theorem, which tells us that the order of any subgroup of a finite group must be a factor of the larger group's order. The *converse* of this theorem would tell us the logic works the other way around: for any factor of the larger group's order, there must be a subgroup of that order. Famously, the converse is *false* - we shall shortly see a counterexample.

Now, whilst the converse of Lagrange's theorem itself is in general false, there is a *partial* converse to Lagrange's theorem. This result is called *Cauchy's theorem*, and it says specifically that for *prime* factors of a group's order, there is always a subgroup of that order.

Theorem 3.9: (Cauchy's theorem) Let G be a finite group. For each prime factor of the group's order, G has a subgroup of that order.

Proof: Let p be a prime factor of the order of the finite group G . Consider the set:

$$S = \{(x_1, \dots, x_p) : x_1 x_2 \dots x_p = e, x_i \in G\}$$

consisting of all lists of p elements in the group whose product is the identity. First, notice that the size of this set is $|S| = |G|^p$, because there are $|G|$ choices of element for each of the p positions on the list. Therefore, p divides the size of S .

Further, notice that if $(x_1, \dots, x_p)$ is in the set S , then any cyclic permutation of this list is also in the set:

$$(x_i, \dots, x_p, x_1, \dots, x_{i-1}) \in S.$$

This is because $x_1 x_2 \dots x_p = e$ implies $x_1 x_2 \dots x_p (x_1 \dots x_{i-1}) = x_1 \dots x_{i-1}$, by multiplication on the right. Multiplying on the left by $(x_1 \dots x_{i-1})^{-1}$, we have $x_i \dots x_p x_1 \dots x_{i-1} = e$.

We can therefore split the set S into two parts:

$$S = \{(x_1, \dots, x_p) : x_1 x_2 \dots x_p = e, x_i \text{ are all equal}\} \cup \{(x_1, \dots, x_p) : x_1 x_2 \dots x_p = e, x_i \text{ are not all equal}\}.$$

The second subset here must have size a multiple of p , because of the cycling property. Since the size of S is a power of p , this implies that p divides the first subset. The first subset contains $(e, \dots, e)$ so is not empty; therefore it contains some list $(x, \dots, x)$ with the property that $x^p = e$ with $x \neq e$. This shows the existence of an order p element, which must generate a cyclic subgroup of order p . $\square$

As an immediate corollary, note that this means that if p is a prime factor of the order of a finite group G , then there is an element of order p contained in the group.

Using Cauchy's theorem, we can also produce a counterexample to the converse of Lagrange's theorem:¹²

Proposition 3.10: The converse of Lagrange's theorem is, in general, false.

Proof: The smallest possible counterexample actually occurs at order 12, and is given by the chiral tetrahedral group. Suppose that the chiral tetrahedral group contains a subgroup of order six; call it H . Then by Cauchy's theorem, H must contain elements x, y of orders two and three respectively.

¹²Here, we carry out the proof using a geometric argument. It is much easier to do it algebraically, by first recognising that the chiral tetrahedral group T is isomorphic to the alternating group A_4 of permutations, which we shall see in Chapter 7. The geometric proof here also relies heavily on the idea of *conjugacy*, which is discussed in Chapter 5.

As we saw in Chapter 1, the element x of order two is necessarily a π rotation about an axis through the midpoint of two opposite edges of the tetrahedron. On the other hand, the element y of order three is necessarily a $2\pi/3$ rotation about an axis through a vertex (or a $4\pi/3$ rotation, but this is equivalent to a $2\pi/3$ in the opposite sense).

The key insight that we need is that the element of order three, y , can be thought of as permuting the axes through the midpoints of two opposite edges, as shown in Fig. 22.

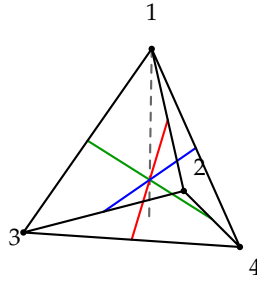


Figure 22: The three axes through the midpoints of the tetrahedron, shown in red, green and blue, are permuted by the action of a 120° rotation through any symmetry axis through a vertex. For example, rotating the base face through $2\pi/3$ sends blue to green, green to red, then red to blue. Further rotating the base face through $4\pi/3$ sends blue to red, red to green, then green to blue.

Importantly, this means that we can interpret the element $y^{-1}xy$ in the following way:

- This element first involves rotation of one of the faces. This permutes the axes through the midpoints of opposite edges.
- Next, we rotate about π through the an axis through the midpoints of opposite edges. But, given we have already swapped these axes in the first step, we are actually rotating through a *different* axis to the one we started with.
- Finally, by rotating a face in the opposite sense to the one we did originally, this restores the axes to their original configuration; albeit, having carried out a π rotation about a different axis.

Overall, $y^{-1}xy$ is a rotation by π around a different axis through midpoints of opposite signs. Similar reasoning implies that $y^{-2}xy^2$ is a rotation by π around yet another distinct axis through midpoints of opposite signs. Overall then, by closure, the subgroup H must contain all the rotations through midpoints of opposite signs.

Consider now the subset K of H given by:

$$K = \{I, R, G, B\},$$

where R, G, B are the rotations through the red, green, and blue axes in Fig. 22. By considering the transformation of the labels, it is true that $RG = B$ (a rotation about the green axis swaps 1, 3 and 2, 4; subsequently rotating through the red axis overall sends 1 to 4, 2 to 3, 3 to 2, and 4 to 1, equivalent to B). Overall then, we have $R^2 = I, G^2 = I$ and $RG = B$; multiplying the final relation by B on the right gives $RGB = I$. We saw on Examples Sheet 1, Question 6, that these relations imply K is a group (isomorphic to D_2).

But then $K \leq H$ is a subgroup of order four of a group of order six, which contradicts Lagrange's theorem. Hence H cannot exist, and the group indeed has no subgroup of order six. $\square$

3.5 Classification of finite groups up to order seven

Lagrange's theorem and Cauchy's theorem give us a lot of power in classifying groups. First, we can use Lagrange's theorem to easily classify all groups of prime order; indeed, all such groups are *cyclic*:

Proposition 3.11: Every group of prime order is isomorphic to a cyclic group; that is, the unique group of order p up to isomorphism is given by C_p .

Proof: Let G have order p . Then every non-identity element of G has order p , by Lagrange, since the only factors of p are 1 and p . Let $g \in G$ be any non-identity element of G . Then the elements of the group can be written as:

$$\{e, g, g^2, \dots, g^{p-1}\}.$$

We cannot have $g^a = g^b$ for $a < b$, since then $g^{b-a} = e$ by the exponent laws. Therefore the group is a cyclic group of order p , isomorphic to C_p . $\square$

We can use this fact, together with a more careful classification of groups of order six, to complete the classification of finite groups up to and including order seven:

Proposition 3.12: The complete list of finite groups, up to isomorphism, of orders five, six, and seven are:

- Order 5: C_5 .
- Order 6: C_6 and D_3 .
- Order 7: C_7 .

Proof: Orders five and seven are done by the previous proposition, since five and seven are prime.

It remains to consider the order six case. If there is an element of order six, then the group is isomorphic to C_6 . Hence, we may assume by Lagrange's theorem that the only element orders are one, two and three.

By Cauchy's theorem, there must be an element m of order two and an element r of order three. Let $H = \{e, r, r^2\}$ be the subgroup generated by r ; note that m is not contained in this set, because r, r^2 both have order three. Now, consider the left cosets:

$$H, \quad mH.$$

Since $m \notin H$, we must have that these cosets are distinct and therefore are unequal and cover the group. To finish, note that the right cosets H, Hm also partition the group, so by comparison with the left cosets, we must have $mH = Hm$, so that:

$$\{m, mr, mr^2\} = \{m, rm, r^2m\}.$$

This implies that one of the following is true:

- Case 1. $mr = rm$. In this case the group is Abelian, and rm has order six (by Examples Sheet 2, Question 11). This is a contradiction.
- Case 2. $mr = r^2m$. This is the defining relation for the dihedral group D_3 . Thus the group is isomorphic to D_3 . $\square$

Chapter 4

Product groups

In the previous chapter, we explored how to find smaller groups hidden inside larger groups. What if we turned this around - is there a way of taking smaller groups, and combining them to produce larger groups?

One way of doing this is to take the *direct product* of the groups (although this is not the unique way of doing this). To combine the groups G, H , we take pairs of elements of the form (g, h) where $g \in G, h \in H$, and define a new multiplication law that combines the pairs together. In this way, we have constructed a larger group, which we call the direct product, written $G \times H$.

In this chapter, we shall explore direct products, and use them to classify all finite Abelian groups and all finite groups up to order eleven. This is quite hard work, so be prepared to reread some of these sections!

In more detail, we shall cover the following in this chapter:

- We begin by defining the direct product of two groups. We give examples, and prove basic properties of the direct product. One of the key examples we explore is the product of cyclic groups; we shall show the isomorphism $C_n \times C_m \cong C_{nm}$ if and only if n, m are coprime.
- Next, we shall ask when a larger group can be written as a direct product of its subgroups. To this end, we shall state and prove the *direct product theorem*, and see some examples of its application.
- To conclude the chapter, we shall attempt two ambitious classification tasks:
 - We shall first classify all finite Abelian groups. We shall show that they are all isomorphic to direct products of cyclic groups; this allows us to write down lists of all possible Abelian groups of a given order extremely efficiently!
 - Finally, we shall use direct products to classify all groups of order up to and including 11. The non-trivial cases are orders eight, nine, and 10, which require some significant work.

4.1 Direct products

We begin with a definition of the *direct product* of two groups: forming a new larger group from a set of pairs drawn from two smaller groups.

Definition 4.1: Let G_1, G_2 be two groups. We define their *direct product* to be the set:

$$G_1 \times G_2 = \{(g_1, g_2) : g_1 \in G_1, g_2 \in G_2\}$$

of all pairs of elements from the groups, together with the operation:

$$(g_1, g_2)(g'_1, g'_2) = (g_1g'_1, g_2g'_2).$$

Indeed, we should verify that this definition does result in a group structure:

Proposition 4.2: The direct product $G_1 \times G_2$ of two groups is a group in its own right, with respect to the given operation on pairs.

Proof: We verify each of the group axioms in turn:

(G1) *Closure.* This is immediate from closure of G_1, G_2 separately. In the product:

$$(g_1, g_2)(g'_1, g'_2) = (g_1g'_1, g_2g'_2),$$

if $g_1, g'_1 \in G_1$, then $g_1g'_1 \in G_1$ by closure. Similarly $g_2g'_2 \in G_2$ by closure.

(G2) *Associativity.* This also follows from the associativity of the individual groups G_1, G_2 . We note that:

$$(g_1, g_2)[(g'_1, g'_2)(g''_1, g''_2)] = (g_1, g_2)[(g'_1g''_1, g'_2g''_2)] = (g_1(g'_1g''_1), g_2(g'_2g''_2)) = (g_1g'_1g''_1, g_2g'_2g''_2),$$

where the last expression is unambiguous because of the associativity of G_1 and the associativity of G_2 . Similarly, we can evaluate $[(g_1, g_2)(g'_1, g'_2)](g''_1, g''_2)$ to get the same result.

(G3) *Identity.* The identity is evidently (e_1, e_2) , where e_1 is the identity of G_1 and e_2 is the identity of G_2 .

(G4) *Inverses.* The inverse of the element (g_1, g_2) is evidently (g_1^{-1}, g_2^{-1}) .

Hence, we indeed have a group structure. $\square$

Having made this abstract definition, let's consider some concrete examples.

- Consider the direct product of cyclic groups $C_2 \times C_2$, where $C_2 = \{e, r\}$. We have four elements, $\{(e, e), (e, r), (r, e), (r, r)\}$. The group table is given below.

*	(e, e)	(e, r)	(r, e)	(r, r)
(e, e)	(e, e)	(e, r)	(r, e)	(r, r)
(e, r)	(e, r)	(e, e)	(r, r)	(r, e)
(r, e)	(r, e)	(r, r)	(e, e)	(e, r)
(r, r)	(r, r)	(r, e)	(e, r)	(e, e)

We see that the table is isomorphic to D_2 (note the leading diagonal is just the identity repeated four times, implying one element of order one, and three of order two), so $C_2 \times C_2 \cong D_2$.

- Consider the direct product of cyclic groups $C_2 \times C_3$, where $C_2 = \{e, r\}$ and $C_3 = \{e, r, r^2\}$. We have six elements, $\{(e, e), (e, r), (e, r^2), (r, e), (r, r), (r, r^2)\}$. Rather than constructing the group table directly, let's consider the element (r, r) in more detail. We note that its powers are given by:

$$\begin{aligned}(r, r)^0 &= (e, e), & (r, r)^1 &= (r, r), & (r, r)^2 &= (e, r^2), & (r, r)^3 &= (r, e), \\ (r, r)^4 &= (e, r), & (r, r)^5 &= (r, r^2), & (r, r)^6 &= (e, e).\end{aligned}$$

Hence, (r, r) generates the entire group $C_2 \times C_3$. It must therefore be isomorphic to the cyclic group C_6 , i.e. $C_2 \times C_3 \cong C_6$.

The interesting behaviour we have seen in the above examples leads us to an interesting question: when is it true that the direct product of two cyclic groups is isomorphic to another cyclic group?

To answer this question, we first analyse the *order* of group elements in a direct product.

Proposition 4.3: (Order of elements in a direct product) Let $g \in G, h \in H$ be group elements which have orders m, n respectively in each group. Then, the order of $(g, h) \in G \times H$ in the direct product is the lowest common multiple of m, n .

Proof: Consider:

$$(g, h)^k = (g^k, h^k).$$

We have $g^k = e$ if and only if k is a multiple of m , and $h^k = e$ if and only if k is a multiple of n . Hence $(g, h)^k = (e, e)$ if and only if k is both a multiple of m, n . This occurs for the first time when k is the lowest common multiple of m, n ; hence, this is the order of (g, h) in the direct product. $\square$

From this proposition, we can easily deduce the following result:

Proposition 4.4: Let C_{mn} be a cyclic group. Then $C_{mn} \cong C_m \times C_n$ if and only if m, n share no common factors except one.

Proof: If m, n share no common factors, then (r, r) has order mn , since this is the lowest common multiple of the order of the first r , namely m , and the second r , namely n . Thus (r, r) generates the entire group $C_m \times C_n$, which must be isomorphic to the cyclic group C_{mn} .

On the other hand, if m, n have a common factor, the lowest common multiple of m, n is strictly less than mn , say $k < mn$. $\square$

This theorem may seem quite unimportant - however, we shall see that it is central to classifying all possible Abelian groups later in this chapter.

There are many other useful properties of direct products, which make their manipulation easier. In particular, the following results will be used constantly throughout the rest of this chapter; you shall prove them as part of Examples Sheet 4:

Proposition 4.5: Let G_1, G_2, G_3 be groups. Then the direct product is commutative up to isomorphism, $G_1 \times G_2 \cong G_2 \times G_1$, and associative up to isomorphism, $G_1 \times (G_2 \times G_3) \cong (G_1 \times G_2) \times G_3$.

Proof: See Examples Sheet 4, Question 3. $\square$

4.2 The direct product theorem

We have now seen how two groups can be ‘multiplied’ together to create a larger group. Combining this idea with the concept of a subgroup, we might naturally ask: when is it possible to write $G \cong H \times K$ for two subgroups H, K of a group G ? That is, when is it possible to ‘decompose’ a group into a direct product of smaller groups?

The answer comes in the form of the *direct product theorem*:

Proposition 4.6: (The direct product theorem, version I) Suppose that $H, K \leq G$ are subgroups of a group G satisfying:

- (i) every element $g \in G$ can be written uniquely in the form $g = hk$, for elements $h \in H, k \in K$;
- (ii) for all $h \in H, k \in K$, we have $hk = kh$.

Then $G \cong H \times K$. Conversely, G is isomorphic to a direct product of groups only if there exist subgroups $H, K \leq G$ satisfying the conditions (i), (ii) above.

Proof: Suppose that there exist subgroups $H, K \leq G$ satisfying conditions (i) and (ii). We define the isomorphism $f : H \times K \rightarrow G$ by:

$$f(h, k) = hk.$$

Since every element of G can be written uniquely in the form hk , this map is evidently invertible. Further, we have:

$$f((h_1, k_1)(h_2, k_2)) = f(h_1h_2, k_1k_2) = h_1h_2k_1k_2 = h_1k_1h_2k_2 = f(h_1, k_1)f(h_2, k_2),$$

so this map is a homomorphism. Hence it is an isomorphism.

Conversely, suppose that $G \cong H' \times K'$ for some group H', K' . Let $f : H' \times K' \rightarrow G$ be an isomorphism, and define:

$$H = \{f(h', e) : h' \in H'\}, \quad K = \{f(e, k') : k' \in K'\}.$$

We claim that H, K are the required subgroups. First, observe that H is a subgroup; verifying the group axioms:

- (G1) *Closure.* Suppose that $f(h'_1, e), f(h'_2, e) \in H$. Then $f(h'_1, e)f(h'_2, e) = f((h'_1, e)(h'_2, e)) = f(h'_1h'_2, e) \in H$, so the subset is closed.
- (G2) *Associativity.* Associativity is inherited, since H is a subset of G .
- (G3) *Identity.* Note that $f(e, e) = e$, since f is an isomorphism to preserves the identity. Hence the identity is in H .
- (G4) *Inverses.* Suppose that $f(h', e) \in H$. Then $f((h')^{-1}, e) \in H$, since H' is closed under inverses, as it is a group. Further, $f(h', e)f((h')^{-1}, e) = f((h', e)((h')^{-1}, e)) = f(h'(h')^{-1}, e) = f(e, e) = e$, and similarly $f((h')^{-1}, e)f(h', e) = e$. Thus $f((h')^{-1}, e)$ is the inverse of $f(h', e)$.

Similarly, we can show that K is a subgroup. Now observe that:

- (i) Since f is an isomorphism, for every $g \in G$, there exists a unique $(h', k') \in H' \times K'$ such that $f(h', k') = g$. But then $g = f((h', e)(e, k')) = f(h', e)f(e, k')$ for elements $f(h', e) \in H$ and $f(e, k') \in K$, proving existence.

Suppose that $f(h'_1, e)f(e, k'_1) = f(h'_2, e)f(e, k'_2)$. Then $f(h'_1, k'_1) = f(h'_2, k'_2)$, implying $(h'_1, k'_1) = (h'_2, k'_2)$ since f is invertible. Thus indeed the representation is unique.

(ii) If $h = f(h', e) \in H$ and $k = f(e, k') \in K$, we have:

$$hk = f(h', e)f(e, k') = f((h', e)(e, k')) = f(h', k') = f((e, k')(h', e)) = f(e, k')f(h', e) = kh.$$

Hence, the converse of the direct product theorem is also true. $\square$

In practice, it can be quite difficult to check the first condition (i) in all but the simplest of cases. Hence, we often refine the statement of the direct product theorem to the following:

Proposition 4.7: (The direct product theorem, version II) Suppose that $H, K \leq G$ are subgroups of a group G satisfying:

- (i) $|H| \cdot |K| = |G|$;
- (ii) $H \cap K = \{e\}$;
- (iii) for all $h \in H, k \in K$, we have $hk = kh$.

Then $G \cong H \times K$. Conversely, G is isomorphic to a direct product of groups only if there exist subgroups $H, K \leq G$ satisfying the conditions (i), (ii), (iii) above.

Proof: We show that the hypotheses (i) and (ii) in this statement of the theorem imply the condition (i) in the original statement of the theorem.

Suppose that $hk = h'k'$ for elements $h, h' \in H$ and $k, k' \in K$. Then $(h')^{-1}h = k'k^{-1}$, with the left hand side in H and the right hand side in K . Since they intersect only in the identity, we must have $(h')^{-1}h = e$, i.e. $h = h'$, and we must have $k'k^{-1} = e$, i.e. $k = k'$. It follows that all elements of the form hk are distinct. Since $|H| \cdot |K| = |G|$, these elements must cover the entire group. This implies the condition (i) from the original statement of the theorem.

The converse also follows from the original proof. Note that with the isomorphism $f : H' \times K' \rightarrow G$ defined in the original proof, and our subgroups H, K , we have $|H'| = |H|$, $|K'| = |K|$, so that $|H||K| = |G|$. Further if:

$$f(h', e) = f(e, k'),$$

we must have $(h', e) = (e, k')$ since f is invertible. Thus $h' = e, k' = e$, implying that $H \cap K = \{e\}$, as required. The third condition (iii) is unchanged. $\square$

Now we have seen two statements of the direct product theorem, let's see some concrete applications:

- Consider the modular multiplication group $\mathbb{Z}_{15}^*$. The elements of the group are:

$$\{1, 2, 4, 7, 8, 11, 13, 14\}.$$

We note that $2^2 = 4, 2^3 = 8, 2^4 = 1$, so there is a cyclic subgroup $H = \{1, 2, 4, 8\}$. We also note that $11^2 = 1$, so there is a cyclic subgroup $K = \{1, 11\}$. These subgroups satisfy the conditions of the direct product theorem (namely $|H| \cdot |K| = 4 \cdot 2 = 8$, and $H \cap K = \{1\}$), and every element in H commutes with every element in K since the group is Abelian, so we have:

$$\mathbb{Z}_{15}^* \cong \{1, 2, 4, 8\} \times \{1, 11\} \cong C_4 \times C_2.$$

- Consider the symmetry group of an equilateral triangular prism. If we label the vertices 1 to 6, then there are six places that vertex 1 can be sent under a symmetry operation. After that, there are two places that the 2 vertex can be sent, requiring it to remain connected to the original vertex. Thus the order of the symmetry group is 12.

It's clear that the symmetry group needs to contain D_3 as a subgroup; this corresponds to performing symmetry operations on the triangular ends (indeed, performing the same operation to both ends at the same time).

Further, there is a reflection R in the plane through the centre of the prism, parallel to the triangular faces at the end of the prism; see Fig. 23 for a visualisation of this reflection. Note that $\{I, R\} \cong C_2$ forms a subgroup, where I is the identity operation.

Observe that these operations commute with one another: performing the reflection R , then performing any of the operations on the equilateral triangle, is equivalent to carrying out these transformations in the other order. These subgroups also intersect trivially, and the product of their sizes is 12.

It follows by the direct product theorem that the symmetry group of an equilateral triangular prism must be isomorphic to a direct product, $D_3 \times C_2$.

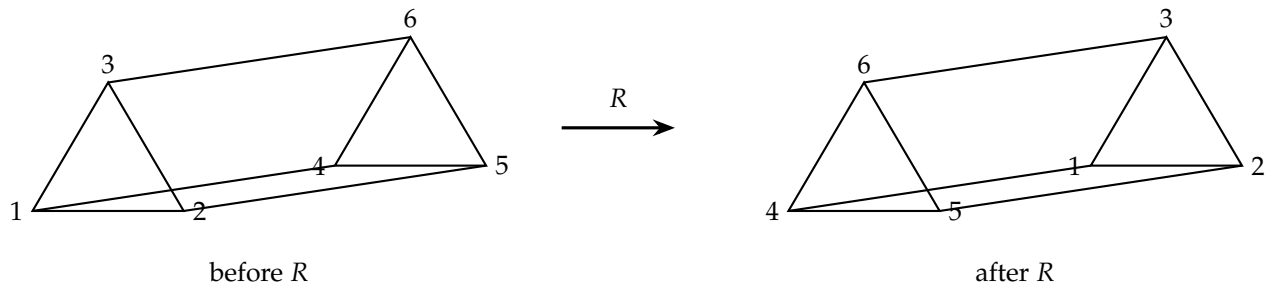


Figure 23: The effect of the reflection R on the triangular prism.

4.3 Classification of finite Abelian groups

Armed with our definition of the direct product, and the direct product theorem, we can attempt something really ambitious: the full classification of *all* finite Abelian groups. This is rather advanced, and very powerful, and hence not typically covered in a first course on group theory! However, we now have all the tools we need to approach it.

The central idea of the proof is as follows. Any finite group has some generating set (as you proved in Examples Sheet 2, Question 14). So for a general finite Abelian group G , choose some generating set for it, $\{x_1, \dots, x_k\}$. Because this is an Abelian group, the general element can then be written as:

$$x_1^{\alpha_1} x_2^{\alpha_2} \dots x_k^{\alpha_k}$$

for some integers $\alpha_1, \alpha_2, \dots, \alpha_k$. Already, this is extremely close to the direct product representation! This is because each of the x_i has finite order, and thus generates some cyclic subgroup. As a result we can write any element in the form

$$g_1 \dots g_k,$$

where g_i is in the cyclic subgroup generated by x_i . Further, all these cyclic subgroups commute since the group is Abelian. The only thing that could fail in the direct product theorem is the *uniqueness of the representation*; that is, there could be some choice of α_i and β_i such that:

$$x_1^{\alpha_1} x_2^{\alpha_2} \dots x_k^{\alpha_k} = x_1^{\beta_1} x_2^{\beta_2} \dots x_k^{\beta_k} \quad \Leftrightarrow \quad x_1^{\alpha_1 - \beta_1} x_2^{\alpha_2 - \beta_2} \dots x_k^{\alpha_k - \beta_k} = e.$$

Equivalently then, there may be some choice of integers γ_i , not all zero, such that:

$$x_1^{\gamma_1} x_2^{\gamma_2} \dots x_k^{\gamma_k} = e.$$

That is, not all of the generators can be regarded as being independent. The main point of the proof is to show that we can always construct a generating set where this cannot happen.

With our firm intuition of *how* the proof should work, we begin in earnest. We start with a useful smaller result that will help us in the main proof; such a result is called a *lemma*:

Lemma 4.8: Let $x_1, \dots, x_k$ be generators of an Abelian group G . If $c_1, \dots, c_k$ are positive integers whose highest common factor is one, then we may construct a new set of generators $y_1, \dots, y_k$ with $y_1 = x_1^{c_1} \dots x_k^{c_k}$.

Proof: We prove the claim by induction on $s = c_1 + \dots + c_k$. If $s = 1$, then the claim is immediate. Now suppose $s > 1$. Then at least two of the c_i are non-zero, and by rearrangement, we may take $c_1 \geq c_2 > 0$. Now note that:

- $\{x_1, x_1 x_2, x_3, \dots, x_n\}$ clearly still generates the group;
- $\gcd(c_1 - c_2, c_2, c_3, \dots, c_n) = 1$;
- $(c_1 - c_2) + c_2 + c_3 + \dots + c_n < s$.

Hence by the induction hypothesis, there is a generating set $y_1, \dots, y_k$ with:

$$y_1 = x_1^{c_1 - c_2} (x_1 x_2)^{c_2} x_3^{c_3} \dots x_n^{c_n} = x_1^{c_1} x_2^{c_2} \dots x_n^{c_n},$$

as required. $\square$

We can now prove the main result:

Theorem 4.9: (The fundamental theorem of finite Abelian groups) Any finite Abelian group is isomorphic to the direct product of cyclic groups.

Proof: Every finite Abelian group has a set of generators; we prove the result by induction on the number of generators. If there is a single generator, the group is cyclic by definition.

Now suppose that there are $k > 1$ generators for the finite Abelian group G , $x_1, \dots, x_k$. Suppose we have also chosen the generating set such that x_1 has the least possible *order* of all the possible generating sets we could have used. We shall show that G is the direct product of the group H generated by x_1 , and the group K generated by $x_2, \dots, x_k$.

Certainly, every element can be written in the form hk for $h \in H$, $k \in K$, by the definition of a generating set. Further, all elements commute. The only part of the direct product theorem that may fail is the *uniqueness* of the representation. Therefore, if G is *not* the direct product of H and K , then there must exist some relation:

$$e = x_1^{m_1} x_2^{m_2} \dots x_k^{m_k},$$

with $0 < m_1 < n$. Define $d = \gcd(m_1, \dots, m_k)$ and $c_i = m_i/d$ for each i . Then $\gcd(c_1, \dots, c_k) = 1$, and hence according to the Lemma there is some generating set $y_1, \dots, y_k$ with $y_1 = x_1^{c_1} \dots x_k^{c_k}$. But note that:

$$y_1^d = x_1^{dc_1} \dots x_k^{dc_k} = e,$$

and $d \leq m_1 < n$, which contradicts the choice of x_1 . Hence the relation cannot hold, and indeed the group is the direct product $G = H \times K \cong C_n \times K$. By induction, we're done. $\square$

As an example application, consider Abelian groups of order 18. The possible products of cyclic groups that would give this order are C_{18} , $C_2 \times C_9 \cong C_{18}$ (using the result from the first section of this chapter, since 2, 9 are coprime), $C_2 \times C_3 \times C_3 \cong C_6 \times C_3$. This is the complete list of Abelian groups of order 18!

4.4 Classification of finite groups up to order 11

Using everything we have learned so far, we can continue with our classification of finite groups of small order; this time we can get up to order 11.¹³

Proposition 4.10: The complete list of groups, up to isomorphism, of orders 7 to 11 are:

- Order 7: C_7 .
- Order 8: C_8 , $C_4 \times C_2$, $C_2 \times C_2 \times C_2$, D_4 and Q_8 .
- Order 9: C_9 and $C_3 \times C_3$.
- Order 10: C_{10} and D_5 .
- Order 11: C_{11} .

Proof: The prime orders 7, 11 are obviously cyclic. The order 10 case is tackled by Exercise 10 from Examples Sheet 3. This leaves only order 8 and 9 to do.

- Order 8: The Abelian case is already tackled by the fundamental theorem of Abelian groups, giving the required list.

For the non-Abelian case, note that by Lagrange's theorem, the possible element orders are 1, 2, 4, 8. If there is an element of order 8, the group is C_8 , which is Abelian. If all elements have order 2, then the group is Abelian by Exercise 16 from Examples Sheet 2, so we have already classified it.

Therefore, there must be an element of order 4, let it be a , and let $H = \{e, a, a^2, a^3\}$ be the subgroup it generates. Let b be any other element of the group not included in H . Since $|H| = 4$, it has only two left cosets in the group, which can be written as:

$$H, \quad bH.$$

Similarly, it has only two right cosets, H, Hb . Note that this implies the elements can all be uniquely written as:

$$\{e, a, a^2, a^3, b, ab, a^2b, a^3b\}.$$

Since the cosets partition the group in both cases, we must have $bH = Hb$. Thus we have:

$$ab = ba^r$$

for some $r = 0, 1, 2, 3$. If $r = 0$, then $a = e$, which is false. If $r = 1$, the group is Abelian, and we have already tackled this case. Finally, the case $r = 2$ cannot occur, else:

$$e = a^4 = (a^2)^2 = (b^{-1}ab)^2 = b^{-1}a^2b \quad \Rightarrow \quad a^2 = e.$$

Therefore the only possibility is:

$$ab = ba^3.$$

¹³At order 12, there are two new types of group which we haven't come across yet: an *alternating group*, A_4 , which we shall learn about later in the course, and a *dicyclic group*, which we shall not. Therefore, we shall stop the classification here.

To finish the classification, we split based on the order of b . We have:

- Case 1. If b is of order two, the group is evidently D_8 .
- Case 2. If b is of order four, then b^2 is of order two and cannot equal any of e, b, ab, a^2b, a^3b (else we would have $b^2 = e$, or $b \in H$). The only possible element is therefore $b^2 = a^2$.

Writing $e = 1, a = i, b = j, ab = k$, and $a^2 = b^2 = -1$, we can then write all group elements as:

$$\{1, i, -1, -i, j, k, -j, -k\}.$$

We have $i^2 = j^2 = -1$. We can also check that $k^2 = (ab)^2 = abab = ba^4b = -1$, and that:

$$ijk = abab = -1,$$

which are the defining relations of the quaternion group, Q_8 .¹⁴

- Order 9: The only possible orders of the elements are 1, 3, 9. If there is an element of order 9, the group is Abelian and cyclic, C_9 . Therefore we may assume that all elements have order 3.

Choose a to be an element of the group of order 3, and let:

$$H = \{e, a, a^2\}$$

be the subgroup it generates. Let b be an element of the group outside of H , and let:

$$K = \{e, b, b^2\}$$

be the subgroup it generates. Observe that we have $|H||K| = 9$.

Now $H \cap K$ is a subgroup of both H, K , so by Lagrange's theorem it has either order one or three. But $b \notin H$, hence $H \cap K = \{e\}$.

To finish, we wish to show that $ab = ba$ so that we can appeal to the direct product theorem, concluding that the group is isomorphic to $H \times K$.

To finish, we use a very clever trick: consider the element bab^{-1} . If $bab^{-1} = a$, the group is Abelian and we are done. Similarly, if $bab^{-1} = a^2$, then we would have $b^2ab^{-2} = ba^2b^{-1} = (bab^{-1})(bab^{-1}) = a^4 = a$, so a, b^2 commute. Then the group is Abelian (by replacing the generator b with a generator b^2) and we are done.

Thus, assume that bab^{-1} is not a power of a . Let L be the group generated by bab^{-1} ; note that $H \cap L = \{e\}$. The cosets of L partition the group as:

$$L, \quad aL, \quad a^2L.$$

Note that b^{-1} must lie in one of these cosets, so $b^{-1} = a^kba^lb^{-1}$ for some k, l . But then $a^kba^l = e$, which gives $b = a^{-k-l}$, which is a contradiction. $\square$

¹⁴The sharp-eyed amongst you will remember that when we defined the quaternions, we implicitly assumed that all 'real' numbers commuted with the base quaternions i, j, k . Technically therefore, we must check that 1 and -1 commute with i, j, k here. The fact that 1 commutes is trivial, because it is the identity. The fact that -1 commutes comes from the defining relations $i^2 = j^2 = k^2 = -1$; for example, $i(-1) = i(i^2) = i^3 = (i^2)i = (-1)i$.

Chapter 5

Conjugacy and normality

In one of the exercises early on in the course (namely Examples Sheet 2, Question 8), we proved that the elements x and xyx^{-1} have the same order. In this chapter, we explore the relationship between the elements x and xyx^{-1} much more closely, discovering that in fact, they share much more in common.

We say that the elements x and xyx^{-1} are called *conjugate elements*, related by the *conjugating element* y . In geometric groups of symmetries, we shall see that x, yxy^{-1} represent the ‘same transformation, viewed by different observers’, hence they possess a great deal more than just order in common. We shall make this idea clearer through many examples.

After introducing the idea of conjugate elements, we introduce the concept of a *normal subgroup*, which contains all its conjugates. Therefore all observers ‘agree’ on the elements of a normal subgroup. Normal subgroups are foundational in allowing us to define the concept of a ‘group modulo a subgroup’ - an extension of the idea of the ‘integers modulo n ’. These ‘groups modulo subgroups’ are called *quotient groups*, and we shall define them in this chapter and see many examples of them, including the cyclic and dihedral groups modulo their normal subgroups.

Finally, we shall discuss the idea of *simple groups*, which are groups containing no proper, non-trivial normal subgroups. We shall briefly mention the incredible fact that *all finite simple groups are completely classified* - this is a huge achievement of modern pure mathematics, and involved an enormous collaborative proof totalling thousands of pages!

In more detail, in this chapter we shall cover the following:

- We begin by introducing the idea of conjugate elements, and explain the geometric intuition behind conjugacy. We see extended examples in D_4 , and we note that in Abelian groups, all elements are self-conjugate.
- Next, we give some properties of conjugate elements. In particular, we shall prove that conjugate elements have the same order (which motivated the introduction to this chapter!), and we shall prove that if two elements are conjugate, then their inverses are conjugate. We shall prove that the conjugacy classes (sets of elements wherein every element is conjugate to every other) *partition* a group, just like the left cosets. We shall also prove that the *size* of a conjugacy class divides the order of a group.
- We then define *normal subgroups*, and give several examples. We state and prove four equivalent conditions for normality, which are useful in various settings.
- With most of the main definitions out of the way, we introduce the idea of a ‘group modulo a normal subgroup’, where all elements in the normal subgroup are considered equivalent to the identity. This is called a *quotient group*. We motivate the idea using D_4 , seeing why we need to use normal subgroups. We then make the formal definition and prove that it works in all settings, before giving some more examples.
- Turning to our favourite groups, we then classify all possible conjugacy classes, normal subgroups, and quotients of the cyclic and dihedral groups. We give geometric interpretations in each case, understanding why things work.
- Finally, we state the definition of *simple groups*, and see some basic examples. We state the classification of finite simple groups, and marvel at it.

5.1 Conjugate elements

We begin with a rather abstract notion, the definition of conjugate elements. We will immediately give it some context by considering geometry though.

Definition 5.1: Let x, y be two elements of a group. We say that y is a *conjugate element* of x if there exists some element g in the group such that:

$$y = gxg^{-1}.$$

We say that g is a *conjugating element*. The set of all elements which are conjugate to a given element x is called the *conjugacy class* of x , and is written $\text{Cl}(x)$.

Conjugacy is a very geometric notion; let us try to understand why. Suppose that we are working with the symmetry group G of some object, and suppose that y is a conjugate of x , through the conjugating element g . That is, we have $y = gxg^{-1}$, as above. We claim that x, y represent the ‘*same symmetry transformation, viewed from a different perspective*’.

An effective way of understanding this is to understand x, y as physical transformations of the object, whilst g represents an unphysical transformation of some ‘observer’. In more detail, notice that we can interpret gxg^{-1} first as a transformation g^{-1} . Instead of transforming the physical object, imagine that an *observer* is instead transformed by the transformation g . From the observer’s perspective, the physical object has indeed undergone the transformation g^{-1} . See Fig. 24 for an example with symmetries of a square.

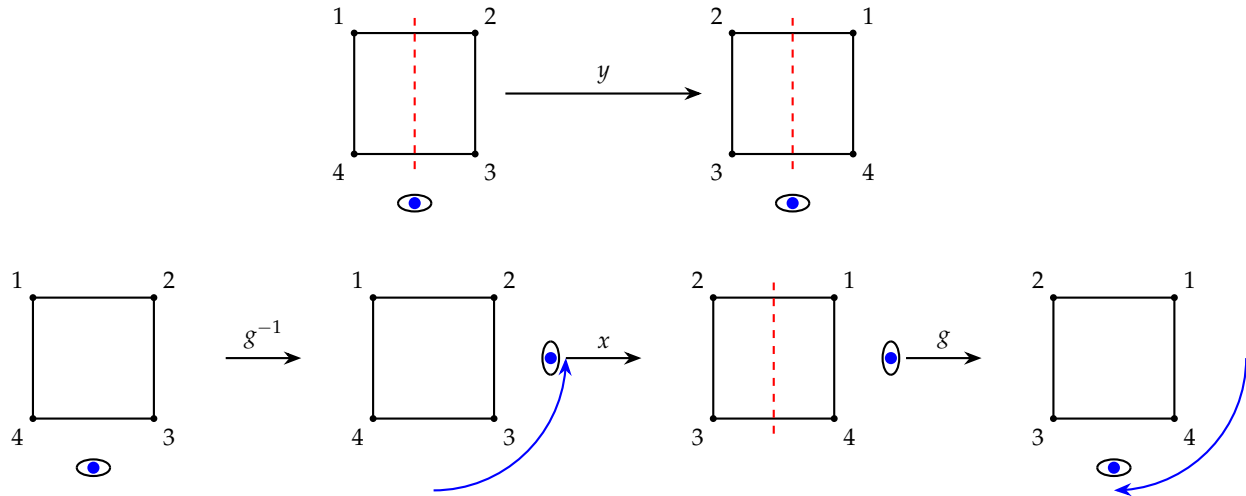


Figure 24: Let x be a horizontal reflection of a square, and y be a vertical reflection of a square. Let g be an anticlockwise rotation by $\pi/2$. One way of viewing the symmetry $y = gxg^{-1}$ is as a direct reflection y (shown in the series of top panels). Alternatively, as shown in the series of bottom panels, we can view the symmetry as rotating an observer - represented by the ‘eye’ looking at the square - by g (from their perspective, the square is rotated by g^{-1}), carrying out the reflection x in the observer’s new frame, and then rotating the observer back by g^{-1} (from their perspective, the square is rotated by g). Both procedures produce the same final configuration.

Next, the transformation x is applied to the physical object. It is applied from the perspective of the observer; in the case of Fig. 24, this means that x appears to be a *vertical reflection*, when it is a *horizontal reflection* in the original configuration.

Finally, the transformation g is applied. Again, instead of transforming the physical object, the observer is transformed by the transformation g^{-1} ; from their perspective, the physical object has undergone the transformation g , completing the overall transformation.

The idea that x, y are conjugate tells us that simply performing y directly is equivalent to this procedure involving ‘changing perspective, then applying a transformation’. But importantly, this tells us x, y are really just the same symmetry if they are conjugate - up to some ‘unphysical’ transformation of an observer.

For emphasis, let’s state this all again:

Conjugate elements in a symmetry group represent the same geometrical transformation, up to some transformation of an observer’s perspective, effected by some symmetry of the group.

Let’s see some more examples in D_4 to help the idea settle more:

- We saw in Fig. 24 that in D_4 , the vertical reflection is conjugate to the horizontal reflection. We can check this algebraically, by noting that m, r^2m represent the horizontal and vertical reflections respectively. Conjugating by r , the $\pi/2$ rotation, we have:

$$rmr^{-1} = r^2m,$$

using the fundamental relation $mr^{-1} = rm$. So indeed, we have verified that the vertical reflection r^2m is a conjugate of the horizontal reflection m .

Notice that the conjugating element is not unique - we could have used something else here. Indeed, if we had instead conjugated by a $3\pi/2$ rotation r^3 , this would also have transformed a horizontal reflection into a vertical reflection, via:

$$r^3mr^{-3} = r^6m = r^2m.$$

- Similarly in D_4 , the reflections in the diagonals are necessarily conjugate to one another. Geometrically, this is because they are related by a $\pi/2$ rotation (or a $3\pi/2$ rotation). Algebraically, we note that rm and r^3m are related via:

$$r(rm)r^{-1} = r^5m = rm,$$

using the fundamental relation $mr^{-1} = rm$ as usual. The diagonal reflections are shown in Fig. 25.



Figure 25: The two diagonal reflection axes of a square, shown as red dashed lines. These are conjugate by a $\pi/2$ rotation, similarly to the vertical/horizontal reflections.

- On the other hand, in D_4 , the only possible conjugate of the π rotation is itself. Geometrically, this is because no other symmetry effects the same kind of geometric transformation. To see it algebraically, suppose that we conjugate the π rotation r^2 by a generic conjugating element $r^\alpha m^\beta$ in D_4 :

$$(r^\alpha m^\beta)r^2(r^\alpha m^\beta)^{-1} = r^\alpha m^\beta r^2 m^\beta r^{-\alpha} = r^\alpha r^{2(-1)^\beta} r^{-\alpha} = r^{2(-1)^\beta} = r^2,$$

since $r^2 = r^{-2}$. Thus the only possible conjugate of the π rotation is the π rotation, as expected.

Some more abstract examples are as follows:

- The only conjugate of the identity e in any group is itself. To see this, let g be any possible conjugating element. Then any possible conjugate must take the form:

$$geg^{-1} = gg^{-1} = e,$$

using the identity and inverse axioms of the group. The result follows.

- In an Abelian group, the only conjugate of any element x is itself. To see this, let g be any possible conjugating element. Then any possible conjugate must take the form:

$$gxg^{-1} = gg^{-1}x = x,$$

since the group is Abelian. The result follows. Consequently, conjugacy is not a very interesting tool in the theory of Abelian groups - it is *very* useful in the theory of non-Abelian groups, though.

5.2 Properties of conjugate elements

Because conjugate elements are really the ‘same’ element, just viewed from different perspectives, we expect them to share many properties. Two basic examples are the following:

Proposition 5.2: If y is a conjugate of x , then:

- (i) x, y have the same order;
- (ii) y^{-1} is a conjugate of x^{-1} (i.e. their inverses represent the ‘same’ transformation).

Proof: (i) This was part of Examples Sheet 2, Question 8. For completeness, we carry out the proof again here. Since y is a conjugate of x , there must exist a conjugating element g satisfying $y = gxg^{-1}$. Now observe that:

$$y^k = (gxg^{-1})^k = (gxg^{-1})(gxg^{-1})\dots(gxg^{-1}) = gx^k g^{-1},$$

for all values of k , by cancelling all the g, g^{-1} terms in the middle of the x ’s. If x has order n , then taking $k = n$ gives $y^n = geg^{-1} = e$, so that y has order at most n . On the other hand, notice that if y had order $m < n$, we could take $k = m$ to give $e = gx^m g^{-1}$; multiplying on the left and right appropriately, we could deduce that $e = x^m$, which is a contradiction. Thus y has exactly order n , equal to the order of x .

(ii) Since y is a conjugate of x , there is some conjugating element g satisfying $y = gxg^{-1}$. Taking the inverse here, we have $y^{-1} = gx^{-1}g^{-1}$, using the result of Examples Sheet 2, Question 2. Hence y^{-1} is a conjugate of x^{-1} , via the same conjugating element g . $\square$

So far, we have been quite careful to say ‘ y is conjugate to x ’. But, it should be fairly obvious from the geometric description that if y is conjugate to x , then x is conjugate to y . We collect these ‘obvious’ results in the following proposition:¹⁵

Proposition 5.3: Conjugate elements satisfy the following properties:

- (i) *Reflexivity.* Any element is conjugate to itself.
- (ii) *Symmetry.* If y is a conjugate of x , then x is a conjugate of y .
- (iii) *Transitivity.* If z is a conjugate of y , and y is a conjugate of x , then z is a conjugate of x .

Proof: (i) Clear, since for any group element x , we have $x = exe^{-1}$, where e is the identity.

(ii) If y is a conjugate of x , there exists a conjugating element g such that $y = gxg^{-1}$. But then $g^{-1}yg = x$, which gives $x = g^{-1}y(g^{-1})^{-1}$, so that x is a conjugate of y via the conjugating element g^{-1} .

(iii) If z is a conjugate of y , we have $z = gyg^{-1}$ for some g . If y is a conjugate of x , we have $y = h x h^{-1}$ for some h . Therefore, $z = gyg^{-1} = ghxh^{-1}g^{-1} = (gh)x(gh)^{-1}$, so that z is a conjugate of x via the conjugating element gh . $\square$

Immediate from this proposition is the fact that conjugacy classes (now understood as sets where every element is conjugate to every other element in the set) *partition* a group, just as we saw left cosets did earlier in the course. Every element belongs to exactly one conjugacy class:

Proposition 5.4: Every element of a group belongs to precisely one conjugacy class; that is, conjugacy classes partition the group.

Proof: By reflexivity, every element is conjugate to itself, so for any x in a group, we have $x \in \text{Cl}(x)$. Thus every element belongs to some conjugacy class.

Suppose now that there is some element lying in the intersection of two conjugacy classes, $z \in C(x) \cap C(y)$. Let $x' \in C(x)$. Then x' is conjugate to x , but also z is conjugate to x , so by symmetry and transitivity, we must have that x' is conjugate to z . Additionally, z is conjugate to y , so by transitivity again we have that x' is conjugate to y . It follows that $x' \in C(y)$, and thus $C(x) \subseteq C(y)$. The same argument works in reverse, so we must have $C(x) = C(y)$. Therefore, if there is any overlap between conjugacy classes, they are in fact the same conjugacy class - so they indeed partition the group. $\square$

As a concrete example, consider D_4 . Geometrically, we know that the conjugacy classes have to be:

$$\{e\}, \quad \{m, r^2m\}, \quad \{rm, r^3m\}, \quad \{r, r^3\}, \quad \{r^2\}.$$

These correspond to doing nothing, a reflection in the vertical/horizontal, a reflection in one of the diagonals, a rotation by $\pi/2$ or $3\pi/2$ (related by a reflection), or a rotation by π .

¹⁵The properties we prove here make conjugacy an example of an *equivalence relation*, a notion of equality in mathematics. You will encounter equivalence relations more formally in courses on set theory.

To conclude this section, we shall prove an interesting result that mirrors Lagrange's theorem for subgroups: the size of a conjugacy class must divide the order of a group. This is not a result we will use in the rest of the text, but it provides a useful check for you when calculating conjugacy classes!

Proposition 5.5: Let G be a finite group, and let $\text{Cl}(x)$ be a conjugacy class in G , with x an element of the group. Then the size of $\text{Cl}(x)$ divides the order of the group.

Proof: Obviously, we are going to try to use Lagrange's theorem. Unfortunately, we cannot use Lagrange's theorem directly, because $\text{Cl}(x)$ is almost never a subgroup (the identity belongs to its own conjugacy class remember, so cannot lie in any other conjugacy classes!). The idea is instead to make a clever subgroup.

Define the set:

$$C(x) := \{g \in G : gxg^{-1} = x\}.$$

That is, $C(x)$ is the set of elements which leave x invariant, when we attempt to use g to transform x via conjugation. We claim two things about this set:

(1) *Claim 1:* $C(x)$ is a subgroup. To prove this, we check each of the group axioms in turn:

(G1) *Closure.* Let $g, h \in C(x)$. Then $gxg^{-1} = x$ and $h x h^{-1} = x$. Consider:

$$(gh)x(gh)^{-1} = ghxh^{-1}g^{-1} = gxg^{-1} = x.$$

We therefore see that $gh \in C(x)$, so the proposed subgroup is closed.

(G2) *Associativity.* This is automatically inherited from the larger group G .

(G3) *Identity.* Note that $exe^{-1} = x$, so $e \in C(x)$.

(G4) *Inverses.* Suppose that $g \in C(x)$. Then $gxg^{-1} = x$, which implies by multiplying on the left and right that $g^{-1}xg = x$, i.e. $g^{-1}x(g^{-1})^{-1} = x$. Hence $g^{-1} \in C(x)$, so the proposed subgroup contains all its inverses.

Therefore, the claim that $C(x)$ is a subgroup is proved.

(2) *Claim 2:* The number of left cosets of $C(x)$ is equal to the number of distinct conjugates of x . This is a bit more of a surprising claim. To prove it, suppose that $gxg^{-1}, h x h^{-1}$ are two conjugates of x . If they are equal, we have:

$$gxg^{-1} = h x h^{-1} \quad \Rightarrow \quad x = h^{-1} g x g^{-1} h = (h^{-1} g) x (h^{-1} g)^{-1},$$

so that $h^{-1}g \in C(x)$. This implies that $h^{-1}g = k$ for some $k \in C(x)$, giving $g = hk \in hC(x)$. Therefore, g, h belong to the same left coset of $C(x)$.

Conversely, suppose that g, h belong to the same left coset of $C(x)$, so that $g = hk$ for some $k \in C(x)$. Then the argument above is reversible, implying that $gxg^{-1} = h x h^{-1}$. It follows that two elements belong to the same coset of $C(x)$ if and only if they give rise to the same conjugate of x , after they are used to transform it via conjugation. Thus the number of left cosets of $C(x)$ is indeed equal to the number of distinct conjugates of x .

Now we can put everything together. By Lagrange's theorem, we have $|G| = n|C(x)|$, where n is the number of left cosets of $C(x)$; thus n divides $|G|$. By the second claim above though, n is equal to the number of distinct conjugates of x , i.e. the size of the conjugacy class. It follows that the size of the conjugacy class divides the order of the group, as required. $\square$

5.3 Normal subgroups

Subgroups which contain all of their conjugates are called *normal subgroups* - we like this kind of subgroup because, geometrically speaking, if an observer changes their perspective on an object, this kind of subgroup necessarily remains the same. Formally:

Definition 5.6: A subgroup of a group is called *normal* (or sometimes *invariant*) if it can be written as the union of conjugacy classes.

The following are simple examples:

- In D_3 , the conjugacy classes are $\{e\}$, $\{r, r^2\}$, $\{m, rm, r^2m\}$, corresponding to the identity transformation, the two rotations, and the three reflections. Therefore $\{e\}$, D_3 are normal subgroups (they are the unions of $\{e\}$, or all the conjugacy classes respectively), and the subgroup of rotations $\{e, r, r^2\}$ is a normal subgroup, since we can write it as:

$$\{e, r, r^2\} = \{e\} \cup \{r, r^2\}.$$

On the other hand, the subgroups of reflections are not normal subgroups. We cannot decompose $\{e, m\}$ into a union of conjugacy classes; this is because all the reflections lie in the same conjugacy class. Geometrically, this corresponds to the reflection subgroup becoming *another* reflection subgroup under conjugation. Indeed, if we transform the whole subgroup by the rotation r for example, we have:

$$r\{e, m\}r^{-1} = \{e, rmr^{-1}\} = \{e, r^2m\}.$$

This is *another* rotation subgroup, but not the original one!

- In an Abelian group, all subgroups are normal. This is because the conjugacy classes in an Abelian group are singleton sets, and any subgroup is equal to the union of all its elements.

Sometimes, there are some simpler conditions that can be used to check for normality; we will use some of these in later sections and chapters.

Proposition 5.7: Let $H \leq G$ be a subgroup of G . The following are equivalent:

- (i) H is a union of conjugacy classes (i.e. is a normal subgroup);
- (ii) $gHg^{-1} \subseteq H$ for all $g \in G$;
- (iii) $gHg^{-1} = H$ for all $g \in G$;
- (iv) $gH = Hg$ for all $g \in G$ (i.e. the left and right cosets of H coincide).

Proof: (i) $\Rightarrow$ (ii). Since H is a union of conjugacy classes, it must contain all its conjugates. Hence if $x \in H$, we have $gxg^{-1} \in H$ for any g . This implies $gHg^{-1} \subseteq H$ for any g , as required.

(ii) $\Rightarrow$ (iii). We already have $gHg^{-1} \subseteq H$ for all $g \in G$. Suppose now that $h \in H$. Given any $g \in G$, we note that $h = g(g^{-1}hg)g^{-1}$. Since $g^{-1}hg \in g^{-1}Hg \subseteq H$ by assumption, we must have $h \in gHg^{-1}$. Thus $H \subseteq gHg^{-1}$, and hence $gHg^{-1} = H$.

(iii) $\Rightarrow$ (iv). Fix some element $g \in G$, and suppose that $gh \in gH$. Then $ghg^{-1}g \in Hg$, since $ghg^{-1} \in H$ by assumption. Thus $gH \subseteq Hg$. Conversely, if $hg \in Hg$, then $gg^{-1}hg \in gH$, since $g^{-1}hg \in H$ by assumption. Thus $gH = Hg$, as required.

(iv) $\Rightarrow$ (i). Let $\text{Cl}(x)$ be the conjugacy class of any element x , and consider $\text{Cl}(x) \cap H$. If this is empty, then the conjugacy class of $\text{Cl}(x)$ is completely disjoint from H . Else, choose $h \in \text{Cl}(x) \cap H$. Then for any $y \in \text{Cl}(x)$, we have that y is conjugate to h , since $h \in \text{Cl}(x)$. Therefore there exists some conjugating element g such that $y = ghg^{-1} = h'gh^{-1} = h' \in H$ using $gh = h'g$ for some h' from the assumption $gH = Hg$. It follows that $\text{Cl}(x) \subseteq H$. Therefore, conjugacy classes are either completely disjoint from H , or completely contained in H , and hence H is the union of conjugacy classes. $\square$

We will use some of these conditions later in the text; indeed, we shall certainly use one in the next chapter on maps between groups.

5.4 Groups modulo subgroups; or, quotient groups

At the very beginning of the course, we described the idea of modular arithmetic. This concept can be extended dramatically to *any* group; the idea is that we can consider the group ‘modulo a subgroup’. Importantly, the subgroup we are removing must be a *normal subgroup* though - we shall see some examples to motivate why.

A first example: D_3 modulo rotations

To motivate this idea, let’s consider D_3 as a concrete example. Suppose that we want to consider elements of D_3 to be ‘equivalent modulo rotations, $\{e, r, r^2\}$ ’. Then D_3 should be partitioned into the cosets:

$$H = \{e, r, r^2\}, \quad mH = \{m, rm, r^2m\},$$

because all the rotations are now considered equivalent, and all the reflections are now considered equivalent (they can all be rotated into one another). Note that, crucially, the representatives of the cosets are not unique here - we have $mH = rmH$, and $H = rH$, for example.

Once we have partitioned the elements ‘modulo rotations’, multiplication in the group is now also ‘modulo rotations’, so we can only talk about ‘a rotation multiplied by a rotation’ or a ‘rotation multiplied by a reflection’ - it is meaningless to specify *which* rotation or *which* reflection we are talking about. This suggests that we should introduce some multiplication of the *cosets* H, mH themselves.

The natural way of doing this is to define $(H)(H)$, or $H(mH)$, etc, as the set of *all possible ways* of multiplying elements from the first set with elements of the second set. For example:

$$HH = \{ee, er, er^2, re, rr, rr^2, r^2e, r^2r, r^2r^2\} = \{e, r, r^2\} = H,$$

where in the second equality, we have evaluated all the products, and removed duplicate elements. Similarly, you can check that with this definition, we have $H(mH) = mH$, $(mH)H = mH$ and $(mH)(mH) = H$.

This suggests that the sets $\{H, mH\}$ form a group under this multiplication in their own right. The group table is:

$\cdot$	H	mH
H	H	mH
mH	mH	H

What has happened intuitively? As intended, all rotations have collapsed into a single element in this group, H . Similarly, all reflections have collapsed into a single element in this group, mH . The new group is a group ‘modulo rotations’ - the multiplication structure is telling us that: ‘any rotation composed with any rotation is a rotation’, and ‘any rotation composed with any reflection is a reflection’, and finally ‘any reflection composed with any reflection is a reflection’.

A first counterexample: D_3 modulo a reflection

Can we construct a similar ‘modular group’, but for D_3 modulo some reflection? Can we consider the elements D_3 to be ‘equivalent modulo reflections through a fixed vertex, $\{e, m\}$ ’? In this case, D_3 would be partitioned into the sets:

$$K = \{e, m\}, \quad rK = \{r, rm\}, \quad r^2K = \{r^2, r^2m\}.$$

Similarly to the previous case, we can try to check the products of left cosets, to see whether we have a group structure. We note, for example, that:

$$(K)(rK) = \{er, erm, mr, mrm\} = \{r, rm, r^2m, r^2\}.$$

Oh no! This is not one of the left cosets we started with. What has gone wrong?

The problem is that K , the subgroup consisting of one of the reflections, is not a *normal* subgroup. We can see this in a couple of ways:

- Geometrically, we would like K to be a set of symmetries which we consider ‘equivalent’ to the identity. But recall that by conjugation, we can ‘transform’ a symmetry into an equivalent symmetry by simply changing ‘observers’. Imagine we take any symmetry g as a conjugating element then, and consider:

$$gKg^{-1},$$

the set of all conjugates of elements in K . Recall that all elements in K are meant to be ‘equivalent’ to the identity - therefore, if we change observers, we should not produce some symmetry which is *inequivalent* to the identity. That is, we should require:

$$gKg^{-1} \subseteq K$$

which is one of the four conditions we proved for normality.

- Algebraically, we require that $(g_1H)(g_2H)$ is itself a left coset. Since $g_1 \in g_1H$ and $g_2 \in g_2H$, the left coset that the product $(g_1H)(g_2H)$ must correspond to is $(g_1g_2)H$. So, in general, we require that for all $h, h' \in H$ and $g_1, g_2 \in G$, there exists some $h'' \in H$ such that:

$$g_1hg_2h' = g_1g_2h''.$$

Rearranging this condition, we see that:

$$g_2^{-1}hg_2 = h''(h')^{-1} \in H.$$

Since this must hold for general h and general g_2 , we see that H contains all its conjugates, so H *must* be a normal subgroup for our idea of a group ‘modulo’ the subgroup to make sense.

This teaches us an important lesson: we can only consider a group ‘modulo’ a subgroup if that subgroup is a *normal subgroup*.

Formal definitions

With all this motivation behind us, we can properly define a *quotient group*: a group where elements are considered equivalent ‘modulo a normal subgroup’.

Definition 5.8: Let $H \trianglelefteq G$ be a normal subgroup of the group G . We define the *quotient group* G/H to be the set of all left cosets of H , together with the operation:

$$(g_1H)(g_2H) := \{xy : x \in g_1H, y \in g_2H\}.$$

It remains to verify that this is always a group, with the given multiplication.

Proposition 5.9: Let $H \trianglelefteq G$ be a normal subgroup of the group G . With the appropriate operation defined above, the quotient group G/H indeed forms a well-defined group.

Proof: We simply need to verify all the group axioms hold. We start with closure:

(G1) *Closure.* Let g_1H, g_2H be left cosets of H . We need to show that:

$$(g_1H)(g_2H) := \{xy : x \in g_1H, y \in g_2H\}$$

is also a left coset of H . First, note that $g_1 \in g_1H$ and $g_2 \in g_2H$, so $g_1g_2 \in (g_1H)(g_2H)$. Therefore, if this set is a left coset, it must be the left coset $(g_1g_2)H$.

Now let g_1h, g_2h' be any two elements in g_1H, g_2H respectively. Observe that:

$$g_1hg_2h' = g_1g_2 \underbrace{(g_2^{-1}hg_2)}_{\in H} h' \in (g_1g_2)H,$$

using the normality of H to identify $g_2^{-1}hg_2$ as a member of H . It follows that $(g_1H)(g_2H) \subseteq (g_1g_2)H$. Conversely, if $g_1g_2h \in (g_1g_2)H$, then we have $g_1g_2h = (g_1)(g_2h) \in (g_1H)(g_2H)$, since $g_1 \in g_1H$ and $g_2h \in g_2H$. Thus $(g_1g_2)H \subseteq (g_1H)(g_2H)$.

Hence, we have shown that $(g_1H)(g_2H) = (g_1g_2)H$ for all $g_1, g_2 \in G$, implying that G/H is closed under this multiplication.

(G2) *Associativity.* This is inherited from the group, since:

$$[(g_1H)(g_2H)](g_3H) = [(g_1g_2)H](g_3H) = ((g_1g_2)g_3)H = (g_1g_2g_3)H,$$

since the product of three elements g_1, g_2, g_3 in the group is unambiguous. Similarly, this is equal to $(g_1H)[(g_2H)(g_3H)]$.

(G3) *Identity.* The identity is clearly H itself, since $H = eH$, so $(gH)(eH) = (ge)H = gH = (eg)H = (eH)(gH)$ for all elements g of the group.

(G4) *Inverses.* The inverse of the left coset gH is $g^{-1}H$, which is also a left coset, since:

$$(gH)(g^{-1}H) = (gg^{-1})H = eH = H, \quad (g^{-1}H)(gH) = (g^{-1}g)H = eH = H.$$

Hence, all the group axioms are satisfied, and we’re done. $\square$

With the formal definition behind us, we can give some more interesting examples:

- Consider the set of integers under addition, $\mathbb{Z}$. A subgroup is the set $n\mathbb{Z}$, which is the set of all integer multiples of n . Since $\mathbb{Z}$ is Abelian, all subgroups of $\mathbb{Z}$ are normal. This means that we can form the quotient group:

$$\mathbb{Z}/n\mathbb{Z}.$$

The multiplication of the cosets satisfies:

$$(n\mathbb{Z})(m\mathbb{Z}) = (n + m)\mathbb{Z}.$$

Therefore, this group is isomorphic to the additive group of integers modulo n ; i.e. $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

- Consider the general linear group $GL(n, \mathbb{R})$, and consider the set of non-zero scalar multiples of the identity matrix:

$$H = \{\lambda I : \lambda \in \mathbb{R}^*\}.$$

We note that H is a subgroup, by verifying each of the group axioms in turn:

- (G1) *Closure*. Given $\lambda I, \mu I \in H$, we have $(\lambda I)(\mu I) = \lambda\mu I$. Since $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, and thus this subset is closed under matrix multiplication.
- (G2) *Associativity*. Matrix multiplication is associative.
- (G3) *Identity*. The identity I is included, taking $\lambda = 1$ in the definition.
- (G4) *Inverses*. The inverse of λI is $(1/\lambda)I$, which is included in the subset since $\lambda \neq 0$.

Further, H is a normal subgroup, since for any element $\lambda I \in H$, we have:

$$A(\lambda I)A^{-1} = \lambda AA^{-1} = \lambda I,$$

for all matrices $A \in GL(n, \mathbb{R})$. It follows that the quotient $GL(n, \mathbb{R})/H$ exists. This is the set of all matrices *modulo* multiplication by an overall scalar. That is, we consider the matrices:

$$\begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

to be the same. This group is called the *projective linear group* $PGL(n, \mathbb{R})$.

5.5 Extended example: cyclic and dihedral groups

As a further extended example, we shall now consider conjugacy, normality, and quotients in cyclic and dihedral groups.

Cyclic groups are particularly straightforward, because they are Abelian:

Proposition 5.10: The conjugacy classes in the cyclic group C_n are all singleton sets. All subgroups are normal. For each factor d of n , if we write $C_{n/d}$ for the unique cyclic subgroup of C_n of size n/d , we have:

$$\frac{C_n}{C_{n/d}} \cong C_d.$$

Proof: This is mostly trivial; we leave it as an exercise to the reader. The final quotient part is Question 8 of Examples Sheet 5. $\square$

The result $C_n/C_{n/d} \cong C_d$ has a nice geometrical interpretation. Since $C_{n/d} \cong \{e, r^{n/d}, r^{2n/d}, \dots, r^{n(d-1)/d}\}$, by taking the quotient we are effectively declaring that a rotation by $r^{n/d}$ is equivalent to doing nothing (we are taking the group C_n modulo $C_{n/d}$). As a result, we can pretend that if we have rotated through a certain number of vertices of the polygon, if we go too far, we suddenly jump back to the vertex we started at. In this way we effectively ‘glue’ together vertices of the polygon. An example with a hexagon is shown in Fig. 26.

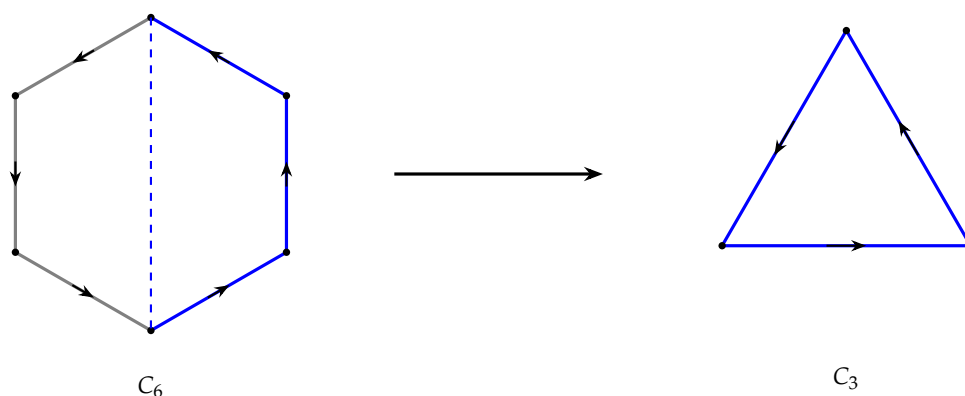


Figure 26: Consider the group C_6 , the rotational symmetries of a directed equilateral hexagon. The subgroup $C_2 \cong \{e, r^3\}$ identifies the π rotation of the hexagon with the identity. This is equivalent to ‘gluing’ two opposite vertices of the hexagon together; shown in the figure by three blue sides of the hexagon and three grey sides of the hexagon, with the vertices we are joining connected by a blue dashed line. When the vertices are glued, the remaining polygon is a directed equilateral triangle, hence the quotient becomes C_3 , as expected.

Dihedral groups are more interesting:

Proposition 5.11: For the dihedral group D_n , generated by a rotation r and reflection m :

- If n is odd, the conjugacy classes are:

$$\{e\}, \quad \{r, r^{n-1}\}, \quad \{r^2, r^{n-2}\}, \quad \dots \quad \{r^{(n-1)/2}, r^{(n+1)/2}\}, \quad \{m, rm, r^2m, r^3m, \dots, r^{n-1}m\}.$$

- If n is even, the conjugacy classes are:

$$\begin{aligned} \{e\}, \quad \{r, r^{n-1}\}, \quad \{r^2, r^{n-2}\}, \quad \dots \quad \{r^{(n-2)/2}, r^{(n+2)/2}\}, \quad \{r^{n/2}\}, \\ \{m, r^2m, r^4m, \dots, r^{n-2}m\}, \quad \{rm, r^3m, r^5m, \dots, r^{n-1}m\}. \end{aligned}$$

For n odd and even, all rotation subgroups are normal; for each divisor d of n , we have a normal subgroup:

$$H = \{e, r^{n/d}, r^{2n/d}, \dots, r^{n(d-1)/d}\},$$

which is cyclic and may be identified with C_d . The corresponding quotient is given by $D_n/H \cong D_{n/d}$.

If n is odd, there is only one normal subgroup containing any reflections, which is D_n itself. This has trivial quotient $D_n/D_n \cong C_1$.

If n is even, there are three normal subgroups containing reflections, given by D_n itself (with quotient $D_n/D_n \cong C_1$), and the two normal subgroups:

$$H_1 = \{e, r^2, r^4, \dots, r^{n-2}, m, r^2m, r^4m, \dots, r^{n-2}m\}, \quad H_2 = \{e, r^2, r^4, \dots, r^{n-2}, rm, r^3m, r^5m, \dots, r^{n-1}m\}.$$

In both cases, these subgroups are isomorphic to $D_{n/2}$. We have the quotients $D_n/H_1 \cong C_2$ and $D_n/H_2 \cong C_2$.

This is a complete list of normal subgroups and their corresponding quotients.

Proof: The conjugacy classes are clear geometrically in both cases. In the odd case, a rotation by angle θ can be viewed as a rotation by $2\pi - \theta$ by reflection, hence conjugating a reflection to its inverse and vice-versa. The reflections are all conjugate transformations, because they all result in a reflection through a vertex of the polygon and the midpoint of its opposite side.

In the even case, we similarly have that a rotation is conjugate to its inverse; this degenerates for the rotation $r^{n/2}$ which forms its own conjugacy class. Now, the reflections form two distinct conjugacy classes. Some involve reflection through an axis which passes through two opposite vertices, whilst some involve reflections through an axis which passes through two opposite sides; these have a distinct character and cannot be transformed into one another by ‘changing observers’. Thus they break into two separate conjugacy classes.

The difference between the characters of the reflections is illustrated in Fig. 27.

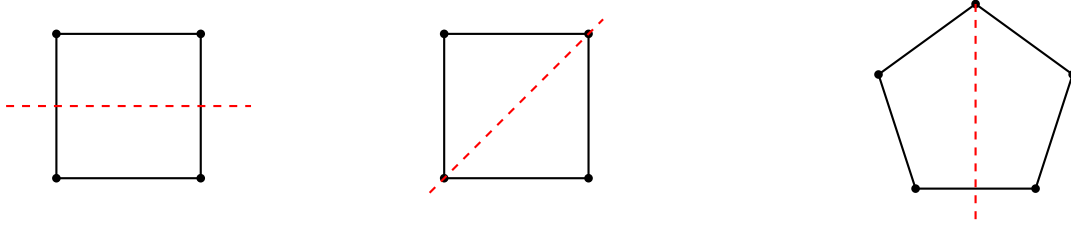


Figure 27: Reflection axes in regular polygons. For an even-sided polygon (illustrated by the square), there are two distinct types of reflection: those passing through opposite edges (left) and those passing through opposite vertices (centre). For an odd-sided polygon (illustrated by the pentagon), every reflection passes through a vertex and the midpoint of the opposite edge, so all reflections are geometrically equivalent.

Regarding normality, we need only check that the subgroups we found in Chapter 3 are unions of the conjugacy classes. Recall that all the cyclic subgroups of rotations take the form:

$$H = \{e, r^{n/d}, r^{2n/d}, \dots, r^{n(d-1)/d}\} = \{e\} \cup \{r^{n/d}, r^{n(d-1)/d}\} \cup \{r^{2n/d}, r^{n(d-2)/d}\} \cup \dots,$$

where all rotations pair up with their inverse in the odd case, and all rotations pair up with their inverse in the even case except $r^{n/2}$, which forms its own conjugacy class. Since all rotation subgroups are clearly unions of conjugacy classes, they must all be normal subgroups. The left cosets are:

$$H, \quad rH, \quad r^2H, \quad r^{n/d-1}H, \quad mH, \quad rmH, \quad r^2mH, \quad \dots, \quad r^{n/d-1}mH,$$

with $(rH)^{n/d} = H$, $(mH)^2 = H$ and $(rH)(mH) = (rm)H = (mr^{-1})H = (mH)(rH)^{-1}$. Hence the left cosets satisfy the defining relations of the dihedral group $D_{n/d}$, giving the stated isomorphism.

From Chapter 3, we know that the other type of subgroup of a dihedral group is another dihedral group; for each factor d of n , we get n/d dihedral groups generated by $r^{n/d}$ and $r^k m$, for each $k = 0, 1, \dots, n/d - 1$. These groups take the form:

$$\{e, r^{n/d}, r^{2n/d}, \dots, r^{n(d-1)/d}, r^k m, r^{k+n/d} m, \dots, r^{k+n(d-1)/d} m\}.$$

In the case that n is odd, the reflections form their own conjugacy class in D_n . As a result, we need to include *all* reflections. In particular, we must take $k = 0$ and $d = n$, giving the only possible subgroup D_n . The quotient D_n/D_n is trivial.

Otherwise, if n is even, the reflections split into two conjugacy classes. If we take $k = 0, d = n$, we recover D_n as before with quotient D_n/D_n . We can also take $k = 0, d = n/2$ if n is even, giving the subgroup:

$$H_1 = \{e, r^2, r^4, \dots, r^{n-2}, m, r^2m, \dots, r^{n-2}m\}.$$

Similarly, we can take $k = 1, d = n/2$ if n is even, giving the subgroup:

$$H_2 = \{e, r^2, r^4, \dots, r^{n-2}, rm, r^3m, \dots, r^{n-1}m\},$$

which completes the list of normal subgroups in the even case. The quotient D_n/H_1 is the set of left cosets:

$$H_1, \quad rH_1,$$

which must be isomorphic to C_2 , since there is only one group of order two up to isomorphism. Similarly $D_n/H_2 \cong C_2$. $\square$

Similarly to the cyclic group, we can try to understand the quotients of the dihedral group geometrically. First, the quotient of D_n via a rotation subgroup $\{e, r^{n/d}, r^{2n/d}, \dots, r^{n(d-1)/d}\}$ is similar to the cyclic group - the only difference is that this time, we don't deal with directed polygons. We identify a vertex with a vertex that is rotated by $r^{n/d}$, and 'glue' them together. The resulting polygon after the gluing then has symmetry group $D_{n/d}$. This is shown in the case of a hexagon in Fig. 28.

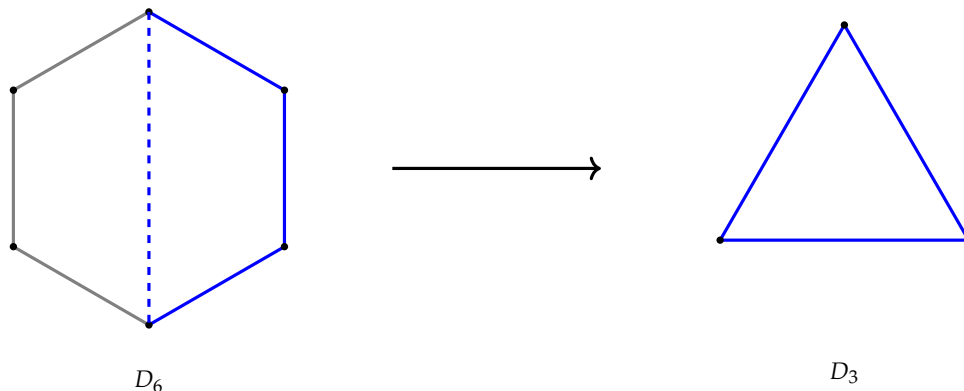


Figure 28: The same as Fig. 26, but for the dihedral group, so the directed edges are removed. Here the dihedral group D_6 of the hexagon is considered modulo the subgroup $\{e, r^3\}$; this effectively identifies ('glues') opposite vertices, leaving the symmetry group of a triangle instead. The quotient is $D_6 / \{e, r^3\} \cong D_3$.

In the case of n even, one of the two normal subgroups of D_n that contains reflections is given by:

$$H_1 = \{e, r^2, r^4, \dots, r^{n-2}, m, r^2m, r^4m, \dots, r^{n-2}m\}.$$

To understand the quotient, we notice that ...

5.6 Simple groups

We have seen in this chapter that if a group G has a normal subgroup H , we can consider all elements of H to be ‘equivalent to the identity’, hence constructing the *quotient group* G/H . This group is usually smaller (if $H \neq \{e\}$), and hence easier to study. Groups that have no proper, non-trivial normal subgroups do not admit quotient groups, so we cannot study them by breaking them down in this way. We call such groups *simple groups*:

Definition 5.12: A *simple group* is a group that has no proper, non-trivial normal subgroups.

Some easy examples are the following:

- The cyclic groups of prime order, C_p , are all simple groups. This is because they have no proper, non-trivial subgroups, so certainly cannot have any proper, non-trivial normal subgroups!
- The group $D_2 = \{e, r, m, rm\}$ is *not* a simple group; it has a normal subgroup $\{e, r\}$.

Rather surprisingly, we have the following result:

Theorem 5.13: (The classification of finite simple groups) All finite simple groups have been completely classified. They fall into one of the following classes:

- the infinite family of cyclic groups of prime order, C_p , where p is prime;
- the infinite family of alternating groups on more than 5 symbols, A_n for $n \geq 5$;
- the infinite families of classical Chevalley groups, and Chevalley groups;
- the infinite families of classical Steinberg groups, and Steinberg groups;
- the infinite family of Suzuki groups;
- the infinite family of Ree groups, together with the ‘special’ Tits group;
- one of 26 special groups, which do not fall into the above infinite families - these are called the *sporadic groups*, and include, for example, the *monster group*.

A very pretty picture of the groups is available as a ‘periodic table’ from Ivan Andrus, at <https://people.math.harvard.edu/~ctm/links/math/periodic-table.pdf>.

Proof: Ask an algebraist. $\square$

Most of the groups in this classification will look unfamiliar - indeed, we shall only encounter cyclic groups and alternating groups in this course (we shall see alternating groups in Chapter 7). Many of the other simple groups in the classification come from the study of *Lie groups* (infinite groups that depend continuously on some parameter), which you are likely to see in an advanced course in your third-year or master’s year. Lie groups are extremely important in theoretical physics, too.

There are also the *sporadic groups* - the ones that don’t fit into an infinite family. No one knows why they exist. The *monster group* is the largest of these; a nice video on the monster group is available at <https://www.youtube.com/watch?v=1bN8EMcOH5o>, starring the wonderful John Conway.¹⁶

¹⁶John Conway was a British mathematician who worked in a huge variety of areas. He was one of the main contributors in the classification of finite simple groups. He discovered three of the sporadic groups, and was also responsible for the *monstrous moonshine* conjecture, which relates group theory to the theory of complex functions - this is, surprisingly, relevant in string theory!

Chapter 6

Maps between groups

In the very first chapter, we introduced the idea of an *isomorphism*: an invertible map between groups that preserves their multiplicative structure. We explained why this concept gave an ‘equivalence’ between groups that we have used throughout the text so far.

Often, the concept of an isomorphism is quite strong. For example, imagine we have a map $f : G_1 \rightarrow G_2$ between groups that is *not* invertible, but still preserves the multiplicative structure. We could use such a map to ‘embed’ a smaller group in a larger group (i.e. map a group G_1 into a subgroup of G_2), or we could use such a map to ‘reduce’ a larger group to a smaller one, meaning we only keep part of its structure when studying it (very similar to quotienting the group by a normal subgroup, as we saw in the previous chapter - this, in fact, is no accident). Such maps are called *homomorphisms*, and are a more relaxed version of isomorphisms. We will study this type of mapping in some detail in this chapter.

Something else that we can notice about isomorphisms is that they are not *unique*; there can be many isomorphisms between two groups. This is measured by the *automorphism group* of a group, which we shall also discuss briefly.

In more detail, in this chapter we shall cover the following:

- We begin by defining homomorphisms formally, and proving their basic properties (e.g. regarding element orders, and generators, under homomorphisms). We introduce the terms *kernel* and *image* of a homomorphism, and prove relevant properties of each.
- Next, we shall prove the central result concerning homomorphisms: the *first isomorphism theorem*. This tells us how homomorphisms can be closely related to quotient groups; the elements that a homomorphism sends to the identity can be considered the elements ‘equivalent to the identity’ in the quotient.
- Finally, we consider isomorphisms between a group and itself, $f : G \rightarrow G$, which are called *automorphisms*. We show that automorphisms form a group in their own right.

6.1 Homomorphisms

We begin by defining a *homomorphism*, which is a map between groups that preserves the group multiplication, but is not necessarily invertible:

Definition 6.1: Let G_1, G_2 be two groups. A *homomorphism* is a map $f : G_1 \rightarrow G_2$ satisfying:

$$f(xy) = f(x)f(y)$$

for all $x, y \in G_1$. A homomorphism is an isomorphism if it is additionally invertible, as we defined in Chapter 1.

Some basic examples include the following:

- Consider the map $f : \mathbb{Z} \rightarrow \mathbb{Q}$, where $\mathbb{Z}, \mathbb{Q}$ are the additive groups of integers and rationals respectively, given by:

$$f(x) = x.$$

This is trivially a homomorphism, because:

$$f(x + y) = x + y = f(x) + f(y).$$

Hence, we see that a homomorphism can be used to ‘embed’ a smaller group inside a larger group.

- Consider the map $f : \mathbb{Z} \rightarrow \mathbb{Z}_2$ given by:

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is odd,} \\ 0, & \text{if } x \text{ is even.} \end{cases}$$

This is a homomorphism, because:

$$f(x + y) = \begin{cases} 1, & \text{if } x + y \text{ is odd,} \\ 0, & \text{if } x + y \text{ is even,} \end{cases} \quad \text{and} \quad f(x) + f(y) = \begin{cases} 0, & \text{if } x, y \text{ are both odd, so } x + y \text{ is even,} \\ 1, & \text{if } x \text{ is odd, } y \text{ is even, so } x + y \text{ is odd,} \\ 1, & \text{if } x \text{ is even, } y \text{ is odd, so } x + y \text{ is odd,} \\ 0, & \text{if } x, y \text{ are both even, so } x + y \text{ is even.} \end{cases}$$

Hence in all cases we have $f(x + y) = f(x) + f(y)$. Thus the group $\mathbb{Z}$ has been ‘compressed’ or ‘reduced’ to the group $\mathbb{Z}_2$. It is no accident that the elements that map to 0 are precisely the even integers - the ones we ‘identify with zero’ in the quotient $\mathbb{Z}/2\mathbb{Z}$ from the previous chapter.

- As a non-Abelian example, consider the map $f : D_3 \rightarrow C_2$ with $D_3 = \{e, r, r^2, m, rm, r^2m\}$ and $C_2 = \{e, s\}$, given by:

$$f(r^k) = e, \quad f(r^k m) = s.$$

for all k . More compactly, we can simply write:

$$f(r^k m^l) = s^l.$$

We can check that this is a homomorphism directly:

$$f(r^\alpha m^\beta r^{\alpha'} m^{\beta'}) = f(r^{\alpha+\alpha'} m^{\beta+\beta'}) = s^{\beta+\beta'} = s^\beta s^{\beta'} = f(r^\alpha m^\beta) f(r^{\alpha'} m^{\beta'}),$$

so indeed we have that f is a homomorphism.

This homomorphism is interesting because it sends all rotations to the identity; this seems to be saying ‘forget about the rotation itself, consider it equal to one - we are only interested in whether the triangle is reflected or not’. This very closely mirrors the quotient group $D_3 / \{e, r, r^2\} \cong C_2$. Again, this is no accident as we shall see.

Now we have seen several examples of group homomorphisms, we can start to prove some of their basic properties. Some are similar to the properties of isomorphisms, that we have already seen:

Proposition 6.2: Let G_1, G_2 be two groups, and let $f : G_1 \rightarrow G_2$ be a homomorphism. Then the following hold:

- (i) $f(e_1) = e_2$, where e_1, e_2 are the identities of G_1, G_2 respectively;
- (ii) $f(x^{-1}) = f(x)^{-1}$ for all $x \in G_1$;
- (iii) the order of $f(x)$ divides the order of x .

Proof: (i) The proof is identical to the one we gave in Chapter 1 for an isomorphism (we didn't use the invertibility there!).

(ii) The proof is the same as the one you used on Examples Sheet 1, Question 12. For completeness, we carry it out here again. For any $x \in G_1$, by part (i), we have:

$$e_2 = f(e_1) = f(xx^{-1}) = f(x)f(x^{-1}),$$

using the homomorphism property in the last equality. Thus $f(x)^{-1} = f(x^{-1})$, as required.

(iii) Suppose that x has order n in G_1 . Consider $e_2 = f(e_1) = f(x^n)$. We have:

$$e_2 = f(e_1) = f(x^n) = f(x)f(x^{n-1}) = \dots = f(x)^n$$

by repeatedly using the homomorphism property. Hence, the order of $f(x)$ is at most n . $\square$

These basic properties can be useful in ruling out possibilities for homomorphisms. For example, we can find all homomorphisms $f : C_4 \rightarrow C_6$ using this result. Write:

$$C_4 = \{e, r, r^2, r^3\}, \quad C_6 = \{e, s, s^2, s^3, s^4, s^5\}.$$

Observe that if $f : C_4 \rightarrow C_6$ is a homomorphism, we must have $f(e) = e$. Further, if we know $f(r)$, then $f(r^2)$ and $f(r^3)$ are determined by the homomorphism property:

$$f(r^2) = f(r)^2, \quad f(r^3) = f(r)^3.$$

It follows that the homomorphism is completely determined by the choice of $f(r)$. Since r has order four, we must have that the order of $f(r)$ divides four by the above result.

The orders of the elements of C_6 are order one for e , order six for s , order three for s^2 , order two for s^3 , order three for s^4 , and order six for s^5 . Thus there are only two possible choices for $f(r)$. We can either take $f(r) = e$, which gives the complete homomorphism:

$$f(e) = e, \quad f(r) = e, \quad f(r^2) = e, \quad f(r^3) = e,$$

or we can take $f(r) = s^3$, which gives the complete homomorphism:

$$f(e) = e, \quad f(r) = s^3, \quad f(r^2) = e, \quad f(r^3) = s^3.$$

This is the complete list of possible homomorphisms $f : C_4 \rightarrow C_6$.

6.2 Homomorphisms, generators, and relations

Often, it can be quite painstaking to check whether a given map $f : G_1 \rightarrow G_2$ is a homomorphism; indeed checking the condition $f(xy) = f(x)f(y)$ can often be quite burdensome.

In practice therefore, it is often useful to come up with an easy test for whether or not a map $f : G_1 \rightarrow G_2$ is a homomorphism in terms of the *generators* and *relations* which describe the group G_1 . In general, we have the following:

Proposition 6.3: Suppose that G_1 is generated by the elements $S = \{g_1, \dots, g_n\}$ subject to m defining relations (which completely specify the group algebra):

$$x_{1,1}x_{2,1}\dots x_{k_1,1} = e, \quad \dots, \quad x_{1,m}x_{2,m}\dots x_{k_m,m} = e,$$

where $x_{i,j}$ is equal to an element in the generating set, or its inverse, in each case (note any relation can be expressed in the form ' $\dots = e$ ' by appropriate multiplication). Given another group G_2 and a function $g : S \rightarrow G_2$, there is a unique map extending g to a homomorphism $f : G_1 \rightarrow G_2$ (i.e. with $f|_S = g$) if and only if:

$$g(y_{1,\alpha})g(y_{2,\alpha})\dots g(y_{k_\alpha,\alpha}) = e$$

for all possible α . That is, the map g need only respect the relations for it to be uniquely extendable to a homomorphism.

Proof: If there is such a homomorphism $f : G_1 \rightarrow G_2$ then for each α , i.e. for each relation, we must have:

$$e = f(x_{1,\alpha}\dots x_{k_\alpha,\alpha}) = f(x_{1,\alpha})\dots f(x_{k_\alpha,\alpha}) = g(x_{1,\alpha})\dots g(x_{k_\alpha,\alpha}),$$

using the fact that f is a homomorphism, and agrees with g on the generating set.

Conversely, suppose that g respects all the relations. We want to construct a homomorphism $f : G_1 \rightarrow G_2$ extending g . Given an arbitrary element $a \in G_1$, we can write:

$$a = x_1^{\alpha_1}x_2^{\alpha_2}\dots x_r^{\alpha_r}, \quad (*)$$

where $\alpha_i \in \mathbb{Z}$ and $x_i \in S$ is a generator in each case. We would like to be able to define:

$$f(a) = g(x_1)^{\alpha_1}g(x_2)^{\alpha_2}\dots g(x_r)^{\alpha_r}, \quad (**)$$

but there may be an issue with this definition: there may be multiple ways of writing a in terms of the generators in (*). Hence, we must check that our definition (**) is independent of this choice.

The defining relations are, by assumption, the only relations between the generators which are otherwise free. Therefore if there are two such expressions for a in terms of the generators, they must be related to one another using the defining relations. For example, in the dihedral group D_n we can write:

$$a = rm = mr^{n-1} = r^n mr^{n-1} = \dots = m^2 r m r m r^n m r^{n-1},$$

which are all equivalent expressions in terms of the generators.

Note however that since g respects every defining relation. Hence, if we represent a with two different expressions in terms of the generators, they must be related by some transformations using the relations. The corresponding expressions for $f(a)$ can be manipulated in exactly the same way, using the same transformations using the relations, since g respects them all. Hence the expression for $f(a)$ can equally be written in the second form too.

Hence, we can conclude that $f : G_1 \rightarrow G_2$ constructed in the above manner is well-defined. It is also the unique potential extension of g to a homomorphism $f : G_1 \rightarrow G_2$. It remains to check that f constructed in this way is indeed a homomorphism. We let:

$$a = a_1^{\alpha_1} a_2^{\alpha_2} \dots a_r^{\alpha_r}, \quad b = b_1^{\beta_1} b_2^{\beta_2} \dots b_s^{\beta_s},$$

where $a_j, b_j \in S$ are generators and $\alpha_j, \beta_j \in \mathbb{Z}$. Then we have:

$$\begin{aligned} f(ab) &= f(a_1^{\alpha_1} a_2^{\alpha_2} \dots a_r^{\alpha_r} b_1^{\beta_1} b_2^{\beta_2} \dots b_s^{\beta_s}) \\ &= g(a_1)^{\alpha_1} g(a_2)^{\alpha_2} \dots g(a_r)^{\alpha_r} g(b_1)^{\beta_1} g(b_2)^{\beta_2} \dots g(b_s)^{\beta_s} && \text{(definition of } f) \\ &= f(a_1^{\alpha_1} a_2^{\alpha_2} \dots a_r^{\alpha_r}) f(b_1^{\beta_1} \dots b_s^{\beta_s}) \\ &= f(a)f(b), \end{aligned}$$

as required. $\square$

This allows us to quite efficiently determine whether a map is a homomorphism simply by considering where the generators are mapped to. For example consider trying to find all homomorphisms $f : C_n \rightarrow C_m$.

Note that C_n is generated by a single element r , so it sufficient to specify $f(r) \in C_m$. The only relation that r satisfies is $r^n = e$. Hence, provided that $f(r)^n = e$, we indeed have a homomorphism. Suppose now that $f(r) = r^k$ for some k . Then we require that:

$$e = f(r)^n = r^{kn}$$

which holds if and only if $kn \equiv 0$ modulo m . Hence there is a homomorphism $f(r) = r^k$ if and only if m divides kn .

6.3 Kernel and image of a homomorphism

The examples we gave in the first section seemed to suggest that the elements that map to the identity are important in a homomorphism. We give them a name: the *kernel* of the homomorphism. The elements that we can access from the homomorphism are also given a name: the *image* of the homomorphism.

Definition 6.4: Let G_1, G_2 be two groups, and let $f : G_1 \rightarrow G_2$ be a homomorphism. We make the following definitions:

- The *kernel* of the homomorphism is the set of elements in G_1 that are mapped to the identity e_2 of G_2 , namely:

$$\text{Ker}(f) := \{x \in G_1 : f(x) = e_2\}.$$

- The *image* of the homomorphism is the set of all elements in G_2 that are accessible from G_1 via f , namely:

$$\text{Im}(f) := \{y \in G_2 : \text{there exists } x \in G_1 \text{ with } f(x) = y\}.$$

In the examples from the first section:

- For the homomorphism $f : \mathbb{Z} \rightarrow \mathbb{Q}$ with $f(x) = x$, we have that the kernel is just $\text{Ker}(f) = \{0\}$, and the image is $\text{Im}(f) = \mathbb{Z}$.

- For the homomorphism $f : \mathbb{Z} \rightarrow \mathbb{Z}_2$ with $f(x) = 0$ if x is even, $f(x) = 1$ if x is odd, we have that the kernel is $\text{Ker}(f) = 2\mathbb{Z}$ (the set of even integers), and the image is $\text{Im}(f) = \mathbb{Z}_2$.
- For the homomorphism $f : D_3 \rightarrow C_2$ given by $f(r^k m^l) = s^l$ with D_3 generated by r, m and C_2 generated by s , we have that the kernel is $\text{Ker}(f) = \{e, r, r^2\}$ and the image $\text{Im}(f) = C_2$ is all of C_2 .

The basic properties of the kernel and the image are that they are subgroups, and the kernel is a *normal* subgroup, as we show below:

Proposition 6.5: Let $f : G_1 \rightarrow G_2$ be a homomorphism. Then the following hold:

- (i) the kernel of the homomorphism is a normal subgroup of G_1 , $\text{Ker}(f) \trianglelefteq G_1$;
- (ii) the image of the homomorphism is a subgroup (not necessarily normal) of G_2 , $\text{Im}(f) \leq G_2$.

Proof: (i) We first verify the group axioms for the kernel.

- (G1) *Closure.* Let k, k' be elements of the kernel. Then $f(kk') = f(k)f(k') = e_2$, since $f(k) = e_2$ and $f(k') = e_2$, where e_2 is the identity of G_2 . Thus kk' is in the kernel, so it is closed.
- (G2) *Associativity.* The kernel is a subset of G_1 , so associativity is inherited from the associativity of G_1 .
- (G3) *Identity.* We proved earlier that $f(e_1) = e_2$, so e_1 is in the kernel, and hence the kernel contains the identity.
- (G4) *Inverses.* We proved earlier that $f(x^{-1}) = f(x)^{-1}$. Hence if k is in the kernel, so that $f(k) = e_2$, we must have $f(k^{-1}) = f(k)^{-1} = e_2$, proving that the inverse of x is also contained in the kernel.

Now, we shall prove that the kernel is in fact a *normal* subgroup. Let $k \in \text{Ker}(f)$ be an element of the kernel, and let g be any element of G_1 . We consider the conjugate gkg^{-1} :

$$\begin{aligned}
 f(gkg^{-1}) &= f(g)f(k)f(g)^{-1} && \text{(homomorphism property, and } f(x^{-1}) = f(x)^{-1}) \\
 &= f(g)e_2f(g)^{-1} && (k \in \text{Ker}(f)) \\
 &= f(g)f(g)^{-1} && \text{(identity in } G_2) \\
 &= e_2 && \text{(inverses in } G_2)
 \end{aligned}$$

Hence $gkg^{-1} \in \text{Ker}(f)$. Thus $\text{Ker}(f)$ contains all its conjugates, and is hence a normal subgroup by definition.

(ii) For the image, we just need to verify the group axioms.

- (G1) *Closure.* Let $x, y \in G_2$ be elements in the image. Then there exist some elements $a, b \in G_1$ such that $f(a) = x$ and $f(b) = y$. By the homomorphism property, $f(ab) = f(a)f(b) = xy$, so xy is in the image. Therefore it is closed under multiplication in G_2 .
- (G2) *Associativity.* The image is a subset of G_2 , so associativity is inherited from the associativity of G_2 .
- (G3) *Identity.* Since $f(e_1) = e_2$, the identity of G_2 is contained in the image.
- (G4) *Inverses.* Let $x \in G_2$ be an element in the image. Then there exists some element $a \in G_1$ such that $f(a) = x$. But then $f(a^{-1})$ is in the image too, and by a property of homomorphisms, we have $f(a^{-1}) = f(a)^{-1} = x^{-1}$. Thus x^{-1} is contained in the image too, and it follows that the image contains the inverse of every one of its elements.

Hence, we're done. $\square$

6.4 The isomorphism theorem

The fact that we have been dancing around in the examples above is that the kernel of a homomorphism can be thought of as a process which ‘identifies’ all elements of the kernel with the identity. This is essentially the same as *quotienting* by the elements of the kernel. This suggests the following fundamental theorem:

Theorem 6.6: (The isomorphism theorem) Let $f : G_1 \rightarrow G_2$ be a homomorphism between the groups G_1, G_2 . Then there is an isomorphism:

$$G_1/\text{Ker}(f) \cong \text{Im}(f),$$

given by the map $q : G_1/\text{Ker}(f) \rightarrow G_2$ defined by $q(g\text{Ker}(f)) = f(g)$, which sends a left coset to the image of a representative of that left coset.

Proof: First, we need to check that the map q makes sense. We observe:

- It has the right domain, and the right image.
- We need to check that if the coset is represented by different representatives, $g\text{Ker}(f) = g'\text{Ker}(f)$, then we get the same answers for $q(g\text{Ker}(f)), q(g'\text{Ker}(f))$. Observe that if $g\text{Ker}(f) = g'\text{Ker}(f)$, then there exists some $k \in \text{Ker}(f)$ such that $g = g'k$. Then $(g')^{-1}g = k \in \text{Ker}(f)$, so that:

$$q(g'\text{Ker}(f)) = f(g') = f(g')f(k) = f(g'k) = f(g) = q(g\text{Ker}(f)).$$

So, q is a well-defined map. We now need to check the properties that q is supposed to have:

- We note that q is invertible, with inverse $q^{-1}(f(g)) = g\text{Ker}(f)$. To check this makes sense, we suppose that $f(g) = f(g')$. Then $f(g(g')^{-1}) = e$, so that $g(g')^{-1} = k \in \text{Ker}(f)$. This implies that $g = kg'$, giving $g\text{Ker}(f) = g'\text{Ker}(f)$.
- We note that q is a homomorphism:

$$q(g\text{Ker}(f)g'\text{Ker}(f)) = q((gg')\text{Ker}(f)) = f(gg') = f(g)f(g') = q(g\text{Ker}(f))q(g'\text{Ker}(f)).$$

Hence, we’re done. $\square$

Applying this to the above examples, we observe:

- For the homomorphism $f : \mathbb{Z} \rightarrow \mathbb{Q}$ with $f(x) = x$, we have $\text{Ker}(f) = \{0\}$, and $\text{Im}(f) = \mathbb{Z}$, and obviously:

$$\frac{\mathbb{Z}}{\{0\}} \cong \mathbb{Z}.$$

- For the homomorphism $f : \mathbb{Z} \rightarrow \mathbb{Z}_2$ with $f(x) = 0$ if x is even, $f(x) = 1$ if x is odd, we have $\text{Ker}(f) = 2\mathbb{Z}$ and $\text{Im}(f) = \mathbb{Z}_2$. We have:

$$\frac{\mathbb{Z}}{2\mathbb{Z}} \cong \mathbb{Z}_2.$$

- Finally, for the homomorphism $f : D_3 \rightarrow C_2$ given by $f(r^k m^l) = s^l$, with D_3 generated by r , and C_2 generated by s , the kernel is $\{e, r, r^2\}$ and the image is C_2 . We see that:

$$\frac{D_3}{\{e, r, r^2\}} \cong C_2.$$

6.5 Automorphisms

We said a very long time ago when we introduced isomorphisms that isomorphisms are *not unique*. Indeed, there are many ways that two groups can be isomorphic.

This idea can be captured in the following way: if there are non-trivial isomorphisms from a group to *itself*. Such a map is called an *automorphism*.

Definition 6.7: An *automorphism* of a group G is an isomorphism $f : G \rightarrow G$.

A classic example of an automorphism is *conjugation*; given a group element $g \in G$, we can define the automorphism:

$$\alpha_g(x) = gxg^{-1}.$$

This automorphism sends an element $x \in G$ to its conjugate via the conjugating element $g \in G$. To show it is an automorphism, we need to check the two properties of an isomorphism:

- Certainly $\alpha_g(x)$ is invertible, with inverse $\alpha_{g^{-1}}(x) = g^{-1}xg$.
- We also have:

$$\alpha_g(xy) = gxyg^{-1} = gxg^{-1}gyg^{-1} = \alpha_g(x)\alpha_g(y),$$

so α_g is a homomorphism.

Thus indeed α_g is an automorphism.

The most basic property of automorphisms is that they themselves form a group (they are the ‘symmetry group of the symmetries of an object’!).

Proposition 6.8: The set of all automorphisms of a group G , written $\text{Aut}(G)$, together with the operation of composition of functions forms a group.

Proof: We simply check the group axioms. We have:

(G1) *Closure.* If $\alpha : G \rightarrow G, \beta : G \rightarrow G$ are both automorphisms, then $\alpha \circ \beta$ is invertible with inverse $\beta^{-1} \circ \alpha^{-1}$, and it is a homomorphism since:

$$(\alpha \circ \beta)(xy) = \alpha(\beta(xy)) = \alpha(\beta(x)\beta(y)) = \alpha(\beta(x))\alpha(\beta(y)).$$

Thus $\alpha \circ \beta$ is an automorphism.

(G2) *Associativity.* Function composition is associative.

(G3) *Identity.* The identity automorphism, $\text{id}_G : G \rightarrow G$ given by $\text{id}_G(x) = x$ is an automorphism.

(G4) *Inverses.* If $\alpha : G \rightarrow G$ is an automorphism, then $\alpha^{-1} : G \rightarrow G$ exists and is invertible by definition. Further, we have:

$$\alpha^{-1}(xy) = \alpha^{-1}(\alpha(x')\alpha(y')) = \alpha^{-1}(\alpha(x'y')) = x'y' = \alpha^{-1}(x)\alpha^{-1}(y),$$

for any two elements $x, y \in G$, with $x = \alpha(x'), y = \alpha(y')$. It follows that $\alpha^{-1} : G \rightarrow G$ is an automorphism.

Hence, we’re done. $\square$

You will explore the structure of automorphism groups further on the Examples Sheet.

Chapter 7

Permutation groups

Many groups can be viewed as groups of *permutations*. Indeed, the symmetries of a triangle can be viewed as permuting the vertices, swapping the labels 1, 2, 3 in all possible ways. Similarly, the symmetries of a square can be viewed as permuting the vertices, swapping the labels 1, 2, 3, 4 in *some* of the possible ways (in particular, we require that 1 remains adjacent to 2, 4 for example).

Remarkably, this is the case for *all* groups; this result is called *Cayley's theorem*, which tells us that any group is isomorphic to some subgroup of a permutation group.

Consequently, understanding permutations could be an exceptionally useful tool in understanding groups generally. We will spend most of the chapter trying to understand permutations better, in order to better understand general groups.

In more detail, we shall cover the following:

- We begin with the definition of a permutation, and show that the permutations of any set (including infinite sets) form a group. We state and prove *Cayley's theorem*, telling us that any group is isomorphic to a subgroup of a permutation group, motivating the study of permutations in the rest of the chapter.
- We then consider permutations of a finite set, relevant in the study of finite groups. We define the *symmetric group on n symbols*, S_n , which is the group of permutations of the finite set $\{1, 2, \dots, n\}$. We then introduce some notation for permutations of finite groups: *Cauchy's two-line notation*. We see how this notation can be used to compose permutations conveniently.
- At this stage in the chapter, we will be able to multiply permutations but in quite a laborious way. We aim to replace Cauchy's two-line notation with a new type of notation called *disjoint cycle permutation*. This allows us to conveniently multiply permutations.
- Next, we study the idea of *even* and *odd* permutations, which builds any permutation from simply *swapping elements*. We show that this is always possible, and hence define the normal subgroup of *even permutations* in the symmetric group, which is called the *alternating group on n symbols*, A_n .
- To conclude, we study *conjugacy* in the permutation groups. In S_n , conjugacy is straightforward to understand. However in A_n , conjugacy is more complicated - classes in S_n can *split* in two in A_n , if a condition is satisfied.

7.1 Permutations and Cayley's theorem

We begin by defining a permutation:

Definition 7.1: Let S be a set. A *permutation* of the set is an invertible function $f : S \rightarrow S$.

In general, a permutation can be regarded as a 'symmetry' of an abstract set - it is some reordering of the points that produces the same set again. It should therefore be unsurprising that:

Proposition 7.2: The permutations of any set S form a group under function composition. This group is written $\text{Sym}(S)$.

Proof: We check the group axioms:

- (G1) *Closure.* If $\alpha, \beta : S \rightarrow S$ are invertible functions, then $\alpha \circ \beta$ is invertible with inverse $\beta^{-1} \circ \alpha^{-1}$. Hence the set is closed under composition.
- (G2) *Associativity.* Function composition is associative.
- (G3) *Identity.* The identity $\text{id} : S \rightarrow S$, given by $\text{id}(x) = x$ is a permutation.
- (G4) *Inverses.* The inverse of $\alpha : S \rightarrow S$, given by $\alpha^{-1} : S \rightarrow S$, is invertible by definition. Hence it is in the set.

Thus the group axioms are satisfied, and indeed the set of permutations forms a group. $\square$

This formalises the idea that we saw earlier that 'symmetries of any object form a group'.

Perhaps much more remarkably, any group can be viewed as a group of permutations. The proof identifying multiplication by some group element g as a permutation of the group elements (this is essentially the *Latin square* property of group tables):

Theorem 7.3: (Cayley's theorem) Any group G is isomorphic to a subgroup of $\text{Sym}(G)$.

Proof: Define the map $\alpha_g : G \rightarrow G$ via $\alpha_g(h) = gh$; this is the *left-multiplication map*. It is obviously invertible, with inverse $\alpha_{g^{-1}} : G \rightarrow G$. Hence, $\alpha_g \in \text{Sym}(G)$.

We now construct an injective homomorphism $f : G \rightarrow \text{Sym}(G)$. We let $f(g) = \alpha_g$. Then the kernel of this map is precisely e , since $\alpha_g(h) = h$ for all h implies that $gh = h$ for all h , giving $g = e$. It follows that f is injective. Further f is a homomorphism; since $\alpha_{gh}(x) = ghx = \alpha_g(\alpha_h(x))$, we have:

$$f(gh) = \alpha_{gh} = \alpha_g \circ \alpha_h = f(g)f(h).$$

Thus by the isomorphism theorem, we have $G \cong G/\{e\} \cong \text{Im}(f) \leq \text{Sym}(G)$. It follows that any group is isomorphic to a subgroup of some permutation group. $\square$

Cayley's theorem is one of the reasons that people say that group theory is the study of symmetry: it shows that any group (including infinite groups) can be viewed as a group of symmetries of *some* object.

7.2 Permutations of a finite set

Often, we are interested in permutations of finite sets (by Cayley's theorem, any group is a subgroup of permutations of a finite set). Any set of size n can be labelled by the numbers $\{1, 2, \dots, n\}$, giving a standard form to the permutations. Hence we make the definition:

Definition 7.4: The permutation group of the set $\{1, 2, \dots, n\}$ is called the *symmetric group*, written S_n .

Some examples of embeddings of finite groups we have seen into S_n are as follows:

- D_3 can permute any of the three vertices of a triangle arbitrarily. If we label them as 1, 2, 3 respectively, we see immediately that $D_3 \cong S_3$.
- The full tetrahedral group T_h can permute any of the vertices of a tetrahedron arbitrarily. Hence $T_h \cong S_4$. It follows that the chiral tetrahedral group T is a subgroup of S_4 .

Given that we have now seen that all finite groups are subgroups of a symmetric group S_n for some n , it makes sense to understand them a bit better. We begin by understanding multiplication of elements in S_n .

The elements of S_n can be conveniently expressed using *Cauchy's two line notation*:

Definition 7.5: *Cauchy's two line notation* for the permutation $\sigma \in S_n$ is:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \dots & \sigma(n) \end{pmatrix}.$$

This notation gives us a convenient way of multiplying permutations (note, since permutations are functions, we multiply them starting on the right and moving to the left). For example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}.$$

To work out this product, we follow where 1 is mapped by each of the permutations. The permutation on the right is applied first, and sends 1 to 1. Then 1 is mapped to 2 by the permutation on the left subsequently; overall 1 is mapped to 2. Similarly, the permutation on the right sends 2 to 3, then this is followed by the permutation on the left which sends 3 to 1; overall 2 is mapped to 1. Thus we have:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

Whilst this facilitates us taking the product of two elements in S_n , it is rather unwieldy. Our first objective therefore will be to streamline this notation into something a bit simpler - the *disjoint cycle representation*.

The disjoint cycle representation

Any permutation $\sigma \in S_n$ can be broken down into some easier to manage permutations - these are called *cycles*. The general idea is as follows. Start by asking where 1 maps under σ ; it goes to some number $\sigma(1)$. Then, ask where $\sigma(1)$ is sent under σ ; it goes to some new number $\sigma^2(1) := \sigma(\sigma(1))$. Since S_n is a group of finite order, σ itself has finite order. Thus there is a least n such that $\sigma^n(1) = 1$ again. We have wrapped around and formed a *cycle*.

Once we have done this, we can repeat the process, choosing anything that didn't lie in our original cycle. Thus the whole permutation can be thought of as cycling through subsets of elements, independently of one another.

We can formalise this whole process with a couple of definitions and propositions:

Definition 7.6: A *cycle* (of length n) is a permutation $\sigma \in S_n$ of the form:

$$\begin{pmatrix} a & \sigma(a) & \sigma^2(a) & \dots & \sigma^{n-1}(a) & \dots \\ \sigma(a) & \sigma^2(a) & \sigma^3(a) & \dots & a & \dots \end{pmatrix},$$

where everything is fixed except $a, \sigma(a), \sigma^2(a), \dots, \sigma^{n-1}(a)$ (the fixed elements are denoted by the dots). Two cycles are called *disjoint* if the sets of elements that are not fixed by the cycles are disjoint.

The cycle above is often written in *cycle notation* as $(a \ \sigma(a) \ \sigma^2(a) \ \dots \ \sigma^{n-1}(a))$; that is, we write only the elements that are permuted in the order of the cycle. We consider the representations:

$$(a \ \sigma(a) \ \sigma^2(a) \ \dots \ \sigma^{n-1}(a)), \quad (\sigma(a) \ \sigma^2(a) \ \sigma^3(a) \ \dots \ \sigma^{n-1}(a) \ a), \quad \dots$$

given by cyclic permutations of the cycle, all equivalent to one another.

For example, a cycle in S_4 which permutes $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ can be written as (123) or as (231) . Similarly a cycle in S_5 which permutes $1 \rightarrow 3 \rightarrow 5 \rightarrow 1$ can be written as (135) or as (351) (however, (531) is a different cycle, for example).

Proposition 7.7: Disjoint cycles commute.

Proof: Obvious; the cycles permute disjoint sets of elements, so can be applied in either order. $\square$

We now prove the main proposition:

Proposition 7.8: Any permutation $\sigma \in S_n$ can be written uniquely as the product of disjoint cycles.

Proof: First, consider the permutation applied to 1 repeatedly. We obtain a list:

$$1, \quad \sigma(1), \quad \sigma^2(1), \quad \dots$$

Since the set $\{1, 2, \dots, n\}$ is finite, not all elements on this list can be distinct. Hence there must exist some $i > j$ such that $\sigma^i(1) = \sigma^j(1)$. Applying the inverse permutation, we have $\sigma^{i-j}(1) = 1$. Thus there exists a positive integer such that $\sigma^n(1)$. Let n_1 be the smallest such integer. Then if we have repeated elements $\sigma^i(1), \sigma^j(1)$, it must be that $i - j \geq n$, giving $i \geq n_1 + j$. In particular, there cannot

be any repeats in the list:

$$1, \quad \sigma(1), \quad \sigma^2(1), \quad \dots, \quad \sigma^{n_1-1}(1).$$

Define now the cycle $\sigma_1 = (1 \ \sigma(1) \ \sigma^2(1) \ \sigma^3(1) \ \dots \ \sigma^{n_1-1}(1))$. Choose any element a not listed in this cycle, and repeat the procedure: consider the permutation applied repeatedly to a , generating a cycle $\sigma_2 = (a \ \sigma(a) \ \sigma^2(a) \ \sigma^3(a) \ \dots \ \sigma^{n_2-1}(a))$. If there was any overlap with this cycle and the previous cycle, we would have, for some i, j :

$$\sigma^i(1) = \sigma^j(a).$$

Multiplying on the left by σ^{-j} , we have $\sigma^{i-j}(1) = a$, giving that a is a power of the permutation σ applied to 1. But this cannot occur, since a was chosen outside of the cycle σ_1 . Thus the cycles are disjoint.

We can consider this procedure until we have exhausted all elements of $\{1, 2, \dots, n\}$; this will happen eventually because this set is finite. Thus we split σ into cycles as $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$ for some k . We have seen the cycles are disjoint, and every element in $\{1, 2, \dots, n\}$ appears in exactly one cycle mapping to the correct image under σ . $\square$

Disjoint cycle notation is extremely convenient, because it is much more compact than the two-line notation, and can be easily obtained by performing the algorithm in the proposition. For example, we have:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 1 & 2 & 9 & 7 & 4 & 5 & 6 & 8 \end{pmatrix} = (132)(4986)(57).$$

Disjoint cycle notation also makes it extremely easy to compute products of permutations. For example, imagine that we want to compute:

$$(123)(32),$$

in the group S_3 . To calculate this, we work from right to left, keeping track of where each element in the list $\{1, 2, 3\}$ is sent by the permutations. First, 1 is fixed by (32) . The next permutation sends $1 \rightarrow 2$, which gives 1 sent to 2 overall by the product of these permutations. Next, 2 is sent to 3 by the cycle (32) . Subsequently, 3 is sent to 1 by the second cycle. Hence 2 is sent to 1 overall. Similarly, 3 is sent to 2 by the cycle (32) . Subsequently, 2 is sent to 3 by the second cycle (123) . Hence 3 is sent to 3 overall. Thus the product is:

$$(123)(32) = (12).$$

Sign of a permutation

In the above section, we proved that any permutation can be represented as the product of disjoint cycles. This is an extremely convenient representation of a permutation, but there are often other convenient forms. One simple alternative is to express a permutation as a *product of transpositions* - that is, write a permutation as a product of *swaps of two elements*, which are performed sequentially.

Starting with a formal definition:

Definition 7.9: A *transposition* or *swap* is a cycle of the form (ab) , where $a \neq b$.

The most important result in this section is the following, which tells us that any permutation may be expressed as an even number of transpositions, or an odd number of transpositions, but not both:

Proposition 7.10: Any permutation $\sigma \in S_n$ can be written as a product of an even number of transpositions, or an odd number of transpositions, but not both.

Proof: We start by considering a cycle of length m ; without loss of generality, we take it to be $(12\dots m)$. First, we observe that this cycle can be expressed as a product of transpositions:

$$(12\dots m) = (1\ m)(1\ m-1)\dots(13)(12).$$

We now know that any cycle of length m can be expressed as a product of $m-1$ transpositions; through the disjoint cycle representation therefore, any permutation may be expressed as a product of transpositions. It remains to show that the *number* of transpositions is either odd or even, but not both. To do this, consider the polynomial function:

$$f(x_1, \dots, x_n) = \prod_{i < j} (x_i - x_j).$$

Note that any transposition $\tau = (ab)$, with $a < b$, swaps a, b . This swaps the factor $(x_a - x_b)$ to $(x_b - x_a) = -(x_a - x_b)$. It also swaps any factor of the form $(x_a - x_k)$ with $a < k \neq b$ to $(x_b - x_k)$; but then:

- If $k < b$, the factor $(x_k - x_b)$ swaps to $(x_k - x_a)$ simultaneously. Overall we have $(x_a - x_k)(x_k - x_b)$ swaps to $(x_b - x_k)(x_k - x_a) = (x_k - x_b)(x_a - x_k)$, so the product of the factors remains invariant.
- If $k > b$, the factor $(x_b - x_k)$ swaps to $(x_a - x_k)$. Then evidently $(x_a - x_k)(x_b - x_k)$ swaps to $(x_b - x_k)(x_a - x_k)$, so the product of the factors remains invariant.

Thus overall, the transposition changes the sign of the function:

$$f(x_{\tau(1)}, \dots, x_{\tau(n)}) = -f(x_1, \dots, x_n).$$

Thus for σ the product of k transpositions, we can repeatedly apply this result to give:

$$f(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = (-1)^k f(x_1, \dots, x_n).$$

In particular, we see that if σ can be expressed as the product of an even number of transposition and an odd number of transpositions, we could write both:

$$f(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = f(x_1, \dots, x_n) = -f(x_1, \dots, x_n).$$

That is, $f(x_1, \dots, x_n) = 0$ for all $x_1, \dots, x_n$. But this is not the case; f is not identically zero (e.g. take all the x_i distinct). It follows that σ can be expressed as an even number of transpositions, or an odd number of transpositions, but not both. $\square$

This allows us to unambiguously make the definition:

Definition 7.11: A permutation $\sigma \in S_n$ is called *even* if it can be written as an even number of transpositions. It is called *odd* if it can be written as an odd number of transpositions.

We can construct a homomorphism, called the ‘sign’ homomorphism, that forgets about all information contained in a permutation except its parity. In this way, we can show that the set of all even permutations forms a normal subgroup of the symmetric group.

Proposition 7.12: The map $\text{sgn} : S_n \rightarrow \{-1, +1\}$ given by:

$$\text{sgn}(\sigma) = \begin{cases} +1, & \text{if } \sigma \text{ is an even permutation,} \\ -1, & \text{if } \sigma \text{ is an odd permutation,} \end{cases}$$

is a group homomorphism.

Proof: Let $\sigma = \tau_1 \dots \tau_n$ and $\sigma' = \tau'_1 \dots \tau'_m$ be two permutations expressed as products of transpositions τ_i, τ'_i . Then:

$$\text{sgn}(\sigma\sigma') = \text{sgn}(\tau_1 \dots \tau_n \tau'_1 \dots \tau'_m) = (-1)^{n+m} = (-1)^n (-1)^m = \text{sgn}(\tau_1 \dots \tau_n) \text{sgn}(\tau'_1 \dots \tau'_m) = \text{sgn}(\sigma) \text{sgn}(\sigma'),$$

hence it is indeed a homomorphism. $\square$

Indeed, as an immediate consequence, we have:

Proposition 7.13: The set of even permutations in S_n forms a normal subgroup, A_n , called the *alternating group*.

Proof: The set of even permutations A_n is the kernel of sgn above, so immediately $A_n \trianglelefteq S_n$. $\square$

7.3 Conjugacy in the permutation groups

We have now spent a lot of time studying how to multiply elements of S_n quickly. We have also identified an important normal subgroup of S_n (namely A_n). As we know from earlier in the course, something else that would be extremely useful to know about S_n is its *conjugacy classes*, which gives us a sense of ‘which elements are the same as one another’.

Conjugacy in S_n

In S_n the conjugacy structure is extremely straightforward:

Proposition 7.14: Two permutations $\sigma_1, \sigma_2 \in S_n$ are conjugate if and only if they have the same *cycle type*: the list of the lengths of the constituent cycles in the disjoint cycle representations of σ_1, σ_2 .

Proof: Suppose that $\sigma_2 = \pi\sigma_1\pi^{-1}$ for some permutation $\pi \in S_n$. Express σ_1 in terms of its disjoint cycle representation as:

$$\sigma_1 = (a_1 \dots a_{n_1})(b_1 \dots b_{n_2}) \dots (c_1 \dots c_{n_k})$$

Now consider $\pi\sigma_1\pi^{-1}$ acting on $\pi(a_i)$. First π^{-1} maps $\pi(a_i)$ to a_i . Then, σ_1 maps a_i to a_{i+1} (modulo n_1 in the subscript). Finally π maps a_{i+1} to $\pi(a_{i+1})$. Similar conclusions hold for the b_i, c_i , etc. Thus the conjugate is:

$$\pi\sigma_1\pi^{-1} = (\pi(a_1) \dots \pi(a_{n_1}))(\pi(b_1) \dots \pi(b_{n_2}))(\pi(c_1) \dots \pi(c_{n_k})).$$

Thus, all conjugates of σ_1 have the same cycle type. Conversely, if a conjugate has the same cycle type, evidently can choose π such that we can map σ_1 to it. Hence, we’re done. $\square$

For example, the elements $(12)(34)$ and $(13)(24)$ are conjugate in S_4 . To conjugate between them, we want to map 2 to 3, and vice-versa, implying we should conjugate by (23) . This gives:

$$(23)(12)(34)(23)^{-1} = (23)(12)(34)(32) = (13)(24),$$

as expected.

Conjugacy in A_n

Now let us consider conjugacy in A_n . The conjugacy classes of a permutation σ in S_n, A_n are by definition given by:

$$\text{Cl}_{S_n}(\sigma) = \{\pi\sigma\pi^{-1} : \pi \in S_n\}, \quad \text{Cl}_{A_n}(\sigma) = \{\pi\sigma\pi^{-1} : \pi \in A_n\}.$$

As a result, it is obvious that $\text{Cl}_{A_n}(\sigma) \subseteq \text{Cl}_{S_n}(\sigma)$, that is the conjugacy class of an element in A_n is at most the conjugacy class of the element in S_n . However, there is no guarantee that the classes will be equal - perhaps an element that we previously could have accessed in S_n is now inaccessible in A_n .

To investigate this further, we can use the fact that the size of a conjugacy class must divide the order of a group. Recall that we have:

$$n! = |S_n| = |\text{Cl}_{S_n}(\sigma)| |C_{S_n}(\sigma)|, \quad \frac{n!}{2} = |A_n| = |\text{Cl}_{A_n}(\sigma)| |C_{A_n}(\sigma)|,$$

where $C_G(\sigma)$ is the *centraliser* of the permutation σ (the set of all permutations which fix σ under conjugation).

Dividing these relations, we see that:

$$2|\text{Cl}_{S_n}(\sigma)||C_{S_n}(\sigma)| = |\text{Cl}_{A_n}(\sigma)||C_{A_n}(\sigma)|.$$

Since $|\text{Cl}_{S_n}(\sigma)| \geq |\text{Cl}_{A_n}(\sigma)|$ and $|C_{S_n}(\sigma)| \geq |C_{A_n}(\sigma)|$, we must have that either:

- $|\text{Cl}_{A_n}(\sigma)| = \frac{1}{2}|\text{Cl}_{S_n}(\sigma)|$ and $|C_{A_n}(\sigma)| = |C_{S_n}(\sigma)|$;
- $|\text{Cl}_{A_n}(\sigma)| = |\text{Cl}_{S_n}(\sigma)|$ and $\frac{1}{2}|C_{A_n}(\sigma)| = |C_{S_n}(\sigma)|$.

Therefore, we have:

Proposition 7.15: The size of the conjugacy class of an element $\sigma \in A_n$ is either equal to the size of the conjugacy class of the element $\sigma \in S_n$, or half the size of the conjugacy class of the element $\sigma \in S_n$. In the latter case, we say that the conjugacy class *splits* inside A_n .

Proof: By the counting argument above. $\square$

A convenient condition for splitting is given as follows:

Proposition 7.16: The conjugacy class of $\sigma \in A_n$ splits if and only if no odd permutation commutes with σ .

Proof: The conjugacy class splits if and only if the centraliser does not split; this occurs if and only if $C_{S_n}(\sigma) = C_{A_n}(\sigma)$. Equivalently, $C_{S_n}(\sigma)$ contains no odd permutations. $\square$

As a detailed example, consider A_4 . We consider the possible cycle types:

- The conjugacy class corresponding to the cycle type $(1, 1, 1, 1)$ in S_n only contains the identity e . But clearly all odd permutations commute with this, so the class does not split in A_n . It is given by $\{e\}$.
- The conjugacy class corresponding to the cycle type $(2, 2)$ in S_n contains, for example, $(12)(34)$. This commutes with (12) , since $(12)(12)(34) = (12)(34)(12)$. Consequently, the class does not split in A_n . It is given by $\{(12)(34), (13)(24), (14)(23)\}$.
- The conjugacy class corresponding to the cycle type $(3, 1)$ in S_n contains, for example, (123) . There are 8 elements of this form (there are 4 choices for the first element of the cycle, then 3 choices for the second element of the cycle, and finally 2 choices for the third element of the cycle, but then all three cycles of this cycle are equivalent, giving 8 in total). Consequently, we have $24 = 8 \cdot |C_{S_4}((123))|$. But then the centraliser is of odd order so cannot be split, hence the conjugacy class itself must split into two conjugacy classes of size 4 and 4 respectively. We have:

$$\text{Cl}_{A_4}((123)) = \{(123), (134), (142), (243)\}, \quad \text{Cl}_{A_4}((132)) = \{(132), (143), (124), (234)\}.$$

Here, we can imagine (123) conjugating to (134) under 2 mapping to 3, and 3 mapping to 4, i.e. the cycle (234) . This is an even permutation, so this is permitted.

Part II

Representation theory

Chapter 8

Permutation representations, or group actions

Using Cayley's theorem, we have seen how every group can be identified with a subgroup of a permutation group. However, the theorem said nothing about *uniqueness* - there may be many ways in which a group can be embedded inside a permutation group, and indeed some choices may be more convenient than others. Any given choice is called a *permutation representation* of the group, or a *group action* to some.

In this chapter we shall study the general properties of permutation representations. We shall introduce the notion of the *stabiliser* and *orbit* of an element in the set being permuted, and prove the *orbit-stabiliser theorem*, an analogue of Lagrange's theorem in this context. We shall see some applications of this result.

All of this will set the stage for our remaining topic: instead of using *permutations* to represent groups, which can be difficult to calculate with, can we use something else instead? The answer is *yes*: we can use tools from linear algebra, a subject so simple that computers can do it. We shall study the theory of *linear representations*, representing group elements as matrices, throughout the remainder of the course.

8.1 Permutation representations: basic definitions and examples

We begin by making the expected definition, and give various examples (most of which will likely be familiar by now).

Definition 8.1: A *permutation representation* or *group action* of the group G is a homomorphism $f : G \rightarrow \text{Sym}(X)$, where X is some set.

A permutation representation $f : G \rightarrow \text{Sym}(X)$ is called *faithful* if it is *injective*: that is, for all $g, h \in G$ with $g \neq h$, we have $f(g) \neq f(h)$. Equivalently, as you know from Examples Sheet 6, the kernel of f is trivial. In this case, the isomorphism theorem gives $G \cong G/\{e\} \cong \text{Im}(f) \leq \text{Sym}(X)$ so that G embeds as a subgroup of $\text{Sym}(X)$.

Basic examples include the following:

- The group D_3 acts on the vertices of a triangle, say $T = \{1, 2, 3\}$, by permuting them, producing a natural permutation representation $f : D_3 \rightarrow \text{Sym}(T)$. Similarly the group D_3 acts on the edges of the triangle, producing a natural permutation representation. Both of these examples are faithful.
- The chiral tetrahedral group T acts on the vertices of a tetrahedron $T = \{1, 2, 3, 4\}$ by permuting them, producing a natural faithful permutation representation $f : D_3 \rightarrow \text{Sym}(T)$. It could also be thought of as acting on the edges, or on the faces.
- In the proof of Cayley's theorem, we used a map $f : G \rightarrow \text{Sym}(G)$, where $f(g) = \alpha_g$ with $\alpha_g : G \rightarrow G$ given by $\alpha_g(h) = gh$. This is called the *left regular action of the group on itself*. Since the kernel is trivial (with $\alpha_g(h) = gh = h$ if and only if $g = e$), we see that this is a faithful representation.
- Similarly, we could define a map $f : G \rightarrow \text{Sym}(G)$ where $f(g) = \beta_g$, with $\beta_g : G \rightarrow G$ given by $\beta_g(h) = hg^{-1}$. This is called the *right regular action of the group on itself*.

To see that this is a valid permutation representation, note that β_g is invertible with inverse $\beta_{g^{-1}}$, so that β_g is indeed a permutation of G . Next, we have $\beta_g(\beta_{g'}(h)) = \beta_g(h(g')^{-1}) = h(g')^{-1}g^{-1} = h(gg')^{-1} = \beta_{gg'}(h)$. So we have:

$$f(gg') = \beta_{gg'} = \beta_g \circ \beta_{g'} = f(g)f(g'),$$

and indeed f is a homomorphism as required.

Similarly, this is a faithful representation, since the kernel is trivial.

- On Examples Sheet 7, you explored the *left regular action on left cosets*. If $H \leq G$ is a subgroup of a group G , we write G/H for the set of left cosets of H in G (note this is *not* a quotient, just convenient notation that mirrors quotient groups). We define $f : G \rightarrow \text{Sym}(G/H)$ by $f(g) = \alpha_g$ where $\alpha_g : G/H \rightarrow G/H$ merely does left multiplication:

$$\alpha_g(g'H) = (gg')H.$$

Clearly α_g is invertible, with inverse $\alpha_{g^{-1}}$ so is a permutation. It is also clear that the homomorphism property for f holds, hence we have a permutation representation on the left cosets of a subgroup.

8.2 Orbits, stabilisers, and the orbit-stabiliser theorem

Similar to the definitions of kernel and image for a homomorphism, which were very useful ideas, we define the ‘set of all elements which fix something’ and the ‘set of all places something can go to’ for a permutation representation:

Definition 8.2: Let $f : G \rightarrow \text{Sym}(X)$ be a permutation representation of a group G . We define the *orbit* of an element $x \in X$ to be the set of elements accessible from x via the permutations in the image of f :

$$\text{Orb}(x) = \{\sigma(x) : \sigma \in \text{Im}(f)\}.$$

We define the *stabiliser* of an element $x \in X$ to be the set of elements in G whose image permutations fix x :

$$\text{Stab}(x) = \{g \in G : \sigma(g)(x) = x\}.$$

Examples include the following:

- For the action of D_3 on the vertices of a triangle $\{1, 2, 3\}$, the orbit of 1 is all of the vertices, $\text{Orb}(1) = \{1, 2, 3\}$. Meanwhile the stabiliser is the set of elements of D_3 that leave 1 untouched, namely just the identity and the vertical reflection (say m), giving $\text{Stab}(1) = \{e, m\}$.
- For the action of the chiral tetrahedral group T on the vertices of a tetrahedron $\{1, 2, 3, 4\}$, the orbit of 1 is all of the vertices again, $\text{Orb}(1) = \{1, 2, 3, 4\}$. Meanwhile, the stabiliser is the set of elements which leave the vertex 1 untouched, namely the identity, and the rotations by $2\pi/3, 4\pi/3$ in the axis through the vertex 1 and the centre of the opposite face.
- For the left regular action $f : G \rightarrow \text{Sym}(G)$, $f(g) = \alpha_g$ with $\alpha_g(h) = gh$, the stabiliser of the element h is given by all g such that $\alpha_g(h) = gh = h$; this occurs if and only if $g = e$, giving a trivial stabiliser in each case. The orbit of the element h is the entire group, since the equation $gh = x$ for given h, x can always be solved to give $g = xh^{-1}$.

The orbits and stabilisers always satisfy the following useful properties, again similar to the kernel and image of a homomorphism:

Proposition 8.3: Let $f : G \rightarrow \text{Sym}(X)$ be a permutation representation. Then:

- (i) the orbits of the elements partition $\text{Sym}(X)$; that is, every element belongs to exactly one orbit;
- (ii) the stabiliser $\text{Stab}(x)$ of any element $x \in X$ is a subgroup of G , $\text{Stab}(x) \leq G$.

Proof: We have the following:

- (i) Obviously every element x belongs to its own orbit, $x \in \text{Orb}(x)$, since $f(e)(x) = \text{id}(x) = x$. Now suppose that $z \in \text{Orb}(x) \cap \text{Orb}(y)$ is in the intersection of two orbits. Let $x' \in \text{Orb}(x)$. Then we have some $g' \in G$ such that $f(g')(x) = x'$. But there is also some element $g_x \in G$ such that $f(g_x)(x) = z$, and some element $g_y \in G$ such that $f(g_y)(y) = z$. This implies that:

$$f(g')f(g_x)^{-1}f(g_y)(y) = f(g')f(g_x)^{-1}(z) = f(g')x = x',$$

so that $x' \in \text{Orb}(y)$. Thus $\text{Orb}(x) \subseteq \text{Orb}(y)$. Similarly, $\text{Orb}(y) \subseteq \text{Orb}(x)$, so the orbits are in fact equal.

(ii) We simply check the group axioms:

- (G1) *Closure*. If $g, h \in \text{Stab}(x)$, then $f(gh)(x) = f(g)(f(h)(x)) = f(g)(x) = x$, so $gh \in \text{Stab}(x)$.
- (G2) *Associativity*. Inherited from the group G .
- (G3) *Identity*. Since f is a homomorphism, $f(e) = \text{id}$ fixes any element x , so $e \in \text{Stab}(x)$.
- (G4) *Inverses*. Since f is a homomorphism, if $f(g)(x) = x$, we have $x = f(g)^{-1}(x) = f(g^{-1})(x)$ giving $g^{-1} \in \text{Stab}(x)$.

This has the following useful consequence, which we have seen a couple of times in the course in different guises:

Theorem 8.4: (The orbit-stabiliser theorem) Suppose that $f : G \rightarrow \text{Sym}(X)$ is a permutation representation of a finite group G . Then for any element $x \in X$, we have:

$$|G| = |\text{Stab}(x)| |\text{Orb}(x)|.$$

Proof: This should remind you of our conjugacy class proof - the idea there was to take all the elements which fix an element under conjugation, and form them into a subgroup called the *centraliser*, which here has been upgraded to the more general *stabiliser*. Similarly, we then said that the cosets of the centraliser were in one-to-one correspondence with the possible conjugates; here, we shall aim to show that the elements of the orbit are in one-to-one correspondence with the cosets of the stabiliser.

With the idea in mind, we now begin the proof. Certainly $|\text{Stab}(x)|$ divides $|G|$ by Lagrange's theorem. Now define the map $p : G/\text{Stab}(x) \rightarrow \text{Orb}(x)$ by, where $G/\text{Stab}(x)$ is the set of left cosets of $\text{Stab}(x)$ inside G , via:

$$p(g\text{Stab}(x)) = f(g)(x).$$

Certainly this has the correct domain and codomain, since $f(g)(x) \in \text{Orb}(x)$ by definition. We further claim that this map is well-defined and invertible:

- To prove the map is well-defined, we let $g\text{Stab}(x) = g'\text{Stab}(x)$, so that the coset has two representatives g, g' . Then we can write $g' = gk$ for some element $k \in \text{Stab}(x)$ which stabilises the element x . This gives:

$$p(g'\text{Stab}(x)) = f(g')(x) = f(gk)(x) = f(g)(f(k)(x)) = f(g)(x) = p(g\text{Stab}(x)),$$

thus indeed the map is well-defined.

- The map is also invertible, since we can define an inverse via $p^{-1} : \text{Orb}(x) \rightarrow G/\text{Stab}(x)$. This is given by $p^{-1}(f(g)(x)) = g\text{Stab}(x)$. To show this is well-defined, we must show that if $f(g)(x) = f(g')(x)$ (i.e. there are two ways of permuting x to the same element), we land back in the same coset. In this case, we have $f(g^{-1}g')(x) = x$ using the homomorphism property of f , which gives $g^{-1}g' = k \in \text{Stab}(x)$, and hence $g' = gk$. Thus $g'\text{Stab}(x) = g\text{Stab}(x)$, as required.

Hence there is a bijection between the left cosets of $\text{Stab}(x)$ and $\text{Orb}(x)$, proving the theorem. $\square$

Chapter 9

Linear representations: definitions, examples, and equivalence

We have now seen that any group can be embedded inside a permutation group, through Cayley's theorem. This was a particularly fruitful thing to do, because it meant that calculations in groups could be converted into calculations involving permutations, which are straightforward.

Can we do any better? One thing we might hope to do is to convert group theory into the theory of *linear algebra*, every mathematician's favourite theory, because of its extreme simplicity. Indeed, linear algebra is so simple that even computers can do it! This might give a way, for example, to represent a group on a computer.

This is the starting point of the subject of *representation theory*, which takes all the elements of a group and maps them to *matrices*. The possible matrix products correspond to the possible products of the group elements.

In this introductory chapter to the subject, we do the following:

- We define a (*linear group*) *representation*, and what it means to be a *faithful group representation*.
- We give many examples of representations, including representations of the additive group $\mathbb{Z}$, one-dimensional examples of representations of the cyclic and dihedral groups, two-dimensional examples of representations of the cyclic and dihedral groups, and finally an example of a two-dimensional representation of the quaternion group Q_8 .
- We show that any group can be represented faithfully through the *left regular representation* (inspired by Cauchy's theorem), and we also show that the representation of a group in terms of matrices is not necessarily unique.
- We introduce the notion of *equivalence of representations*, which allows us to identify when two representations are really the same as one another (similar to isomorphism of groups). We give examples of equivalent representations.
- Finally, we ask when two representations are inequivalent. Similar to group isomorphisms, it is quite difficult to tell when two representations are inequivalent, because there are so many possible ways in which they can be equivalent - we introduce the *character* of a representation as a useful test for telling representations apart.

9.1 Definition of a representation

We start with the definition of a representation, motivated by the introduction to this chapter:

Definition 9.1: An n -dimensional complex linear representation of a group G is a homomorphism from G to the complex general linear group, $D : G \rightarrow GL(n, \mathbb{C})$.

A representation assigns to each element of the group $g \in G$ a corresponding matrix $D(g)$.

We can similarly define *real representations*, but given computers can handle complex numbers, and our motivation for representation theory is to simulate groups on computers, we shall ignore this case.¹⁷

Our definition of a representation is via a homomorphism. We know already that this gives us the opportunity to ‘reduce a group to the parts we care about’. However, often we want to simulate a group directly, suggesting the definition:

Definition 9.2: A representation $D : G \rightarrow GL(n, \mathbb{C})$ of a group G is called *faithful* if D is an *injective* map; that is, for all $g_1, g_2 \in G$, we have $D(g_1) \neq D(g_2)$.

In this case, the group G is isomorphic to a subgroup of $GL(n, \mathbb{C})$, by the isomorphism theorem. This is because for an injective homomorphism, the kernel is trivial, so $G \cong G/\{e\} \cong \text{Im}(D) \leq GL(n, \mathbb{C})$.

9.2 Examples of representations

Armed with the definition of a representation we can give lots of examples.

- **The trivial representation.** Every group G has a *trivial representation of dimension n* , for each n , given by $D : G \rightarrow GL(n, \mathbb{C})$ with:

$$D(g) = I,$$

where I is the identity matrix.

- **Representation of $\mathbb{Z}$.** Consider the additive group of integers $\mathbb{Z}$. The map $D : \mathbb{Z} \rightarrow \mathbb{C}^*$ (note that $GL(1, \mathbb{C}) = \mathbb{C}^*$, since the set of invertible 1×1 complex matrices is just the set of non-zero complex numbers) given by:

$$D(n) = 2^n$$

is a homomorphism. This is because:

$$D(n + m) = 2^{n+m} = 2^n 2^m = D(n)D(m).$$

This is faithful since the map is injective. Equally, we could have picked $D(n) = 3^n$, etc, so we see immediately that the one-dimensional faithful representations of $\mathbb{Z}$ are not unique.

¹⁷Indeed, the theory of real representations is actually quite subtle. Groups that have certain complex representations may not have corresponding real representations. This would be discussed in detail in a more comprehensive course on representation theory.

We can also construct two-dimensional faithful representations of $\mathbb{Z}$. For example $D : \mathbb{Z} \rightarrow GL(2, \mathbb{C})$ given by:

$$D(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

is a homomorphism, since:

$$D(n)D(m) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & n+m \\ 0 & 1 \end{pmatrix} = D(n+m).$$

- **One-dimensional representations of cyclic groups.** Every cyclic group $C_n = \{e, r, r^2, \dots, r^{n-1}\}$ has a faithful 1-dimensional representation $D : C_n \rightarrow \mathbb{C}^*$ given by:

$$D(r^k) = e^{2\pi i k/n}.$$

This is a homomorphism because:

$$D(r^k r^l) = D(r^{k+l}) = e^{2\pi i (k+l)/n} = e^{2\pi i k/n} e^{2\pi i l/n} = D(r^k) D(r^l).$$

Further, all $D(r^k)$ are distinct as k varies, since we are dealing with the n th roots of unity.

We can actually easily classify all the one-dimensional representations of C_n :

Proposition 9.3: The one-dimensional representations $D : C_n \rightarrow \mathbb{C}^*$ of C_n are generated by the choice $D(r) = e^{2\pi i k/n}$, for each integer $0 \leq k < n$. Therefore there are n distinct one-dimensional representations of C_n .

Proof: Since we require a homomorphism $D : C_n \rightarrow \mathbb{C}^*$, it is sufficient to specify where the generator r is mapped to, $D(r)$. The only relation that needs to be satisfied is $D(r)^n = 1$. Therefore there are n choices for $D(r)$, given by the n th roots of unity. $\square$

We can also come up with two-dimensional representations of cyclic groups; before doing so, though, we shall recap an important fact about matrices.

A fact about matrices. In general, a matrix A can be viewed as a transformation of vectors, $\mathbf{v} \mapsto A\mathbf{v}$. For example:

$$\begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

One very useful property of matrices that you may or may not have seen before is the *columns of a matrix represent the images of the original coordinate basis vectors under that transformation*. Indeed we see this exactly in the example above.

This is a completely general property. If we let $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, \dots, 0)$, ..., $\mathbf{e}_n = (0, \dots, 0, 1)$, where $\mathbf{e}_i$ is the vector which is zero except at the i th entry, which is a one, we have the columns of the matrix A are precisely:

$$A = (A\mathbf{e}_1 \quad A\mathbf{e}_2 \quad \dots \quad A\mathbf{e}_n).$$

This can help us produce transformation matrices simply. For example, to construct the two-dimensional matrix that represents a reflection in the y -axis, we consider what happens to the coordinate basis vectors $\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1)$. The vector $\mathbf{e}_1 = (1, 0)$ is mapped to $(-1, 0)$, whilst the vector $\mathbf{e}_2 = (0, 1)$ remains invariant. Hence we have:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Similarly, if we want a matrix that gives a $\pi/2$ rotation, we can think about where the coordinate basis vectors are mapped to. The basis vector $\mathbf{e}_1 = (1, 0)$ is mapped to $(0, 1)$, and the basis vector $\mathbf{e}_2 = (0, 1)$ is mapped to $(-1, 0)$. Consequently, the matrix that effects this transformation is given by:

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- **Two-dimensional representations of cyclic groups.** The cyclic groups also have faithful 2-dimensional representations $D : G \rightarrow GL(2, \mathbb{C})$, which are inspired by rotation matrices:

$$D(r^k) = \begin{pmatrix} \cos(2\pi k/n) & -\sin(2\pi k/n) \\ \sin(2\pi k/n) & \cos(2\pi k/n) \end{pmatrix}.$$

This is motivated by the above property of matrices: if we rotate the vector $(1, 0)$ by an angle θ anticlockwise about the origin it is mapped to a vector $(\cos(\theta), \sin(\theta))$. Meanwhile, if we rotate the vector $(0, 1)$ by an angle θ anticlockwise about the origin it is mapped to a vector $(-\sin(\theta), \cos(\theta))$. This is shown in Fig. 29.

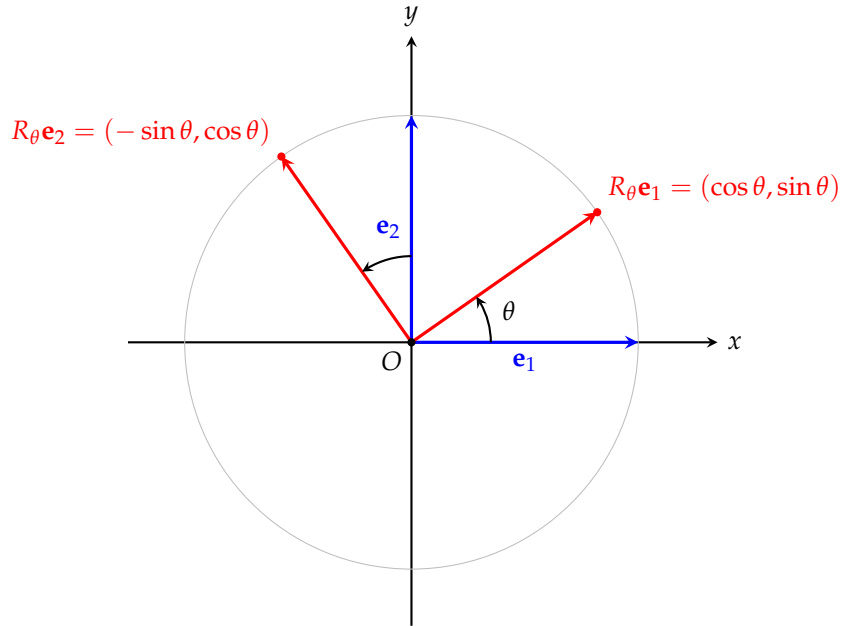


Figure 29: A rotation anticlockwise by an angle θ about the origin can be represented by a matrix. To figure out the columns of the matrix, we consider the action on the coordinate basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$, which are mapped respectively to $(\cos(\theta), \sin(\theta))$ and $(-\sin(\theta), \cos(\theta))$.

Checking the homomorphism property, we have:

$$D(r^k r^l) = D(r^{k+l}) = \begin{pmatrix} \cos(2\pi(k+l)/n) & -\sin(2\pi(k+l)/n) \\ \sin(2\pi(k+l)/n) & \cos(2\pi(k+l)/n) \end{pmatrix},$$

whilst also:

$$\begin{aligned} D(r^k)D(r^l) &= \begin{pmatrix} \cos(2\pi k/n) & -\sin(2\pi k/n) \\ \sin(2\pi k/n) & \cos(2\pi k/n) \end{pmatrix} \begin{pmatrix} \cos(2\pi l/n) & -\sin(2\pi l/n) \\ \sin(2\pi l/n) & \cos(2\pi l/n) \end{pmatrix} \\ &= \begin{pmatrix} \cos(2\pi k/n)\cos(2\pi l/n) - \sin(2\pi k/n)\sin(2\pi l/n) & -\cos(2\pi k/n)\sin(2\pi l/n) - \cos(2\pi l/n)\sin(2\pi k/n) \\ \sin(2\pi k/n)\cos(2\pi l/n) + \sin(2\pi l/n)\cos(2\pi k/n) & -\sin(2\pi k/n)\sin(2\pi l/n) + \cos(2\pi k/n)\cos(2\pi l/n) \end{pmatrix}, \end{aligned}$$

which is equal to $D(r^{k+l})$ by standard compound angle identities.

- **One-dimensional representations of dihedral groups.** An easy example of a one-dimensional representation $D : D_n \rightarrow \mathbb{C}^*$ of D_n simply sends all the rotations to $+1$ and all the reflections to -1 , that is:

$$D(r) = 1, \quad D(m) = -1.$$

This extends to a homomorphism since $D(r)^n = 1$, $D(m)^2 = 1$ and $D(r)D(m) = D(m)D(r)^{-1}$, so all the relations are satisfied.

- **Two-dimensional representations of dihedral groups.** As a two-dimensional example, consider defining $D : D_n \rightarrow GL(2, \mathbb{C})$ by specifying its action on the generators:

$$D(r) = \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}, \quad D(m) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

To understand why we have picked these transformation matrices, we note that the rotation representative $D(r)$ corresponds to sending $(1, 0)$ to $(\cos(2\pi/n), \sin(2\pi/n))$ and $(0, 1)$ to $(-\sin(2\pi/n), \cos(2\pi/n))$. Meanwhile, the reflection representative $D(m)$ corresponds to a reflection in the x -axis, sending $(1, 0)$ to $(1, 0)$ and $(0, 1)$ to $(0, -1)$.

This extends uniquely to a homomorphism, because all the relations are satisfied:

$$D(r)^n = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D(m)^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and finally:

$$\begin{aligned} D(m)D(r) &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \\ &= \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ -\sin(2\pi/n) & -\cos(2\pi/n) \end{pmatrix} \\ &= \begin{pmatrix} \cos(2\pi/n) & \sin(2\pi/n) \\ -\sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= D(r)^{-1}D(m). \end{aligned}$$

- **Representations of the quaternion group Q_8 .** Earlier in the course, we described the quaternion group in terms generators $-1, i, j, k$, but in fact this generating set can be reduced. A minimal generating set for Q_8 is i, j , satisfying the relations:

$$i^4 = 1, \quad i^2 = j^2, \quad ij = ji^{-1}.$$

This recovers the usual algebra of quaternions as one can readily check. The idea is to define $-1 = i^2 = j^2$ and $k = ij$. Then the third relation implies $k^2 = (ij)^2 = ijij = ji^{-1}ij = j^2 = -1$. The third relation also gives $ijk = ijij = ji^{-1}ij = j^2 = -1$, which are the usual relations for the quaternions in terms of the four generators $-1, i, j, k$.

Therefore, to get a faithful representation, we seek a two-dimensional matrix whose fourth power is the identity, but whose second power is not the identity. The following choice works:

$$D(i) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad D(j) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},$$

which we can check as follows:

$$D(i)^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(i)^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and:

$$D(j)^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = D(i)^2.$$

Finally, we have:

$$D(i)D(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$$

and:

$$D(j)D(i)^{-1} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}.$$

We have now seen many examples of representations of familiar groups. Is it always possible to construct a faithful representation of any group? Cayley's theorem guarantees that this will be the case.

Proposition 9.4: Every finite group G of size $|G|$ has a faithful representation of dimension $|G|$.

Proof: Consider the left regular representation $f : G \rightarrow \text{Sym}(G)$ given by $f(g) = \alpha_g$, where $\alpha_g : G \rightarrow G$ is the permutation $\alpha_g(h) = gh$.

Now let $n = |G|$, and introduce an arbitrary correspondence between the coordinate basis vectors of $\mathbb{C}^n$, $\mathbf{e}_1, \dots, \mathbf{e}_n$ and the elements of G , $g_1, \dots, g_n$ (it doesn't matter how we do the correspondence). Then every permutation α_g can be thought of instead as a permutation of the coordinate basis vectors. This suggests that any permutation α_g can be represented as a *permutation matrix* P_g of size $n \times n$, which is a matrix transforming the coordinate basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ to the corresponding permuted coordinate basis vectors.

Certainly such permutation matrices will compose in the correct way, so that $f : G \rightarrow GL(|G|, \mathbb{C})$ given by mapping $f(g) = P_g$ is a representation. Further, it is a faithful representation, because all the permutations are distinct. $\square$

Definition 9.5: The representation constructed in the above proposition is called the *left regular representation* of a group G .

As an example, consider $D_2 = \{e, r, m, rm\}$. Introduce the correspondence $e \leftrightarrow \mathbf{e}_1$, $r \leftrightarrow \mathbf{e}_2$, $m \leftrightarrow \mathbf{e}_3$, and $rm \leftrightarrow \mathbf{e}_4$. Then:

- The map $\alpha_e(g) = g$ just sends every group element to itself. Thus the permutation sends every coordinate basis vector to itself, giving the permutation matrix:

$$P_e = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- The map α_r acts as $\alpha_r(e) = r$, $\alpha_r(r) = e$, $\alpha_r(m) = rm$ and $\alpha_r(rm) = m$. This gives the permutation $\mathbf{e}_1 \rightarrow \mathbf{e}_2$, $\mathbf{e}_2 \rightarrow \mathbf{e}_1$, $\mathbf{e}_3 \rightarrow \mathbf{e}_4$ and $\mathbf{e}_4 \rightarrow \mathbf{e}_3$. Thus the corresponding permutation matrix is:

$$P_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

- The map α_m acts as $\alpha_m(e) = m$, $\alpha_m(r) = rm$, $\alpha_m(m) = e$, and $\alpha_m(rm) = r$. This gives the permutation $\mathbf{e}_1 \rightarrow \mathbf{e}_3$, $\mathbf{e}_2 \rightarrow \mathbf{e}_4$, $\mathbf{e}_3 \rightarrow \mathbf{e}_1$ and $\mathbf{e}_4 \rightarrow \mathbf{e}_2$. Thus the corresponding permutation matrix is:

$$P_m = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

- Finally, the map α_{rm} acts as $\alpha_{rm}(e) = rm$, $\alpha_{rm}(r) = m$, $\alpha_{rm}(m) = r$, and $\alpha_{rm}(rm) = e$. This gives the permutation $\mathbf{e}_1 \rightarrow \mathbf{e}_4$, $\mathbf{e}_2 \rightarrow \mathbf{e}_3$, $\mathbf{e}_3 \rightarrow \mathbf{e}_2$ and $\mathbf{e}_4 \rightarrow \mathbf{e}_1$. Thus the corresponding permutation matrix is:

$$P_{rm} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

9.3 Equivalence of representations

Just as two groups may be isomorphic, it may be the case that two representations are really the ‘same’. Motivated by the idea of conjugacy from earlier on in the course, a family of representing matrices $f_1(g)$ may be the same as another family $f_2(g)$, just ‘viewed from a different perspective’, suggesting the definition:

Definition 9.6: Let $f_1, f_2 : G \rightarrow GL(n, \mathbb{C})$ be two n -dimensional representations of a group G . We say that f_1, f_2 are *equivalent representations* if there exists some $S \in GL(n, \mathbb{C})$ such that:

$$f_1(g) = S^{-1}f_2(g)S$$

for all $g \in G$. The matrix S is called an *intertwiner* of the two representations.

This definition is telling us that S is some transformation of an ‘observer’. What does this mean in the context of vectors and matrices? The idea is that S is transforming our standard coordinate basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ into some other set of coordinate basis vectors $\mathbf{f}_1, \dots, \mathbf{f}_n$. We can therefore view the transformation as:

- $f_1(g)$ simply transforms the standard coordinate basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$;
- $S^{-1}f_2(g)S$ performs the following sequence of transforms. First, S maps the coordinate basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ to a new set of coordinate basis vectors $\mathbf{f}_1, \dots, \mathbf{f}_n$. We then apply $f_2(g)$, which is the ‘transformation $f_1(g)$, but written in the new coordinates’. Finally, we map the coordinates $\mathbf{f}_1, \dots, \mathbf{f}_n$ back to $\mathbf{e}_1, \dots, \mathbf{e}_n$ by S^{-1} .

More concretely, consider the following example:

- Suppose that we have the representations f_1, f_2 of C_4 , given by:

$$f_1(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f_1(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad f_1(r^2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f_1(r^3) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and also:

$$f_2(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f_2(r) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad f_2(r^2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f_2(r^3) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The first set of matrices could be viewed as being defined relative to some coordinates $\{\mathbf{e}_1 = (1, 0), \mathbf{e}_2 = (0, 1)\}$. Then, it consists of $\pi/2, \pi, 3\pi/2$ rotations about the origin anticlockwise in these coordinates.

The second set of matrices appear to be *clockwise* rotations about the origin. But if we were to define our coordinates as $\{\mathbf{f}_1 = (1, 0), \mathbf{f}_2 = (0, -1)\}$ instead, then this would change: the rotations would appear to be *anticlockwise* in these new coordinates. Thus these must transform the representations into one another. We take:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = S.$$

For the diagonal matrices, it is obvious that $f_1(e) = S^{-1}f_2(e)S$ and $f_1(r^2) = S^{-1}f_2(r^2)S$. On the other hand, for the non-diagonal matrices we can check that:

$$Sf_2(r)S^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = f_1(r).$$

Similarly, we can check that $Sf_2(r^3)S^{-1} = f_1(r^3)$.

As a less concrete, but extremely useful example, consider:

- Two one-dimensional representations $f_1, f_2 : G \rightarrow \mathbb{C}^*$ are equivalent if and only if there exists some $s \in \mathbb{C}^*$ such that $f_1(g) = s^{-1}f_2(g)s = f_2(g)$ for all $g \in G$. Therefore, if f_1, f_2 are equivalent if and only if they are the same map, $f_1 = f_2$.

In practice, it can be quite laborious to check the equivalence of a representation with another representation. We can make this slightly quicker by considering generators and relations.

Proposition 9.7: Suppose that G is generated by $\{g_1, \dots, g_n\}$, and let $f_1 : G \rightarrow GL(n, \mathbb{C})$, $f_2 : G \rightarrow GL(n, \mathbb{C})$ be two representations. Then f_1, f_2 are equivalent if and only if there exists a matrix S such that:

$$f_1(g) = S^{-1}f_2(g)S$$

for all $g \in \{g_1, \dots, g_n\}$.

Proof: Any $g \in G$ can be written in the form $g = x_1^{\alpha_1} \dots x_k^{\alpha_k}$ where $x_i \in \{g_1, \dots, g_n\}$ and $\alpha_i \in \mathbb{Z}$. Hence we have:

$$S^{-1}f_2(g)S = S^{-1}f_2(x_1^{\alpha_1} \dots x_k^{\alpha_k})S = S^{-1}f_2(x_1)^{\alpha_1} \dots f_2(x_k)^{\alpha_k}S.$$

Inserting $S^{-1}S$ freely, we see that this is equivalent to:

$$(S^{-1}f_2(x_1)S)^{\alpha_1} \dots (S^{-1}f_2(x_k)S)^{\alpha_k} = f_1(x_1)^{\alpha_1} \dots f_1(x_k)^{\alpha_k} = f_1(g),$$

so we have $f_1(g) = S^{-1}f_2(g)S$ for all $g \in G$, if it holds on the generating set. $\square$

For example, this makes equivalent representations of the dihedral group much easier to check. We have:

9.4 Inequivalence of representations: a first look at characters

Much like group theory, it is easy to tell when two representations f_1, f_2 are equivalent: we just find a matrix S which conjugates all the matrices $f_1(g)$ to the matrices $f_2(g)$ simultaneously. On the other hand, proving two representations are *inequivalent* is a difficult task: we must show that there *is no such matrix*, which is far trickier given the extremely large number of possibilities.

A common way of tackling this problem is to introduce the idea of *invariants* of representations - things that stay the same between equivalence representations. This is very similar to how we tackled non-isomorphism of groups; our invariants were things like the group being Abelian (a property preserved by isomorphism), or the order of the elements of the group, or the sizes of the conjugacy classes.

Unlike groups, there is just one important invariant of a representation, and it is extremely easy to calculate: it is called the *character* of the representation. In order to define it, we first define a property of matrices called the *trace*:

Definition 9.8: Let A be an arbitrary $n \times n$ complex matrix. We define the *trace* of A to be the sum of the diagonal entries of A . For a 1×1 matrix, the trace is just equal to the entry of the matrix itself. The trace is written as $\text{Tr}(A)$.

For example, we have:

$$\text{Tr} \left(\begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix} \right) = 1 + (-1) = 0.$$

Having defined trace, we can define the *character*:

Definition 9.9: Let $f : G \rightarrow GL(n, \mathbb{C})$ be a representation of the group G . We define the *character* of G to be the map $\chi_f : G \rightarrow \mathbb{C}$ given by:

$$\chi_f(g) = \text{Tr}(f(g)).$$

For example, consider the representation of the cyclic group C_4 given by $f : G \rightarrow GL(2, \mathbb{C})$ where:

$$f(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad f(r^2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad f(r^3) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The character of this representation is given by:

$$\chi_f(e) = 2, \quad \chi_f(r) = 0, \quad \chi_f(r^2) = -2, \quad \chi_f(r^3) = 0.$$

As promised, the character is an invariant. In order to prove this, we need one result about traces:

Lemma 9.10: For any two matrices, we have $\text{Tr}(AB) = \text{Tr}(BA)$. This result is called *cyclicity of the trace*.

Proof: Best done with some index or suffix notation; see a linear algebra course. The 2×2 case is given on the Examples Sheet (and asks you to extend to the $n \times n$ case, if you are able). $\square$

Proposition 9.11: Let $f_1, f_2 : G \rightarrow GL(n, \mathbb{C})$ be equivalent representations. Then the characters satisfy $\chi_{f_1} = \chi_{f_2}$.

Proof: Since f_1, f_2 are equivalent, there exists a matrix S such that $f_1(g) = Sf_2(g)S^{-1}$. Thus for any $g \in G$, we have:

$$\chi_{f_1}(g) = \text{Tr}(f_1(g)) = \text{Tr}(Sf_2(g)S^{-1}) = \text{Tr}(f_2(g)S^{-1}S) = \text{Tr}(f_2(g)) = \chi_{f_2}(g).$$

Hence, we're done. $\square$

This gives an easy test for inequivalence of representations: we evaluate their characters, and if they are not equal, the representations are inequivalent. For example, consider the trivial two-dimensional representation of C_4 :

$$f(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f(r) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f(r^2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f(r^3) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The character of this representation is $\chi_f(g) = 2$ for all g . This is clearly unequal to the character of the two-dimensional representation we found above, so they must be inequivalent representations.

Quite remarkably, the converse of the above proposition is also true for a finite group. That is, if the characters of two representations f_1, f_2 of a finite group G are equal $\chi_{f_1} = \chi_{f_2}$, then they are equivalent representations! We shall take steps towards proving this for much of the remainder of the course.

Chapter 10

Reducibility to direct sums

On a computer, we would like to represent groups through matrices using the most efficient means possible - that is, we shouldn't be able to take a particular representation and 'reduce' its size somehow. This motivates the idea of *reducible* and *irreducible* representations, which we discuss in detail in this chapter.

Generally speaking, a *reducible* representation is one where there is some vector subspace V of lower dimension than the ambient vector space to which the matrices could be reasonably restricted, and for the representation still to make sense. This is rather more complicated than just saying $\mathbb{C}^2 \subseteq \mathbb{C}^3$, because the subspace could somehow be 'tilted' - some sort of interesting hyperplane inside a larger space. If we can perform this restriction, we can produce a representation of smaller dimension from a given representation.

This motivates the idea of *irreducible representations*, which cannot be restricted in this way. Irreducible representations are very nice computationally as a result. But, even more surprisingly, we shall see in this chapter that *any* representation of a finite group can be represented as a 'direct sum' of irreducible representations in a particularly nice way - all representations are equivalent to 'block diagonal' representations of irreducibles.

In more detail, this chapter covers the following:

- We begin with reviewing some essential linear algebra. First, we introduce the notion of a vector subspace. We then discuss the idea of the complex scalar product, and abstract scalar products, and how we can use the Gram-Schmidt orthogonalisation procedure to always choose orthogonal bases of vector subspaces with respect to abstract scalar products. We conclude this review with a discussion of unitarity, and unitary matrices, which will be essential going forward.
- Next, we introduce the notion of reducibility via invariant subspaces and subrepresentations, with some examples. We then prove the equivalence to the criterion that a representation is equivalent to upper triangular form.
- Subsequently, we introduce the idea of a *direct sum*, adding matrices in block diagonal form. Direct sums are reducible in two obvious ways, because of their block diagonal nature.
- Finally, we state and prove the main results of the chapter. Using an abstract scalar product, we show that we can always choose our representation to be 'unitary' in some sense, preserving the idea of orthogonality. This means that invariant subspaces always have orthogonal partner invariant subspaces, allowing the reduction of an arbitrary representation to block diagonal form, with irreducibles along the diagonal.

10.1 Complex vector spaces, scalar products, and unitary matrices

Before we get stuck in to the matrix theory in this chapter, we will need several facts about the complex vector space $\mathbb{C}^n$, the set of n -dimensional complex vectors.

We start with the definition of a *vector subspace*, which generalises the idea of a line or plane in three dimensions to n dimensions:

Definition 10.1: A subset $V \subseteq \mathbb{C}^n$ is called an *m -dimensional subspace* of $\mathbb{C}^n$ if there exist m vectors $\mathbf{f}_1, \dots, \mathbf{f}_m \in V$ such that every element $\mathbf{v} \in V$ can be written uniquely in the form:

$$\mathbf{v} = \alpha_1 \mathbf{f}_1 + \dots + \alpha_m \mathbf{f}_m,$$

for some complex coefficients α_i , and all vectors of this form are in V .

Some examples are the following:

- The set of complex vectors of the form:

$$\begin{pmatrix} z \\ 0 \\ w \end{pmatrix} = z \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + w \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

forms a two-dimensional vector subspace of $\mathbb{C}^3$, with basis $\{(1, 0, 0), (0, 0, 1)\}$.

- The plane $z_1 + 2z_2 = 0$ in the complex space of vectors (z_1, z_2, z_3) is a vector subspace. We can solve the condition $z_1 + 2z_2 = 0$ to give $z_1 = -2z_2$, and z_3 is free, suggesting that every vector in the space can be written as:

$$\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \lambda \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Another important concept from vector geometry that it is useful to generalise to complex geometry is the idea of the *scalar product* - that is, how should we measure lengths and angles of complex vectors.

In real geometry, the scalar product of two vectors $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ in three dimensions is defined as:

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3.$$

One of the nice properties of this scalar product is that it encodes the length of a vector squared, by taking:

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + v_3^2.$$

Another useful property is that it measures the *angle* between vectors, so should satisfy $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$, as the angle doesn't care about the order. It is also true that $\mathbf{v} \cdot (\mathbf{w} + \mathbf{u}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{u}$, which one can see through a nice picture.

How should this idea be extended to complex vectors? Now the idea of angle doesn't make much sense anymore, but we can still take about length. The problem is that if we use the standard real scalar product, we can get lengths that are *negative*:

$$\begin{pmatrix} 1 \\ 2i \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2i \\ 0 \end{pmatrix} = 1 - 4 = -3.$$

This is very bad if we want to talk about how far apart things are. As a result, we make a slight tweak to the definition of the real scalar product when dealing with complex vectors: we take a complex conjugate first. This gives:

$$\mathbf{v} \cdot \mathbf{w} = v_1^* w_1 + v_2^* w_2 + v_3^* w_3$$

for complex vectors $\mathbf{v}, \mathbf{w}$. We now guarantee that $\mathbf{v} \cdot \mathbf{v} = |v_1|^2 + |v_2|^2 + |v_3|^2$ does indeed represent a length in a complex space.

Meanwhile, the property of symmetry has changed: we now have $\mathbf{v} \cdot \mathbf{w} = (\mathbf{w} \cdot \mathbf{v})^*$, since swapping the order changes which element we conjugate. The property $\mathbf{v} \cdot (\mathbf{w} + \mathbf{u}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{u}$ still holds with this new definition, though.

More generally, we can do this operation in n dimensions in the natural way. This leads to our definition of the *standard complex scalar product*:

Definition 10.2: Given two complex vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ in n dimensions, we define their *standard complex scalar product* (usually shortened to scalar product or dot product) via:

$$\mathbf{v} \cdot \mathbf{w} = v_1^* w_1 + v_2^* w_2 + \dots + v_n^* w_n.$$

However, even this will not be enough for us. For representation theory, we will need to abstract things further: we will need the concept of an ‘abstract’ scalar product. We take all the properties of the standard complex scalar product that we know and love (at least, have been acquainted with in the past five minutes), and we make them axioms instead:

Definition 10.3: An *abstract scalar product* on $\mathbb{C}^n$ is a function of two variables, $(\cdot, \cdot) : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ satisfying the following axioms:

- (S1) *Positive definiteness.* $(\mathbf{v}, \mathbf{v}) \geq 0$ for all vectors $\mathbf{v}$, with equality if and only if $\mathbf{v} = \mathbf{0}$.
- (S2) *Conjugate symmetry.* $(\mathbf{v}, \mathbf{w}) = (\mathbf{w}, \mathbf{v})^*$ for all vectors $\mathbf{v}$.
- (S3) *Linearity in the second argument.* $(\mathbf{v}, \lambda \mathbf{w} + \mu \mathbf{u}) = \lambda(\mathbf{v}, \mathbf{w}) + \mu(\mathbf{v}, \mathbf{u})$ for all vectors $\mathbf{v}, \mathbf{w}, \mathbf{u}$ and scalars λ, μ .

An easy example is a *weighted scalar product*, which generalises the formulae we found above. For example, we could define:

$$(\mathbf{v}, \mathbf{w}) = v_1^* w_1 + 2v_1^* w_2 + 3v_1^* w_3.$$

This satisfies all of the properties (S1), (S2), (S3), as you will verify on the Examples Sheet.

The Gram-Schmidt algorithm

The most important result concerning abstract scalar products for us is that if we have *some* abstract scalar product, we can always transform a set of basis vectors $\{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ into an *orthogonal* set with respect to this scalar product, without changing the object that they span.

Proposition 10.4: (The Gram-Schmidt algorithm) Given an abstract scalar product $(\cdot, \cdot)$ on $\mathbb{C}^n$, and a basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ of a vector subspace V , we can construct a new basis $\{\mathbf{f}'_1, \dots, \mathbf{f}'_m\}$ such that:

$$(\mathbf{f}'_i, \mathbf{f}'_j) = 0.$$

That is, the vectors are all *orthogonal* with respect to this abstract scalar product. Further, the subspaces described by $\{\mathbf{f}_1, \dots, \mathbf{f}_r\}$ and $\{\mathbf{f}'_1, \dots, \mathbf{f}'_r\}$ are identical.

Proof: The proof is by the *Gram-Schmidt procedure*. We define $\mathbf{f}'_1 = \mathbf{f}_1$; certainly these describe the same subspaces. Now we define a second vector $\mathbf{f}'_2$ by taking $\mathbf{f}_2$, then subtracting any components it had in the orthogonal directions we have already identified:

$$\mathbf{f}'_2 = \mathbf{f}_2 - \frac{(\mathbf{f}'_1, \mathbf{f}_2)}{(\mathbf{f}'_1, \mathbf{f}'_1)} \mathbf{f}'_1.$$

We have:

$$(\mathbf{f}'_1, \mathbf{f}'_2) = (\mathbf{f}'_1, \mathbf{f}_2) - \frac{(\mathbf{f}'_1, \mathbf{f}_2)}{(\mathbf{f}'_1, \mathbf{f}'_1)} (\mathbf{f}'_1, \mathbf{f}'_1) = 0,$$

using linearity in the second argument. Thus these vectors are orthogonal. Further, we have that:

$$\begin{aligned} \alpha_1 \mathbf{f}_1 + \alpha_2 \mathbf{f}_2 &= \alpha_1 \mathbf{f}'_1 + \alpha_2 \left(\mathbf{f}'_2 + \frac{(\mathbf{f}'_1, \mathbf{f}_2)}{(\mathbf{f}'_1, \mathbf{f}'_1)} \mathbf{f}'_1 \right) \\ &= \left(\alpha_1 + \alpha_2 \frac{(\mathbf{f}'_1, \mathbf{f}_2)}{(\mathbf{f}'_1, \mathbf{f}'_1)} \right) \mathbf{f}'_1 + \alpha_2 \mathbf{f}'_2. \end{aligned}$$

The coefficients on the right hand side can be picked independently, so the vector subspaces described by $\mathbf{f}'_1, \mathbf{f}'_2$ and $\mathbf{f}_1, \mathbf{f}_2$ are identical.

The proof continues inductively; the details are left as an exercise to the reader. $\square$

Unitary matrices

One final ingredient we will need is the idea of a *unitary matrix*, which will become key towards the end of this section. First, we make the definition:

Definition 10.5: A complex matrix U is called *unitary* if it preserves the standard complex scalar product; that is, we have:

$$U\mathbf{v} \cdot U\mathbf{w} = \mathbf{v} \cdot \mathbf{w}.$$

for all vectors $\mathbf{v}, \mathbf{w}$.

An important equivalent condition is the following:

Proposition 10.6: A complex matrix U is unitary if and only if its inverse is equal to its *Hermitian conjugate* $U = U^\dagger$. The Hermitian conjugate is the dual action of taking the transpose of the matrix (swapping columns and rows) and also taking the complex conjugate of all entries, i.e. $U^\dagger = (U^T)^*$.

Proof: Consider the action of U on the standard coordinate basis vectors $\mathbf{e}_i$. We must be able to write:

$$U\mathbf{e}_i = \sum_j U_{ji}\mathbf{e}_j,$$

since the image of the i th coordinate basis vector corresponds to the i th column of the matrix, which we can express as a sum of the original standard coordinate basis vectors. Now by the unitary condition, for any pair $\mathbf{e}_a, \mathbf{e}_b$ of the standard coordinate basis vectors we must have:

$$\mathbf{e}_a \cdot \mathbf{e}_b = U\mathbf{e}_a \cdot U\mathbf{e}_b = \left(\sum_j U_{ja}\mathbf{e}_j \right) \cdot \left(\sum_k U_{kb}\mathbf{e}_k \right) = \sum_j \sum_k U_{ja}^* U_{kb} \mathbf{e}_j \cdot \mathbf{e}_k = \sum_k U_{ka}^* U_{kb}.$$

But $U_{ka}^* = U_{ak}^\dagger$, giving precisely the condition that $U^\dagger U = I$, since $\mathbf{e}_a \cdot \mathbf{e}_b = 1$ for $a = b$ and 0 otherwise. This is the condition that its inverse is equal to its Hermitian conjugate.

The converse simply follows the argument backwards, giving the result. $\square$

This result is very useful, because it makes taking the inverse of a unitary matrix very efficient. It does not remain true for an abstract scalar product, but it is *almost* still true. Making the relevant definition first:

Definition 10.7: A complex matrix U is called *unitary with respect to an abstract scalar product* $(\cdot, \cdot)$ if for all $\mathbf{v}, \mathbf{w}$ we have:

$$(U\mathbf{v}, U\mathbf{w}) = (\mathbf{v}, \mathbf{w}).$$

This gives:

Proposition 10.8: A matrix U that is unitary with respect to an abstract scalar product $(\cdot, \cdot)$ is conjugate to a matrix which is unitary with respect to the standard complex scalar product, $U' = S^{-1}US$.

Proof: Let $\{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ be a basis of vectors for $\mathbb{C}^n$ that is orthogonal with respect to the abstract scalar product, and such that $(\mathbf{f}_i, \mathbf{f}_i) = 1$ for all i (this can be guaranteed simply by rescaling the vectors by their lengths with respect to the abstract scalar product). By the same argument as before, we may write:

$$U\mathbf{f}_i = \sum_j \bar{U}_{ji} \mathbf{f}_j,$$

where $\bar{U}_{ji}$ are not the matrix entries of U this time, because we are not using the standard basis, but are the matrix entries ‘with respect to the basis $\mathbf{f}_1, \dots, \mathbf{f}_n$ ’. Then the same argument as before gives:

$$\sum_k \bar{U}_{ak}^\dagger \bar{U}_{kb} = (\mathbf{f}_a, \mathbf{f}_b).$$

Now let the change of basis matrix be:

$$S = (\mathbf{f}_1 \quad \mathbf{f}_2 \quad \dots \quad \mathbf{f}_n).$$

Then for $\mathbf{e}_a, \mathbf{e}_b$ standard coordinate basis vectors, we have:

$$\begin{aligned} S^{-1}US\mathbf{e}_a \cdot S^{-1}US\mathbf{e}_b &= S^{-1}U\mathbf{f}_a \cdot S^{-1}U\mathbf{f}_b \\ &= \sum_j \sum_k \left(S^{-1}\bar{U}_{ja}\mathbf{f}_j, S^{-1}\bar{U}_{kb}\mathbf{f}_k \right) \\ &= \sum_j \sum_k \bar{U}_{ja}^* \bar{U}_{kb} \left(S^{-1}\mathbf{f}_j \cdot S^{-1}\mathbf{f}_k \right) \\ &= \sum_j \sum_k \bar{U}_{ja}^* \bar{U}_{kb} \mathbf{e}_j \cdot \mathbf{e}_k \\ &= \sum_j \bar{U}_{aj}^\dagger \bar{U}_{jb} \\ &= \begin{cases} 0 & \text{if } a \neq b, \\ 1, & \text{if } a = b. \end{cases} \\ &= \mathbf{e}_a \cdot \mathbf{e}_b. \end{aligned}$$

Thus $S^{-1}US$ is unitary with respect to the standard complex scalar product. $\square$

Overall then, we have seen that if a matrix is unitary with respect to *some* abstract scalar product, then it can be conjugated to a matrix which is unitary with respect to the *usual* complex scalar product, taking the form U with $U^\dagger = U^{-1}$.

10.2 Reducibility, invariant subspaces, and subrepresentations

Now, back to representation theory. Much like subgroups in group theory, quite often we can find a representation 'hidden' inside another representation. This can be used to reduce the dimension of a representation, which is extremely important computationally - the smaller the matrices we use to represent a group, the more efficient calculations will be on a computer.

Making this idea precise:

Definition 10.9: A representation $D : G \rightarrow GL(n, \mathbb{C})$ is called *reducible* if there exists a proper, non-trivial vector subspace $V \leq \mathbb{C}^n$ such that:

$$D(g)\mathbf{v} \in V$$

for all $\mathbf{v} \in V$. Such a vector subspace V is called an *invariant subspace*. We say that D is *irreducible* otherwise.

The reason that this makes sense is that we can define a *subrepresentation* on the invariant space V of smaller size:

Definition 10.10: Let V be an invariant subspace of the representation $D : G \rightarrow GL(n, \mathbb{C})$. Let a basis for the invariant subspace be $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ with $1 \leq m < n$. We note that for each $\mathbf{f}_i \in V$, we must have:

$$D(g)\mathbf{f}_i \in V \quad \Rightarrow \quad D(g)\mathbf{f}_i = \sum_j D'_{ji}(g)\mathbf{f}_j,$$

for some coefficients, $D'_{ab}(g)$. Defining a map $D' : G \rightarrow GL(m, \mathbb{C})$ gives a *subrepresentation* D' of D on V , with respect to the basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$.

Checking that this works, we note that:

$$D(g)D(h)\mathbf{f}_i = D(g) \left(\sum_j D'_{ji}(h)\mathbf{f}_j \right) = \sum_k \left(\sum_j D'_{ji}(h)D'_{kj}(g) \right) \mathbf{f}_k.$$

Now the matrix product as the coefficient of $\mathbf{f}_k$ is given by:

$$\sum_j D'_{ji}(h)D'_{kj}(g) = \sum_j D'_{kj}(g)D'_{ji}(h) = [D'(g)D'(h)]_{ki}.$$

Hence, $D'(gh) = D'(g)D'(h)$ as required.

Consider the following examples:

- An excellent example is the left regular representation of C_2 , which is given by:

$$D(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D(r) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This has a one-dimensional invariant subspace V , given by $z_1 = z_2$, the 'line through the origin, inclined equally to the z_1, z_2 axes'. It can be spanned by a basis $\{(1, 1)\}$. This means that D is reducible.

What is a subrepresentation? We consider acting with $D(g)$ on each of the proposed basis vectors:

$$D(e) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad D(r) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

This implies that we can construct a one-dimensional subrepresentation $D' : G \rightarrow \mathbb{C}^*$ given by $D'(g) = 1$ (the coefficient of the basis element $(1, 1)$ in each of the images is 1).

We can similarly find another one-dimensional invariant subspace V , given by $z_1 = -z_2$. This can be spanned by a basis $\{(1, -1)\}$. A subrepresentation on this subspace can be constructed by acting with $D(g)$ on each of the proposed basis vectors:

$$D(e) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad D(r) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = - \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

This gives the subrepresentation $D'(e) = 1, D'(r) = -1$.

- As another brief useful example, consider one-dimensional representations $D : G \rightarrow \mathbb{C}^*$. All one-dimensional representations are irreducible; indeed there are no proper non-trivial subspaces of $\mathbb{C}^*$, only $\{0\}$ and $\mathbb{C}^*$ itself.

It is often useful to develop another criterion for spotting reducibility in terms of the matrices themselves, rather than just having invariant subspaces. The key criterion here is the ability to *reduce to upper triangular form*:

Proposition 10.11: A representation $D : G \rightarrow GL(n, \mathbb{C})$ is reducible to an m -dimensional subrepresentation $D' : G \rightarrow GL(m, \mathbb{C})$ if and only if it is equivalent to a representation of the form:

$$\tilde{D}(g) = \begin{pmatrix} D'(g) & \alpha(g) \\ 0 & \beta(g) \end{pmatrix},$$

where $\alpha(g), \beta(g)$ are some block matrices (of sizes $m \times (n - m)$ and $(n - m) \times (n - m)$ respectively).

Proof: Suppose that there is an equivalence:

$$S^{-1}D(g)S = \begin{pmatrix} D'(g) & \alpha(g) \\ 0 & \beta(g) \end{pmatrix}.$$

Let $S = (\mathbf{f}_1, \dots, \mathbf{f}_n)$ be the columns of the intertwiner S . Then acting on $\mathbf{e}_i$ for $1 \leq i \leq m$, the right hand side maps $\mathbf{e}_i$ to an element in the space V with basis $\{\mathbf{e}_1, \dots, \mathbf{e}_m\}$. Hence V with basis $\{\mathbf{e}_1, \dots, \mathbf{e}_m\}$ is invariant for the right hand side. This implies that:

$$S^{-1}D(g)S\mathbf{e}_i \in V \quad \Rightarrow \quad D(g)\mathbf{f}_i \in SV.$$

But SV is a space with basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ by construction. So SV is an invariant subspace of D , and hence D is reducible. Since we now know that SV is invariant, in the above calculation we can write:

$$S^{-1}D(g)S\mathbf{e}_i = S^{-1}D(g)\mathbf{f}_i = S^{-1} \left(\sum_j d_{ji}(g)\mathbf{f}_j \right) = \sum_j d_{ji}(g)\mathbf{e}_j$$

for some coefficients $d_{ji}(g)$. But comparing with the right hand side, we recognise that $d_{ji}(g) = D'_{ji}(g)$, so indeed we have the subrepresentation.

Conversely, suppose that $D : G \rightarrow GL(n, \mathbb{C})$ is reducible to an m -dimensional subrepresentation $D' : G \rightarrow GL(m, \mathbb{C})$. Let V be an invariant subspace with basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, and define a change of basis matrix $S = (\mathbf{f}_1, \dots, \mathbf{f}_m, \mathbf{e}_{m+1}, \dots, \mathbf{e}_n)$ with the appropriate columns. Then for the i th coordinate basis vector $\mathbf{e}_i$ with $1 \leq i \leq m$, we have:

$$S^{-1}D(g)S\mathbf{e}_i = S^{-1}D(g)\mathbf{f}_i = S^{-1} \left(\sum_j D'_{ji}(g)\mathbf{f}_j \right) = \sum_j D'_{ji}(g)\mathbf{e}_j.$$

This implies that the i th column of $S^{-1}D(g)S$ consists of the i th column of entries of the matrix $D'(g)$. Meanwhile, for the i th coordinate basis vector $\mathbf{e}_i$ with $m+1 \leq i \leq n$, it really does not matter where things are sent. We have shown that for some $\alpha(g), \beta(g)$, we have:

$$S^{-1}D(g)S = \begin{pmatrix} D'(g) & \alpha(g) \\ 0 & \beta(g) \end{pmatrix},$$

hence we're done. $\square$

This gives us another easy example. Recall the example of the additive group of integers $\mathbb{Z}$ with the representation $D : \mathbb{Z} \rightarrow GL(2, \mathbb{C})$ given by:

$$D(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

This is upper triangular for all n , so this is reducible to the subrepresentation $D : \mathbb{Z} \rightarrow \mathbb{C}^*$ given by $D(n) = 1$.

10.3 Direct sums

Just as we have subrepresentations, similar to subgroups in group theory, we have the idea of *direct sums*, similar to direct products in group theory. We can take two representations, and stick them together to make a larger representation without them talking to one another.

Definition 10.12: Let $D_1 : G \rightarrow GL(n, \mathbb{C})$, $D_2 : G \rightarrow GL(m, \mathbb{C})$ be representations, which may be of different dimension. We define their *direct sum* to be the representation $D : G \rightarrow GL(n+m, \mathbb{C})$ given by:

$$D(g) = \begin{pmatrix} D_1(g) & 0 \\ 0 & D_2(g) \end{pmatrix};$$

that is, we take $D(g)$ to be a block diagonal matrix with the upper left hand corner equal to $D_1(g)$, and the lower right hand corner equal to $D_2(g)$. The direct sum is written as $D = D_1 \oplus D_2$.

It is trivial to check that this constitutes a representation, and is left as an exercise to the reader.

Importantly, we note that direct sums are *reducible*, because they are in upper triangular form. Even more impressively though, they are reducible to *two* different subrepresentations - both D_1 and D_2 . This is because $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\{\mathbf{e}_{n+1}, \dots, \mathbf{e}_{n+m}\}$ are both independently invariant subspaces.

This suggests that given a representation D , we could try to break it down as a direct sum. Then we could treat the summands D_1, D_2 as completely independent new representations, and try to break them down too. We can try to carry on doing this until we get to irreducible representations.

The only difficulty with this procedure, is that sometimes a representation might be reducible, but it might not be able to be broken down into a direct sum; that is, it could be of *block upper triangular form*, but not *block diagonal form*. Therefore, we give a special name to the case when the algorithm of ‘break down into direct sums’ successfully reaches irreducibles:

Definition 10.13: A representation D is called *completely reducible* if it is equivalent to a direct sum of irreducible representations D_i :

$$D \cong D_1 \oplus \dots \oplus D_n.$$

In the rest of this chapter, we will prove the remarkable result that *any* representation of a finite group is completely reducible. That is, the algorithm for breaking down representations using subrepresentations always works.

10.4 Unitarity and Maschke’s theorem

The main result we shall try to prove for the remainder of this chapter is the result that every representation of a finite group is completely reducible. The smart trick we shall use is a cleverly chosen abstract scalar product.

First, we define what it means for a representation to ‘play nicely’ with an abstract scalar product; we make this definition in the usual way, according to our understanding of an abstract scalar product:

Definition 10.14: Let $D : G \rightarrow GL(n, \mathbb{C})$ be a representation of the group G . We say that D is a *unitary* representation with respect to an abstract scalar product $(\cdot, \cdot)$ on $\mathbb{C}^n$ if:

$$(D(g)\mathbf{v}, D(g)\mathbf{w}) = (\mathbf{v}, \mathbf{w})$$

for all vectors $\mathbf{v}, \mathbf{w}$. That is, the action of the representation ‘respects’ the scalar product.

Unitary representations already have the desired property of complete reducibility:

Lemma 10.15: If $D : G \rightarrow GL(n, \mathbb{C})$ is a reducible unitary representation with respect to some abstract scalar product $(\cdot, \cdot)$ on $\mathbb{C}^n$, then it is equivalent to a direct sum of representations:

$$D \cong D_1 \oplus D_2.$$

Iterating this procedure, we see that D is completely reducible.

Proof: Since D is reducible, it has an invariant subspace V . Let $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ be a basis for the vector subspace V , and extend this basis to a basis for all of $\mathbb{C}^n$ with some vectors $\{\mathbf{f}_{m+1}, \dots, \mathbf{f}_n\}$. By the Gram-Schmidt procedure applied to all these vectors, we may assume that they are all mutually orthogonal to one another. As a result, we can define a matrix:

$$S = (\mathbf{f}_1 \quad \dots \quad \mathbf{f}_m \quad \mathbf{f}_{m+1} \quad \dots \quad \mathbf{f}_n).$$

Consider now the action of the matrix $S^{-1}D(g)S$ on the standard coordinate basis $\mathbf{e}_1, \dots, \mathbf{e}_n$. For $1 \leq i \leq m$, we have:

$$S^{-1}D(g)S\mathbf{e}_i = S^{-1}D(g)\mathbf{f}_i \in \{S^{-1}\mathbf{v} : \mathbf{v} \in V\}.$$

using the invariance of V . This is the pre-image of the space V under the matrix, and thus can only have non-zero components in the $\mathbf{e}_1, \dots, \mathbf{e}_m$ directions.

Now for $m+1 \leq i \leq n$, we have:

$$S^{-1}D(g)S\mathbf{e}_i = S^{-1}D(g)\mathbf{f}_i.$$

Observe that if $1 \leq j \leq n$, we have:

$$(D(g)\mathbf{f}_i, \mathbf{f}_j) = (D(g^{-1})D(g)\mathbf{f}_i, D(g^{-1})\mathbf{f}_j) = (\mathbf{f}_i, D(g^{-1})\mathbf{f}_j).$$

But since $D(g^{-1})\mathbf{f}_j \in V$ by invariance, this scalar product must give zero, since $\mathbf{f}_i$ is orthogonal to the vectors in V . Therefore, $D(g)\mathbf{f}_i$ is orthogonal to V , and hence can be expressed in terms of $\mathbf{f}_{m+1}, \dots, \mathbf{f}_n$. In particular, the inverse image only gives non-zero components in the $\mathbf{e}_{m+1}, \dots, \mathbf{e}_n$ directions. In particular, this implies that $S^{-1}D(g)S$ takes the form:

$$S^{-1}D(g)S = \begin{pmatrix} D_1(g) & 0 \\ 0 & D_2(g) \end{pmatrix}.$$

This is a direct sum representation, since:

$$\begin{aligned} \begin{pmatrix} D_1(g)D_2(g) & 0 \\ 0 & D_1(h)D_2(h) \end{pmatrix} &= \begin{pmatrix} D_1(g) & 0 \\ 0 & D_2(g) \end{pmatrix} \begin{pmatrix} D_1(h) & 0 \\ 0 & D_2(h) \end{pmatrix} \\ &= S^{-1}D(g)SS^{-1}D(h)S \\ &= S^{-1}D(g)D(h)S \\ &= S^{-1}D(gh)S \\ &= \begin{pmatrix} D_1(gh) & 0 \\ 0 & D_2(gh) \end{pmatrix}. \end{aligned}$$

Hence $D_1(g)D_1(h) = D_1(gh)$ and $D_2(g)D_2(h) = D_2(gh)$, so that this is the direct sum of some subrepresentations. $\square$

The clever part of the remainder of this chapter is therefore: can we find an abstract scalar product for which an arbitrary representation D is unitary? The answer is *yes*, and it is given by *Weyl's unitary trick*:

Lemma 10.16: For any finite representation $D : G \rightarrow GL(n, \mathbb{C})$, there exists an abstract scalar product $(\cdot, \cdot)$ on $\mathbb{C}^n$ with respect to which $D(g)$ is unitary. As an immediate corollary, we may always choose an equivalent representation D' to D such that all matrices in the representation satisfy $D'(g)^\dagger = D'(g)^{-1}$, i.e. they are unitary with respect to the standard complex scalar product.

Proof: Let $(\cdot, \cdot)$ be the standard complex scalar product. Then we claim we can define the abstract scalar product via ‘averaging’ over the group elements:

$$[\mathbf{v}, \mathbf{w}] = \sum_{g \in G} (D(g)\mathbf{v}, D(g)\mathbf{w}).$$

We merely need to check the axioms:

(S1) *Positivity.* $[\mathbf{v}, \mathbf{v}] = \sum_{g \in G} (D(g)\mathbf{v}, D(g)\mathbf{v}) \geq 0$, by positivity of the standard complex scalar product. Further, we have equality if and only if $(D(g)\mathbf{v}, D(g)\mathbf{v}) = 0$ for each $g \in G$, which occurs if and only if $D(g)\mathbf{v} = \mathbf{0}$ for each $g \in G$, which occurs if and only if $\mathbf{v} = \mathbf{0}$.

(S2) *Conjugate symmetry.* We have:

$$[\mathbf{v}, \mathbf{w}]^* = \sum_{g \in G} (D(g)\mathbf{v}, D(g)\mathbf{w})^* = \sum_{g \in G} (D(g)\mathbf{w}, D(g)\mathbf{v}) = [\mathbf{w}, \mathbf{v}],$$

as required.

(S3) *Linearity in the second argument.* We have:

$$\begin{aligned} [\mathbf{v}, \lambda \mathbf{w} + \mu \mathbf{u}] &= \sum_{g \in G} (D(g)\mathbf{v}, \lambda D(g)\mathbf{w} + \mu D(g)\mathbf{u}) \\ &= \lambda \sum_{g \in G} (D(g)\mathbf{v}, D(g)\mathbf{w}) + \mu \sum_{g \in G} (D(g)\mathbf{v}, D(g)\mathbf{u}) \\ &= \lambda [\mathbf{v}, \mathbf{w}] + \mu [\mathbf{v}, \mathbf{u}]. \end{aligned}$$

Finally, we need to show that this new abstract inner product makes D unitary. We have:

$$\begin{aligned} [D(h)\mathbf{v}, D(h)\mathbf{w}] &= \sum_{g \in G} (D(g)D(h)\mathbf{v}, D(g)D(h)\mathbf{w}) \\ &= \sum_{g \in G} (D(gh)\mathbf{v}, D(gh)\mathbf{w}) \\ &= \sum_{g \in G} (D(g)\mathbf{v}, D(g)\mathbf{w}) \\ &= [\mathbf{v}, \mathbf{w}]. \end{aligned}$$

In going from the second to the third lines, we did something a bit scary - we just magically dropped the h 's. This is fine, because right multiplication by some fixed group element h just permutes all the other group elements - so, summing over g is equivalent to summing over gh . Thus, D is unitary with respect to $[\cdot, \cdot]$ and we're done. $\square$

This immediately gives the final result:

Theorem 10.17: (Maschke's theorem) Every representation $D : G \rightarrow GL(n, \mathbb{C})$ of a finite group is equivalent to a direct sum of irreducible representations.

Proof: This is an immediate consequence of the above lemmas, together with induction. $\square$

Chapter 11

The great orthogonality theorems

11.1 Schur's lemmas

One of the basic results about irreducible representations...

Lemma 11.1: (Schur's lemmas) Suppose that $D_1, D_2 : G \rightarrow GL(n, \mathbb{C})$ are irreducible representations of a group G . Let S be a matrix satisfying:

$$SD_1(g) = D_2(g)S,$$

for all $g \in G$. Then:

- (i) if D_1, D_2 are inequivalent representations, we have $S = 0$;
- (ii) if $D_1 = D_2 = D$, we have $S = \lambda I$ for some complex number $\lambda \in \mathbb{C}$.

Proof:

- (i) In the first case, we consider the *kernel* and *image* of the matrix S . We define these as follows:

$$\text{Ker}(S) = \{\mathbf{v} : S\mathbf{v} = \mathbf{0}\}, \quad \text{Im}(S) = \{\mathbf{w} : \mathbf{w} = S\mathbf{v} \text{ for some } \mathbf{v}\}.$$

Both are vector subspaces, as we can readily check.

Suppose that $\mathbf{v} \in \text{Ker}(S)$. Then using the relation $SD_1(g) = D_2(g)S$ we have $SD_1(g)\mathbf{v} = \mathbf{0}$, from which it follows that $D_1(g)\mathbf{v} \in \text{Ker}(S)$. Thus $\text{Ker}(S)$ is an invariant subspace of D_1 , implying that either $\text{Ker}(S) = \{\mathbf{0}\}$ or $\text{Ker}(S) = \mathbb{C}^n$, since D_1 is irreducible.

If $\text{Ker}(S) = \mathbb{C}^n$, then S sends every vector to zero, so must itself be the zero matrix. Otherwise, the kernel of S is trivial.

Now consider the image of S . If $\mathbf{w} \in \text{Im}(S)$, there exists a vector $\mathbf{v}$ such that $\mathbf{w} = S\mathbf{v}$. Therefore we have:

$$D_2(g)\mathbf{w} = D_2(g)S\mathbf{v} = S(D_1(g)\mathbf{v}) \in \text{Im}(S),$$

so $\text{Im}(S)$ must be an invariant subspace of D_2 . Therefore $\text{Im}(S) = \{\mathbf{0}\}$ or $\text{Im}(S) = \mathbb{C}^n$, since D_2 is irreducible.

If $\text{Im}(S) = \{\mathbf{0}\}$, then S sends every vector to zero, so must itself be the zero matrix. Otherwise, we now have that kernel of S is $\{\mathbf{0}\}$ and the image of S is $\mathbb{C}^n$, which is enough to conclude that the matrix S must be an invertible matrix.¹⁸

But then $SD_1(g) = D_2(g)S$, giving $D_1(g) = S^{-1}D_2(g)S$ implying the equivalence of the representations, which is a contradiction. It follows that S is the zero matrix, as required.

- (ii) In the second case, we consider the *eigenspaces* of the matrix S . Every complex matrix has at least one complex eigenvalue λ , satisfying $S\mathbf{v} = \lambda\mathbf{v}$ for some non-zero $\mathbf{v}$. The *eigenspace* is the vector space V of all vectors $\mathbf{v}$ satisfying this equation (called *eigenvectors*).

Let λ be an eigenvalue of S . Then if $\mathbf{v} \in V$ is in the eigenspace, we have:

$$SD(g)\mathbf{v} = D(g)S\mathbf{v} = \lambda D(g)\mathbf{v},$$

so that $D(g)\mathbf{v} \in V$. Thus V is an invariant subspace of D , so must be $V = \{\mathbf{0}\}$ or $V = \mathbb{C}^n$ because D is irreducible. But $V \neq \{\mathbf{0}\}$, since every complex matrix has at least one complex eigenvalue with at least one non-zero complex eigenvector.

Thus, we conclude that $V = \mathbb{C}^n$, and it follows that *every* vector is an eigenvector of S . This gives $S\mathbf{v} = \lambda\mathbf{v}$ for all $\mathbf{v}$, implying that $S = \lambda I$. $\square$

An easy consequence of Schur's lemma is the following:

Proposition 11.2: Suppose that G is an Abelian group. Then all irreducible representations of G are one-dimensional.

Proof: Let $D : G \rightarrow GL(n, \mathbb{C})$ be an irreducible representation of G . Fix any $g \in G$, and consider any other $h \in G$. By the commutativity of the group, we must have:

$$D(g)D(h) = D(h)D(g)$$

for all $h \in G$. But then Schur's lemma implies that $D(g) = \lambda_g I$ for some complex constant λ_g (which can depend on the fixed g). This is irreducible if and only if we are working in one dimension. $\square$

11.2 The great orthogonality theorem

The main application of Schur's lemmas in this course will be to the *great orthogonality theorem*, which tells us interesting things about the matrices that can be used in irreducible representations, and their corresponding characters. We begin by stating and proving the main result:

Theorem 11.3: (The great orthogonality theorem) Suppose that $D_1, D_2 : G \rightarrow GL(n, \mathbb{C})$ are irreducible representations of the group G ; assume without loss of generality that they are unitary. Then:

(i) if D_1, D_2 are inequivalent representations, we have:

$$\sum_g D_1(g)_{ij} D_2(g)_{kl}^{-1} = 0$$

for all elements $D_1(g)_{ij}$ and $D_2(g)_{kl}$ of the matrices $D_1(g), D_2(g)$ respectively;

(ii) if $D_1 = D_2 = D$, we have:

$$\sum_g D(g)_{ij} D(g)_{kl}^{-1} = \frac{|G|}{n} \delta_{il} \delta_{jk}.$$

Here, the symbol δ_{il} is the *Kronecker delta*. This is equal to one if $i = l$, and is equal to zero otherwise.

Proof: Let X be an arbitrary matrix, which we shall choose later. Define the matrix:

$$S = \sum_{h \in G} D_1(h) X D_2(h)^{-1}.$$

Then we have:

$$\begin{aligned}
D_1(g')S &= \sum_{g \in G} D_1(g')D_1(g)XD_2(g)^{-1} \\
&= \left(\sum_{g \in G} D_1(g'g)XD_2(g)^{-1}D_2(g')^{-1} \right) D_2(g') \\
&= \left(\sum_{g \in G} D_1(g'g)XD_2(g'g)^{-1} \right) D_2(g') \\
&= SD_2(g').
\end{aligned}$$

Note that we can change the summation variable from $g'g$ back to g in the sum, because $g'g$ is just a permutation of the group elements. Now by Schur's lemma, since the above identity holds for all $g' \in G$, we must have:

$$\sum_{g \in G} D_1(g)XD_2(g)^{-1} = \begin{cases} 0, & \text{if } D_1, D_2 \text{ are inequivalent,} \\ \lambda I, & \text{if } D_1 = D_2. \end{cases}$$

Now choose X to be the matrix which has zeroes everywhere, and a non-zero entry at the (j, k) th entry. Then we have that the (i, l) th entry of the matrix $D_1(g)XD_2(g)^{-1}$ is given by:

$$[D_1(g)XD_2(g)^{-1}]_{il} = D_1(g)_{ij}D_2(g)^{-1}_{kl}.$$

Hence, we have established that:

$$\sum_{g \in G} D_1(g)_{ij}D_2(g)^{-1}_{kl} = \begin{cases} 0, & \text{if } D_1, D_2 \text{ are inequivalent,} \\ \lambda \delta_{il}, & \text{if } D_1 = D_2. \end{cases}$$

It remains to determine the constant λ in the case $D_1 = D_2 = D$. Taking the trace of the matrices in (i, l) space on both sides, we have:

$$\sum_{i=1}^n \sum_{g \in G} D(g)_{ij}D(g)^{-1}_{ki} = \lambda n.$$

But note that $D(g)^{-1}_{ki}D(g)_{ij}$ corresponds to multiplication of the matrix $D(g)^{-1}$ with the matrix $D(g)$, as we are summing over a changing column in the first matrix (with a fixed row k) and a changing row in the second matrix (with a fixed column j). Since $D(g)^{-1}D(g) = I$, this gives the (k, j) th entry of the identity matrix, δ_{jk} . Hence we can conclude that:

$$\sum_{g \in G} \delta_{jk} = \lambda n \quad \Rightarrow \quad \lambda = \frac{|G|}{n} \delta_{jk},$$

as required. $\square$

What does the great orthogonality theorem gives us? One of the incredible facts it gives us immediately is that a finite group may only have finitely many irreducible representations, and there is a tight constraint

on their sizes.

Corollary 11.4: A finite group G only has finitely many irreducible representations (up to equivalence), say $D_1, \dots, D_n$, with dimensions $d_1, \dots, d_n$ satisfying:

$$d_1^2 + d_2^2 + \dots + d_n^2 \leq |G|.$$

Proof:

We have a similar result for characters which comes from the above:

Theorem 11.5: (The great orthogonality theorem for characters) Let $\chi_{D_1}, \chi_{D_2} : G \rightarrow \mathbb{C}$ be the characters of two irreducible representations D_1, D_2 . Then we have the following:

(i)

(ii)

Proof:

11.3 Character tables

Definition 11.6: The *character table* of a finite group G is a table with header row indexed by the elements $g \in G$, and rows indexed by the characters of the irreducible representations of G . The (χ_D, g) entry is given by $\chi_D(g)$.

We now know the following facts about the character tables:

- There is always a trivial representation along the first row, with ones in every entry.
- The number of rows is equal to the number of conjugacy classes of the group.
- The rows must be orthogonal vectors.
- The columns must be orthogonal vectors.
- The sums of the entries of the rows, accounting for the size of the conjugacy classes, excluding the first row must equal zero.

This information is often enough to completely determine the character table of a group. We consider the following examples:

- The character table of C_3 is determined from this.

	e	r	r^2
χ_{D_1}	1	1	1
χ_{D_2}	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
χ_{D_3}	1	$e^{4\pi i/3}$	$e^{2\pi i/3}$

- The character table of D_3 is determined from this. There are three conjugacy classes, so the character table takes the form:

	e	r	r^2	m	rm	r^2m
χ_{D_1}	1	1	1	1	1	1
χ_{D_2}	1	1	1	-1	-1	-1
χ_{D_3}	2	-1	-1	0	0	0

11.4 Exercises

Reducibility

1. By identifying invariant subspaces, show that the following representations are reducible. Construct subrepresentations in each case.
 - (a)
 - (b)
 - (c)
 - (d)
2. Prove that the left regular representation of a group G is never irreducible.

Schur's lemmas

3. Let D_1, D_2 be two equivalent irreducible representations, and suppose that S, T are two intertwiners satisfying:

$$D_1(g) = S^{-1}D_2(g)S, \quad D_1(g) = T^{-1}D_2(g)T.$$

Show that $S = \lambda T$ for some non-zero complex scalar λ .

4. Suppose that we define a two-dimensional representation D of C_n by extending from the generator:

$$D(r) = \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}.$$

Find all matrices S such that $SD(g) = D(g)S$ for all $g \in G$. [Hint: consider the cases $n \leq 2, n \geq 3$ separately.] Does Schur's lemma apply here? If not, why not?

The great orthogonality theorem

- 5.

Character tables

6. Construct the character tables for D_4 and Q_8 using the consequences of the great orthogonality theorem.

Bonus questions

7. (*) Show that if G is a finite group with a faithful irreducible complex representation, then the centre $Z(G)$ is cyclic.

Chapter 12

Direct sums and Maschke's theorem

12.1 Direct sums

Now we have identified how to reduce the size of a representation, we can ask how to stick representations together to produce larger representations. This is analogous to the idea of a direct product of groups: the *direct sum* of representations:

Definition 12.1: Let $f_1 : G \rightarrow GL(n, \mathbb{C})$ and $f_2 : G \rightarrow GL(m, \mathbb{C})$ be representations of a group G . We define the *direct sum* of the representations to be the map $f_1 \oplus f_2 : G \rightarrow GL(n + m, \mathbb{C})$ given by:

$$(f_1 \oplus f_2)(g) = \begin{pmatrix} f_1(g) & 0 \\ 0 & f_2(g) \end{pmatrix}.$$

We claim that this is still a representation.

Proposition 12.2: The direct sum $f_1 \oplus f_2$ of the representations $f_1 : G \rightarrow GL(n, \mathbb{C}), f_2 : G \rightarrow GL(m, \mathbb{C})$ is itself an $n + m$ dimensional representation.

Proof: An easy exercise for the reader. The matrices are block diagonal in this case, which makes the multiplication straightforward. $\square$

Consider the following examples of direct sums of representations:

-
-

Since characters are useful invariants, it is useful to ask how they behave under direct sums. We have the following proposition:

Proposition 12.3: Let $f_1 : G \rightarrow GL(n, \mathbb{C})$ and $f_2 : G \rightarrow GL(m, \mathbb{C})$ be representations of a group G . The character of the direct sum $f_1 \oplus f_2$ satisfies:

$$\chi_{f_1 \oplus f_2}(g) = \chi_{f_1}(g) + \chi_{f_2}(g).$$

Proof: Trivial from the definition of the character. $\square$

12.2 Inner products and unitary representations

Definition 12.4: A complex inner product on $\mathbb{C}^n$ is a map $(\cdot, \cdot) : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ satisfying:

- (a) *Positive definiteness.* For all vectors $\mathbf{v}$, we have $(\mathbf{v}, \mathbf{v}) \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$.
- (b) *Conjugate symmetry.* For all vectors $\mathbf{v}, \mathbf{w}$, we have $(\mathbf{v}, \mathbf{w})^* = (\mathbf{w}, \mathbf{v})$.
- (c) *Linearity in the second argument.* For all vectors $\mathbf{v}, \mathbf{w}, \mathbf{u}$ and scalars λ, μ , we have:

$$(\mathbf{v}, \lambda \mathbf{w} + \mu \mathbf{u}) = \lambda (\mathbf{v}, \mathbf{w}) + \mu (\mathbf{v}, \mathbf{u}).$$

Proposition 12.5: Let $(\cdot, \cdot)$ be a complex inner product on $\mathbb{C}^n$. For any vector subspace $V \leq \mathbb{C}^n$, there exists a basis which is

12.3 Maschke's theorem

When we were studying groups, we initially studied subgroups and direct products... Unfortunately, we saw that it is not always the case that if you can find a subgroup $H \leq G$, then there is some other subgroup $K \leq G$ such that $G \cong H \times K$. Therefore, subgroups and direct products turn out not to be a particularly useful tool in classifying *all* group structure.

Fantastically, representations are much more well-behaved. It turns out this *is* always the case (for finite groups, at least). This result is called *Maschke's theorem*, which we state and prove in this chapter.

The complex inner product, and unitary representations

Definition 12.6: Let $\mathbf{v} = (v_1, \dots, v_n)$, $\mathbf{w} = (w_1, \dots, w_n)$ be two complex vectors of dimension n . We define the *complex inner product* to be:

$$\mathbf{v} \cdot \mathbf{w} = v_1^* w_1 + v_2^* w_2 + \dots + v_n^* w_n.$$

Definition 12.7: A representation $f : G \rightarrow GL(n, \mathbb{C})$ is called *unitary* if it preserves the complex inner product; that is, for all $g \in G$ we have:

$$\mathbf{v} \cdot \mathbf{w} = f(g)\mathbf{v} \cdot f(g)\mathbf{w}.$$

Proposition 12.8: Suppose that $f : G \rightarrow GL(n, \mathbb{C})$ is a unitary representation, and suppose that $f_1 : G \rightarrow GL(m, \mathbb{C})$ is a proper non-trivial subrepresentation. Then there exists some representation $f_2 : G \rightarrow GL(n - m, \mathbb{C})$ such that f is equivalent to the direct sum of f_1, f_2 :

$$f \cong f_1 \oplus f_2.$$

Proof: Since f has a proper non-trivial subrepresentation f_1 , there exists some m -dimensional invariant vector subspace V of $\mathbb{C}^n$. Let $\mathbf{v}_1, \dots, \mathbf{v}_m$ be an orthogonal basis for V , and extend this to an orthogonal basis for all of $\mathbb{C}_n$ as:

$$\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n.$$

Define the intertwining matrix to be the matrix whose columns are these basis vectors:

$$S = (\mathbf{v}_1 \quad \dots \quad \mathbf{v}_m \quad \mathbf{w}_{m+1} \quad \dots \quad \mathbf{w}_n).$$

Then we claim that:

$$S^{-1}f(g)S = \begin{pmatrix} f_1(g) & 0 \\ 0 & f_2(g) \end{pmatrix}$$

for some $f_2 : G \rightarrow GL(n - m, \mathbb{C})$. To see this, observe that:

- If the matrix $S^{-1}f(g)S$ acts on $\mathbf{e}_i$, the i th coordinate basis vector, then $S\mathbf{e}_i = \mathbf{v}_i$, $f(g)S\mathbf{e}_i \in V$, since V is an invariant subspace, and $S^{-1}f(g)S\mathbf{e}_i$ is therefore in the span of the first m coordinate basis vectors. Hence it has no component in the directions $\mathbf{e}_{m+1}, \dots, \mathbf{e}_n$.
- Meanwhile, since $\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n$ are all orthogonal, we have:

$$f(g)\mathbf{v}_i \cdot f(g)\mathbf{w}_j = 0,$$

by the unitarity of f . It follows that there is no overlap between the image of ...

Theorem 12.9: (Maschke's theorem) Every representation $f : G \rightarrow GL(n, \mathbb{C})$ of a finite group G is equivalent to a direct sum of irreducible representations,

$$f \cong f_1 \oplus f_2 \oplus \dots \oplus f_n.$$

Proof: Let $f : G \rightarrow GL(n, \mathbb{C})$ be a representation of a finite group G . If f is irreducible, we are done. Else there exists an intertwiner S such that:

$$Sf(g)S^{-1} = \begin{pmatrix} f'(g) & \alpha(g) \\ 0 & \beta(g) \end{pmatrix}$$

for all $g \in G$, where $f' : G \rightarrow GL(n', \mathbb{C})$ is a representation with $n' < n$, and $\alpha(g), \beta(g)$ are matrices. We must show that there is an intertwiner that removes the top right part of this block matrix.

It is sufficient to show that f is equivalent to a unitary representation.